

Chapter5

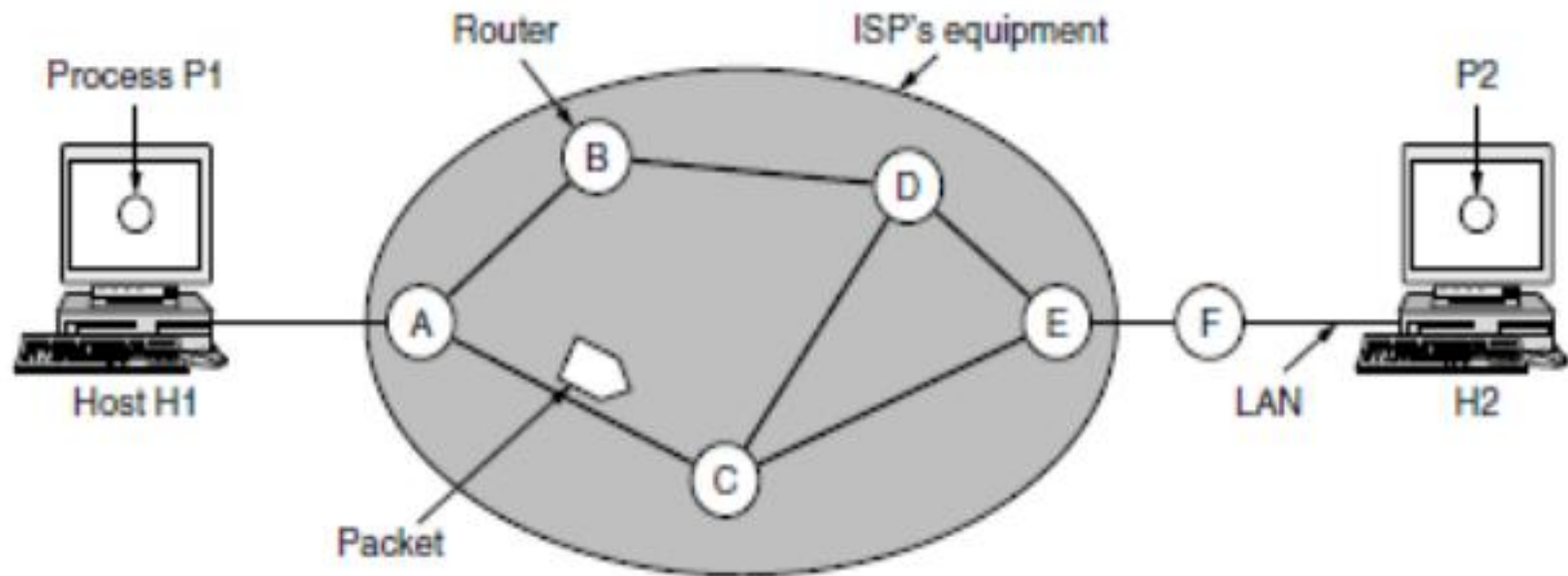
The Network Layer

(around 14 hours)

Outline

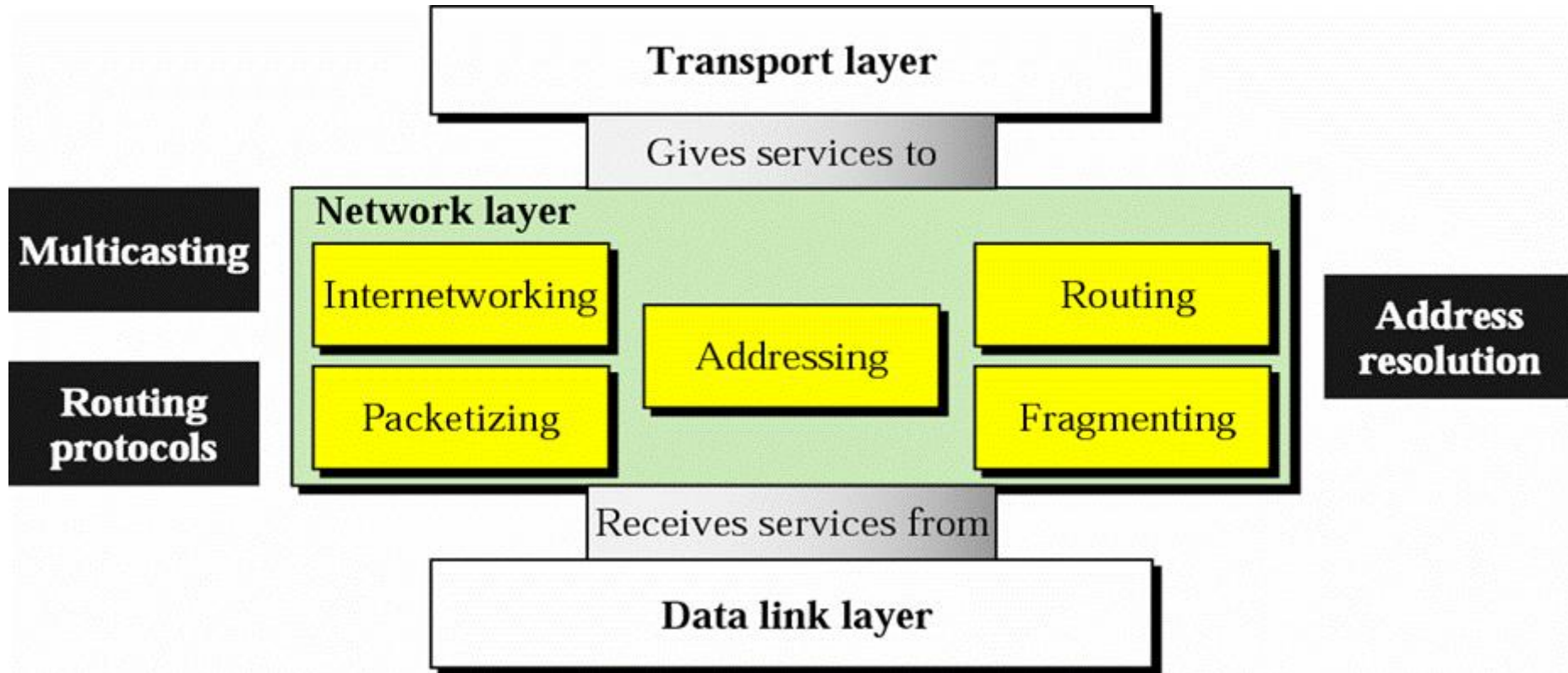
- 5.1 Introduction to Network Layer
- 5.2 Routing Algorithms
- 5.3 Congestion Control
- 5.4 Quality of Service
- 5.5 Internetworking
- 5.6 The network layer in Internet

5.1.1 Environment of Network Layer Protocols



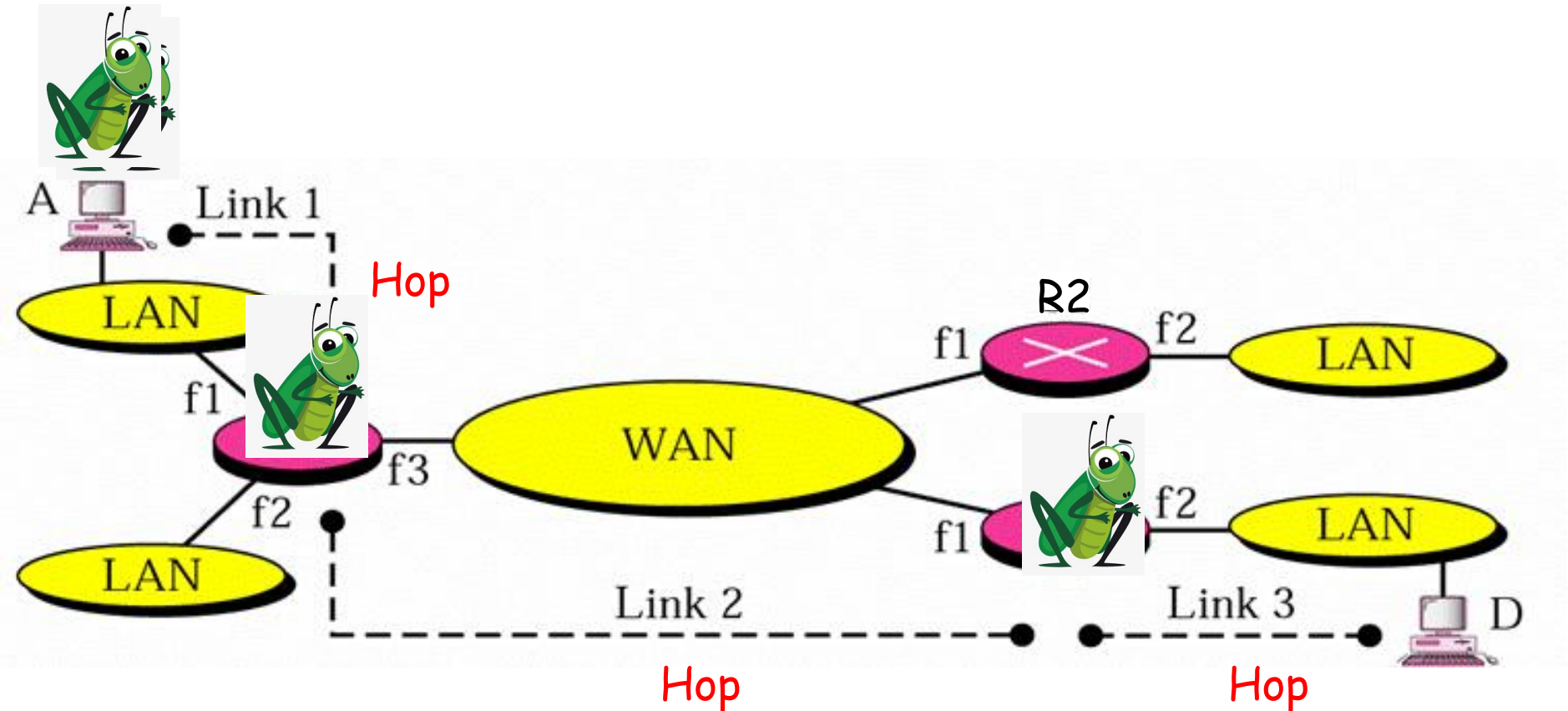
Store-and-Forward Packet Switching
(存储转发 分组交换)

(1) Positions and Functions of Network Layer



Network layer is the lowest layer that deals with **end-to-end (host-to-host)** transmission (端到端传输)

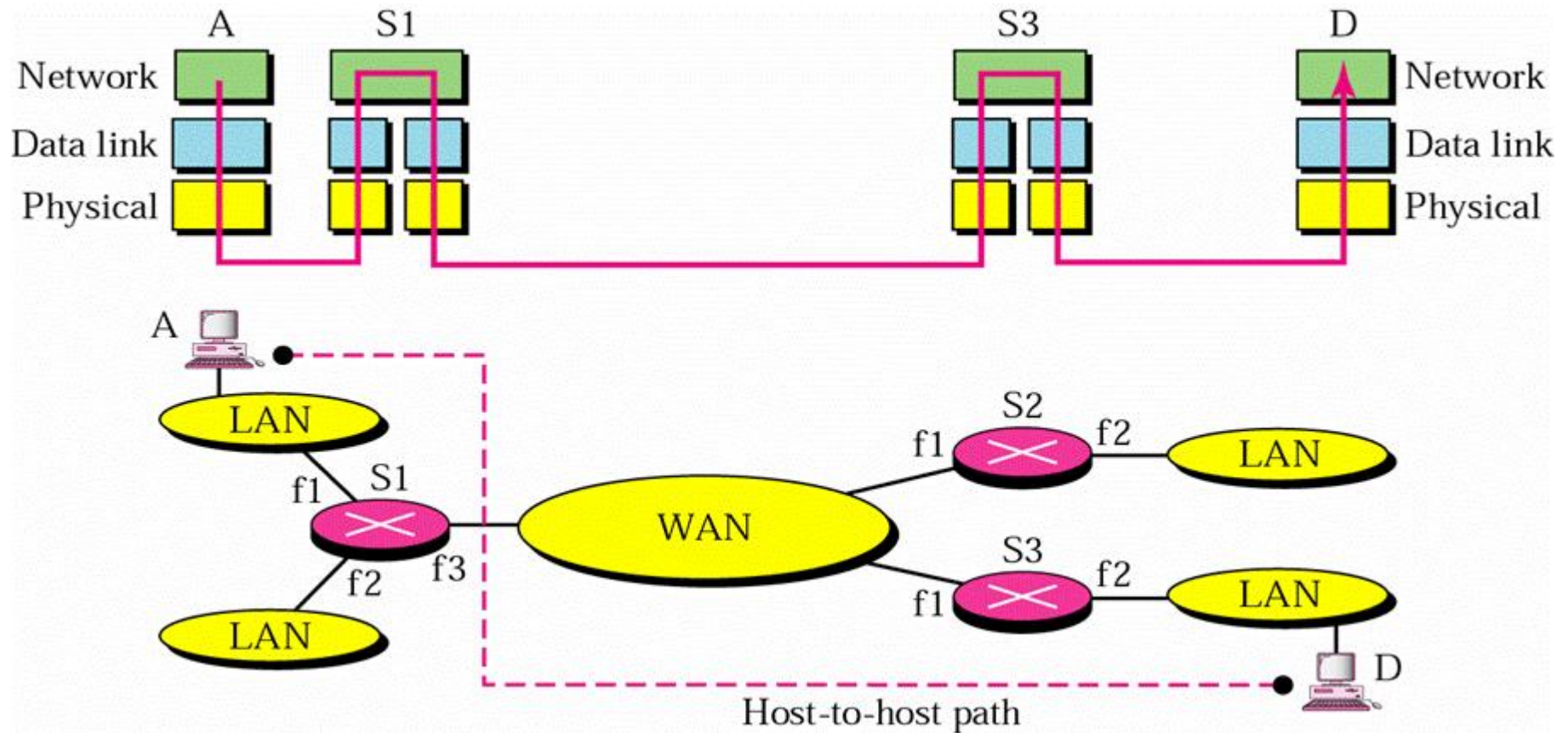
(2) The concept of Hop(跳)



经过一个路由器就是1跳

Next Hop: 下一跳

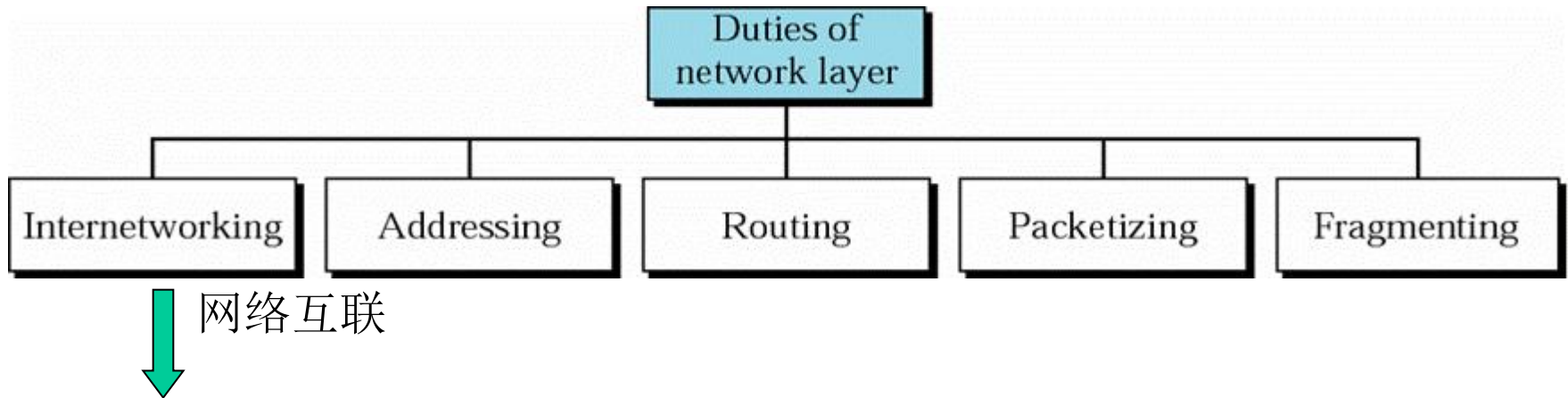
Network Layer in an Internetwork: Host-to-Host Delivery(主机-主机传输)



主机-主机（端-端）

[Forouzan]

(3) Network Duties



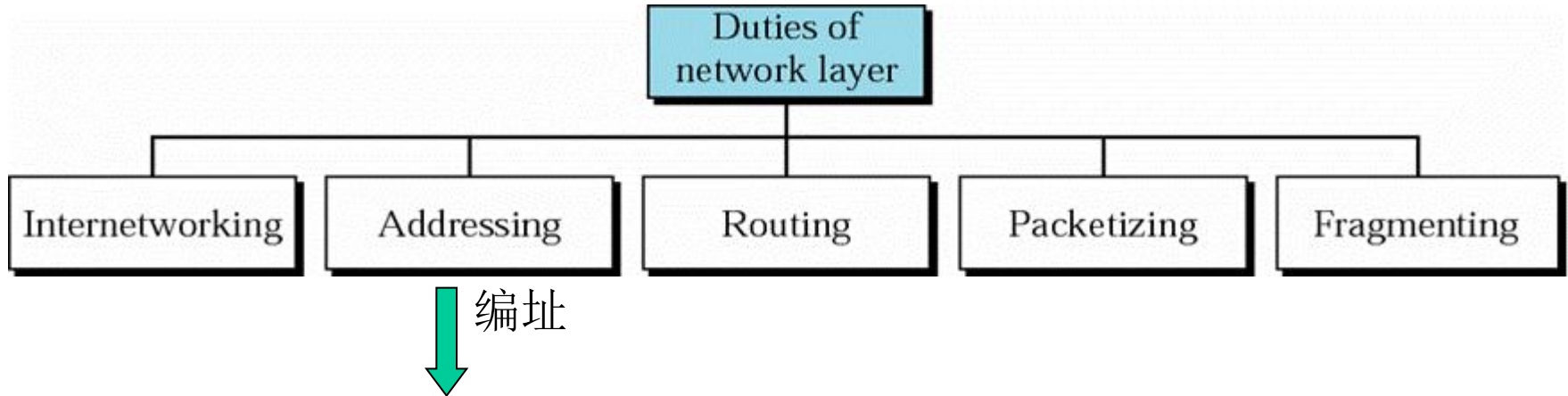
Interconnecting **different** networks (LAN or WAN) and making them look the **same** to the transport layer.

不同的网络：网络地址不同、实现技术可能不同、包格式可能不同

The transport layer should not be worried about the underlying physical network! (网络的个数及其具体技术对于传输层透明)

[Forouzan]

(3) Network Duties(cont.)

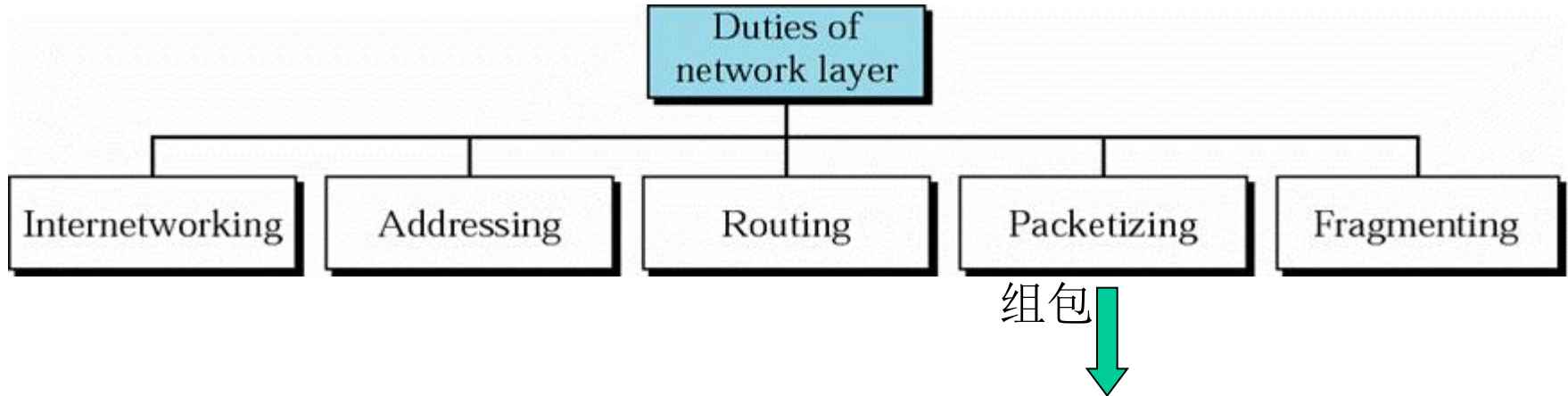


The addresses of a host/router must be **uniquely**(数值唯一) and **universally**(格式统一)

Two devices on the internet can never have the same address. (Address per interface to network, 接口地址唯一)

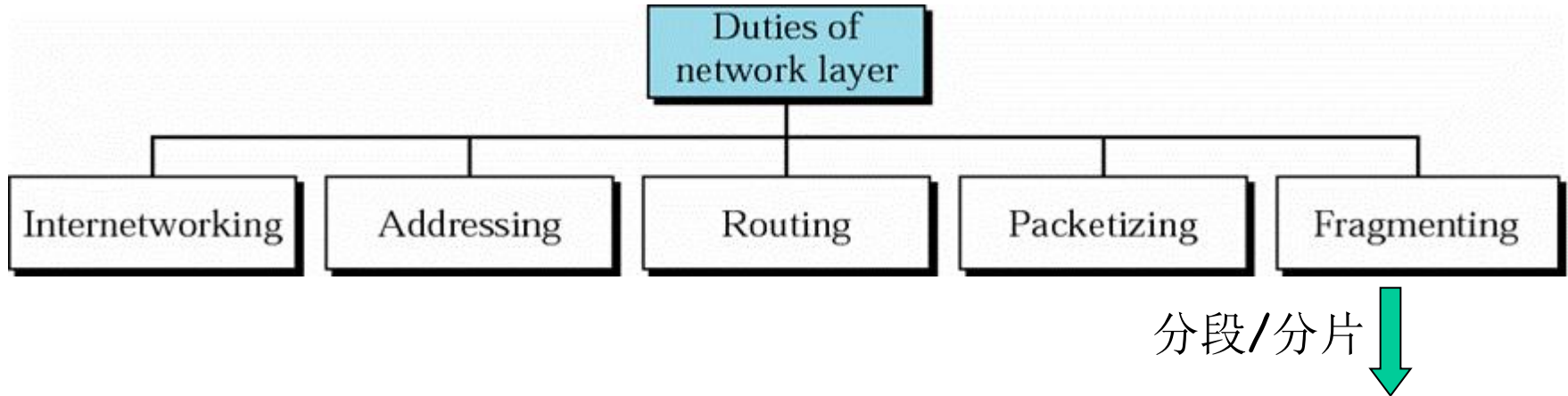
[Forouzan]

(3) Network Duties(cont.)



Encapsulated message received from upper layer and forming **packets**, a task common to all layers. In Internet, packetizing is done by **IP**.

(3) Network Duties(cont.)



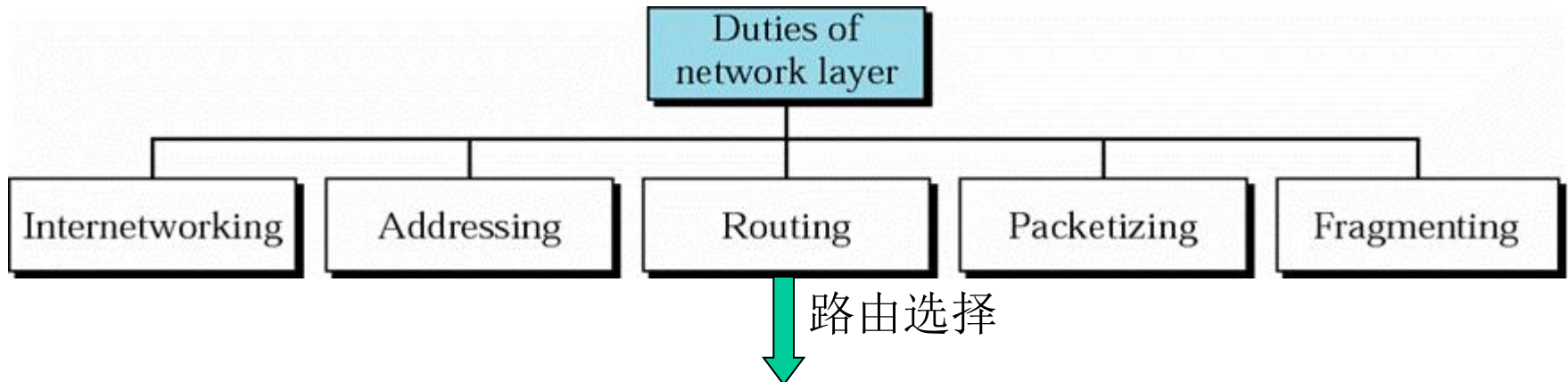
The network layer must be able to operate on top of **any** data-link layer technology.

All these technologies can handle a different packet length. (不同的数据链路层网络支持的最大数据长度不同)

The network layer must be able to **fragment** transport layer PDUs into smaller units so that they can be transferred over various data-link layer technologies.

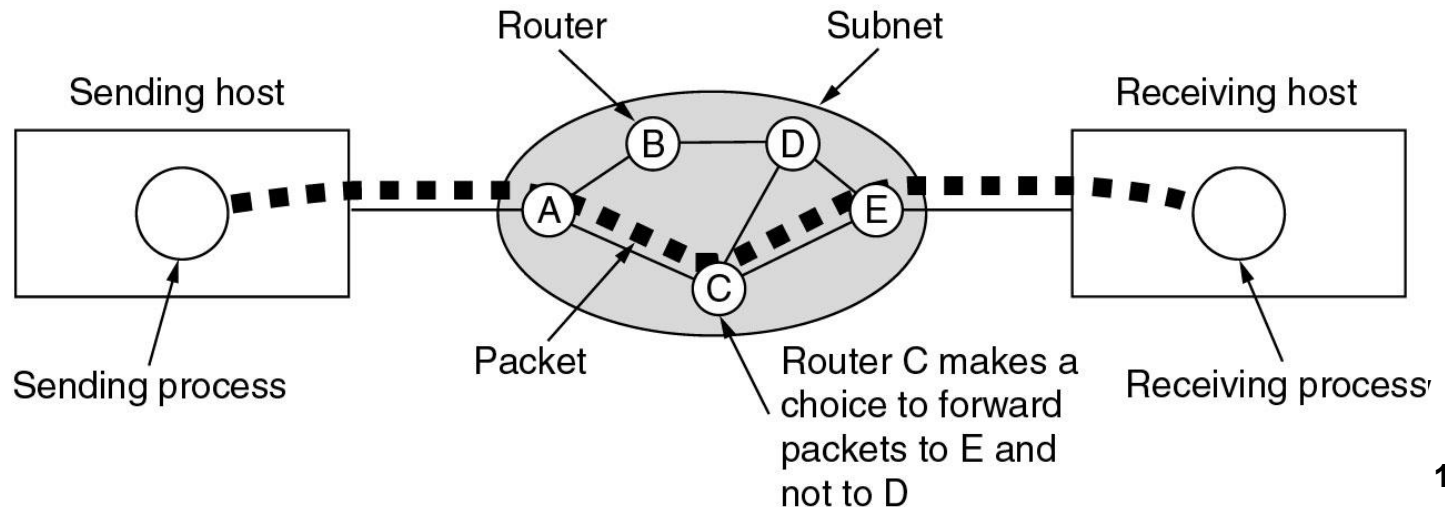
[Forouzan]

(3) Network Duties(cont.)

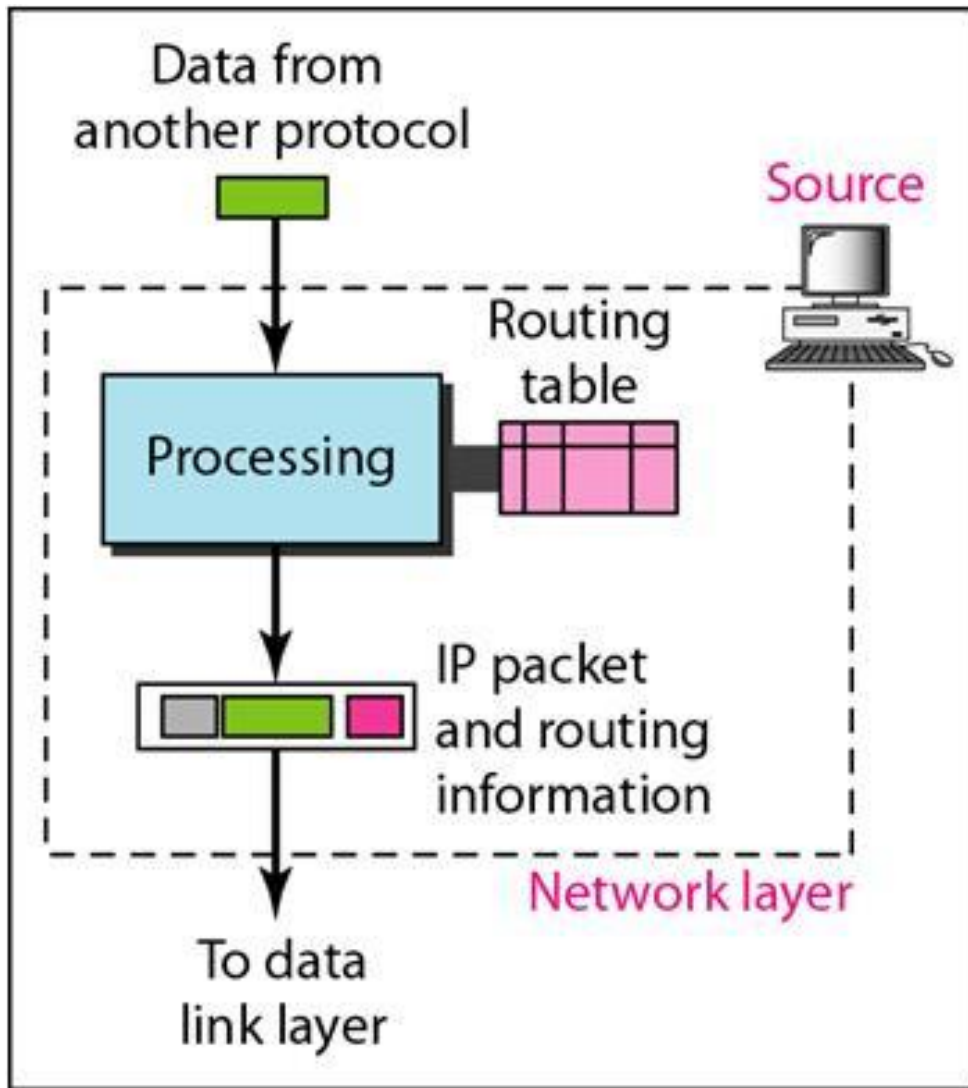


To send the packet to destination, who is the **next hop** (下一跳) ?

Packet cannot choose the route; the routers connecting the LANs/WANs makes this decision.



(4) Network Layer at the Source



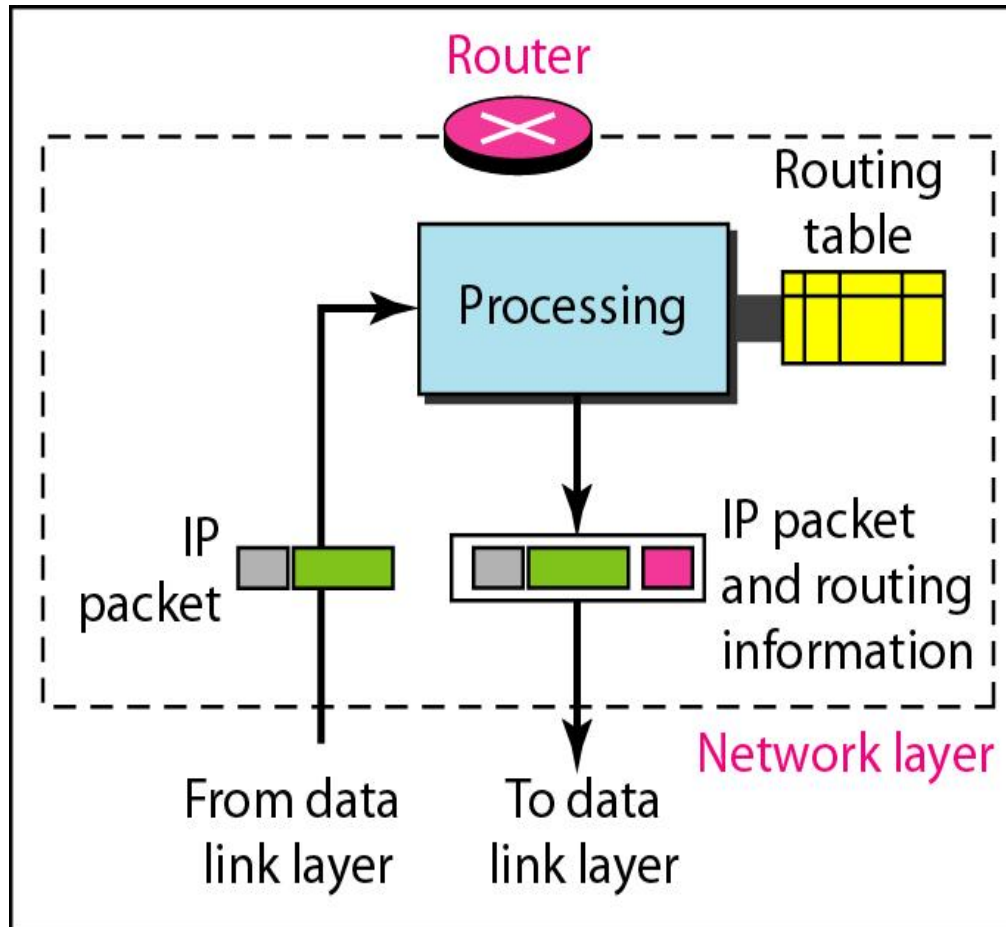
The source network layer **create a packet** carrying addresses of source host and destination host.

The source also checks routing table for find the **routing information** (e.g. outgoing interface, 输出接口, such as Ethernet or wifi).

[Forouzan]

(4) Network Layer at the Router

Responsible for **routing** the packet



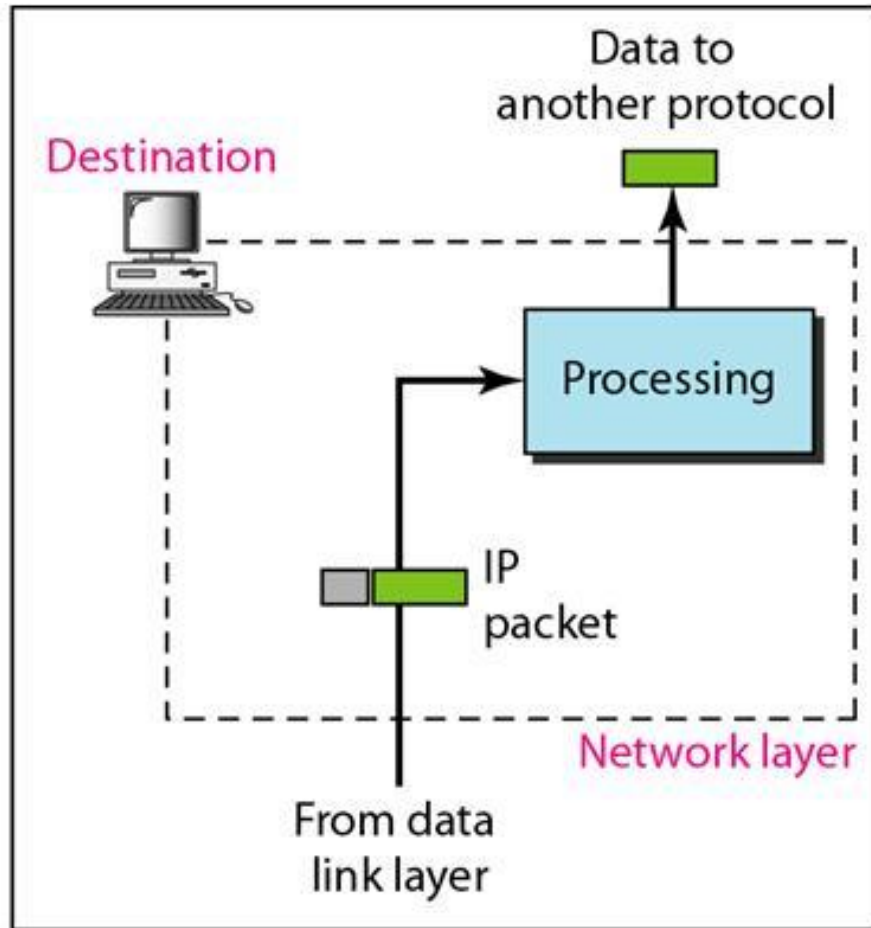
When a packet arrives, the router checks its routing table and finds the **interface** from which the packet must be sent using **routing table**.

c. Network layer at a router

[Forouzan]

(4) Network Layer at the Destination

Responsible for **address verification**



b. Network layer at destination

The destination makes sure that the **destination address** on the packet is the same as the address of the host.

[Forouzan]

5.1.2 Services Provided to Transport Layer

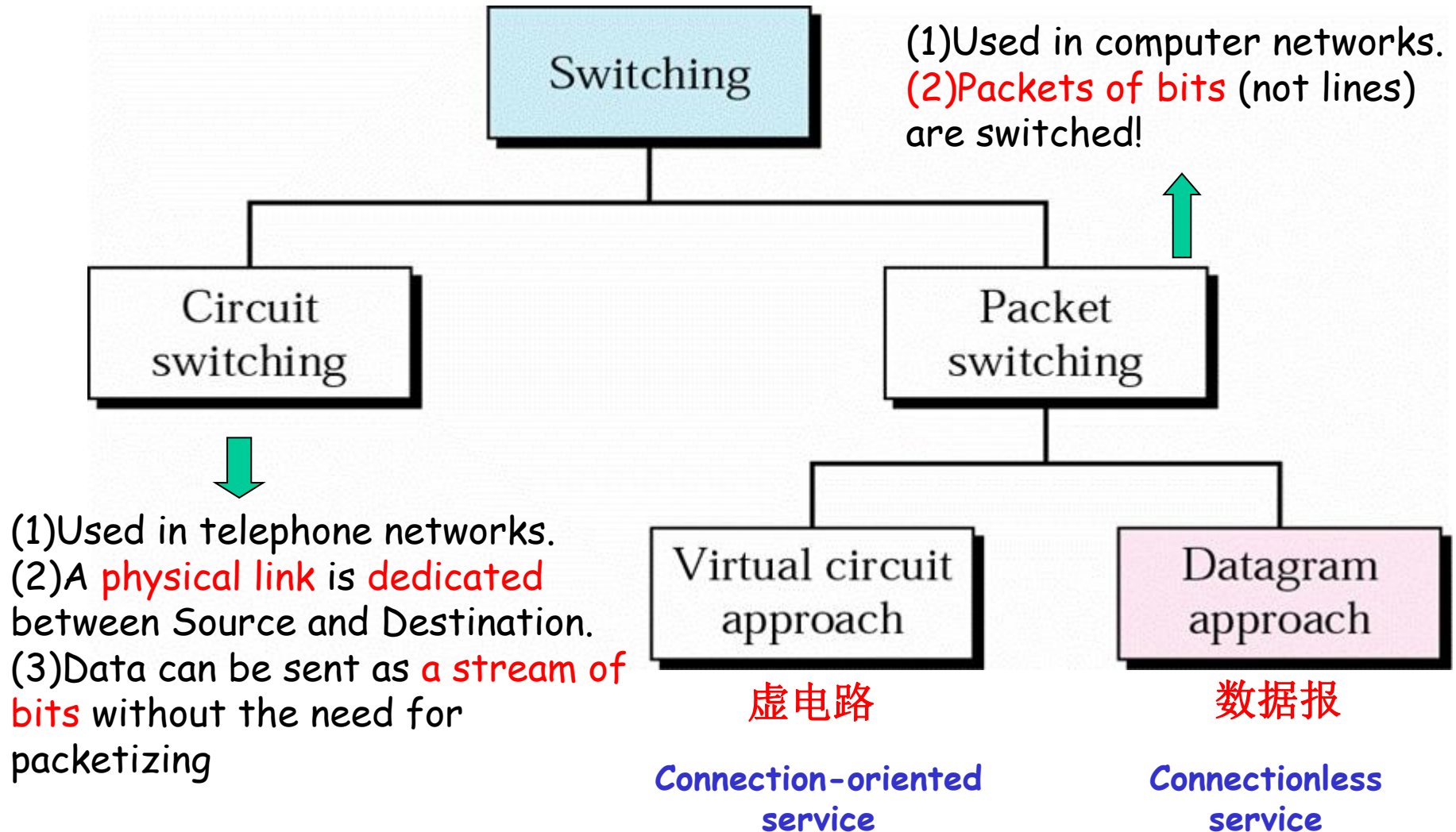
■ Design Goal

- ◆ The services should be independent of the router technology.
- ◆ The transport layer should be shielded from the number, type, and topology of the routers present
- ◆ The network addresses made available to the transport layer should use a uniform(统一) numbering plan, even across LANs and WANs

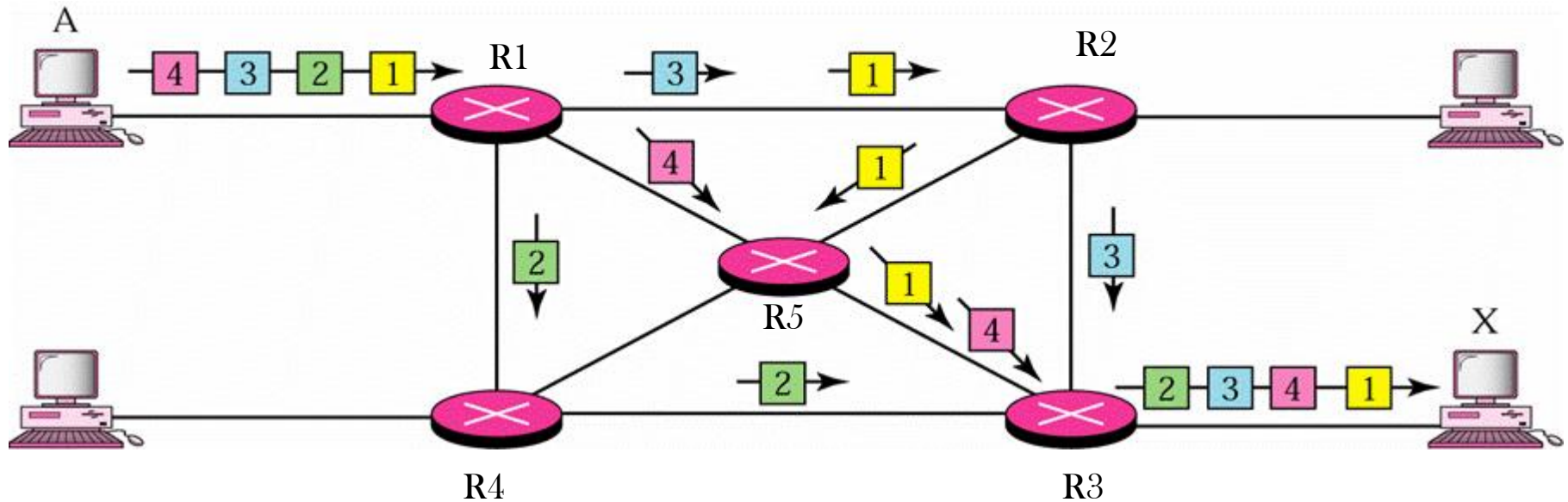
■ Two classes of service

- ◆ Connection-Oriented Service (虚电路—Virtual Circuit)
- ◆ Connectionless Service (数据报--datagram)

Switching



5.1.3 Implementation of Connectionless Service: Datagram(数据报) Network



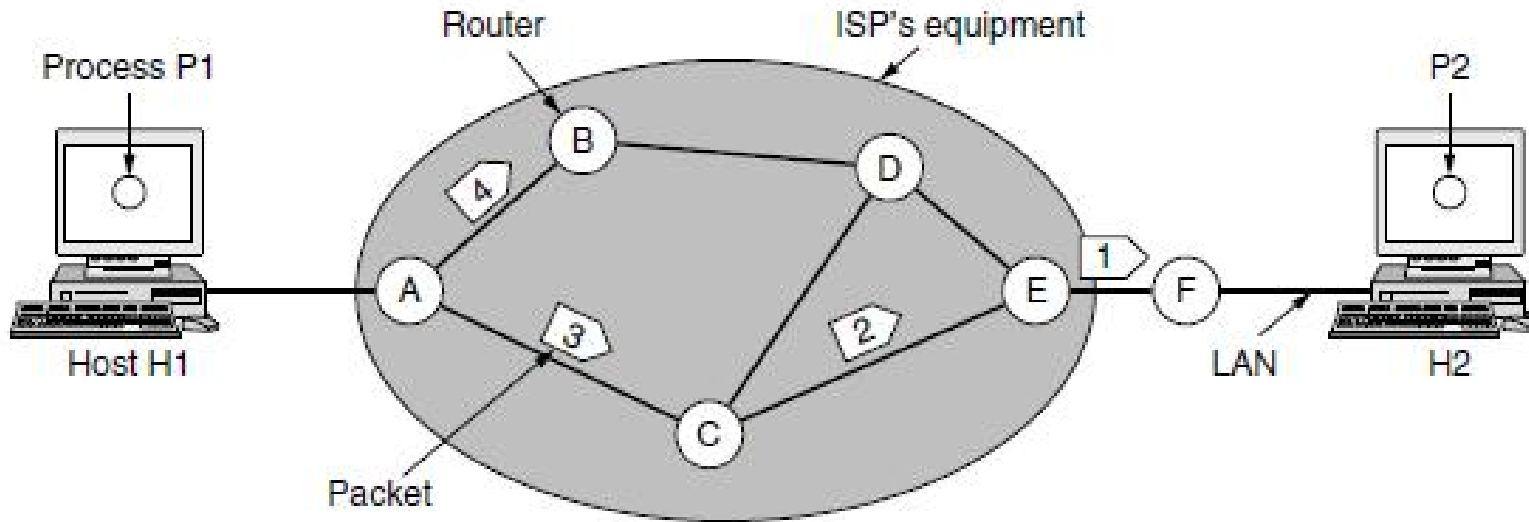
Routing: where to forward the packet?

Routing Table of R1

Dest.	Line(Next Hop, 下一跳)
R1	---
R2	R2
R3	R5
R4	R4
R5	R5

[Forouzan]

Routing within a Datagram Subnet



A's table (initially)

A	-
B	B
C	C
D	B
E	C
F	C

Dest. Line

A's table (later)

A	-
B	B
C	C
D	B
E	B
F	B

C's table

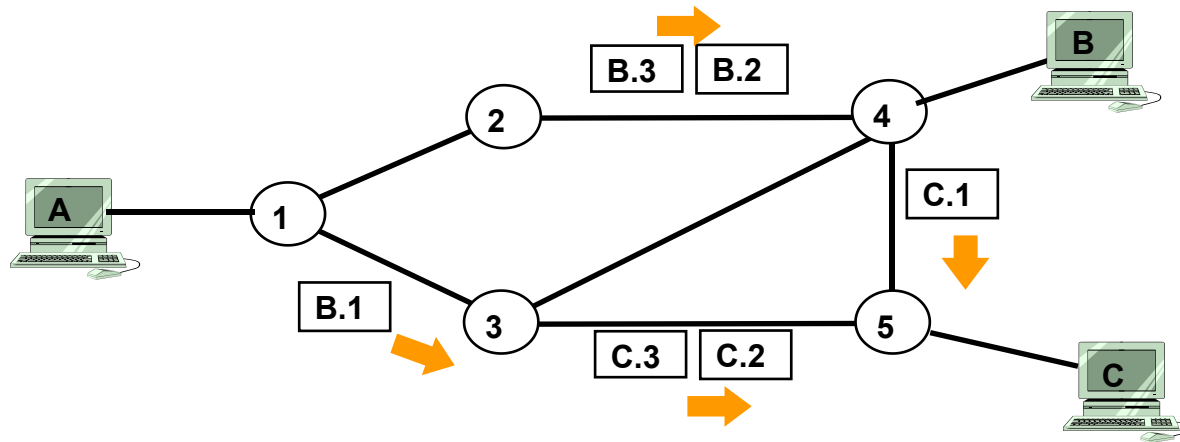
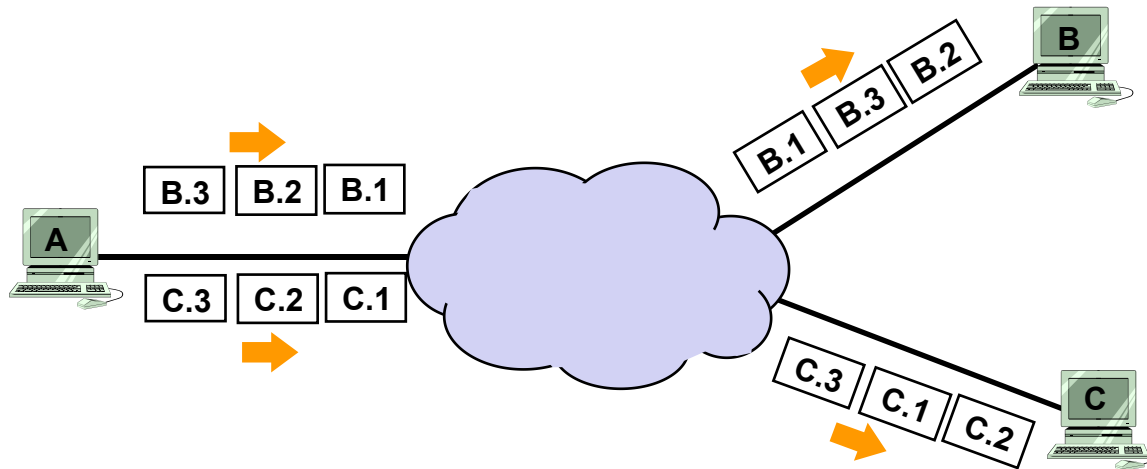
A	A
B	A
C	-
D	E
E	E
F	E

E's table

A	C
B	D
C	C
D	D
E	-
F	F

The routing tables are **dynamic** and are **updated** periodically.

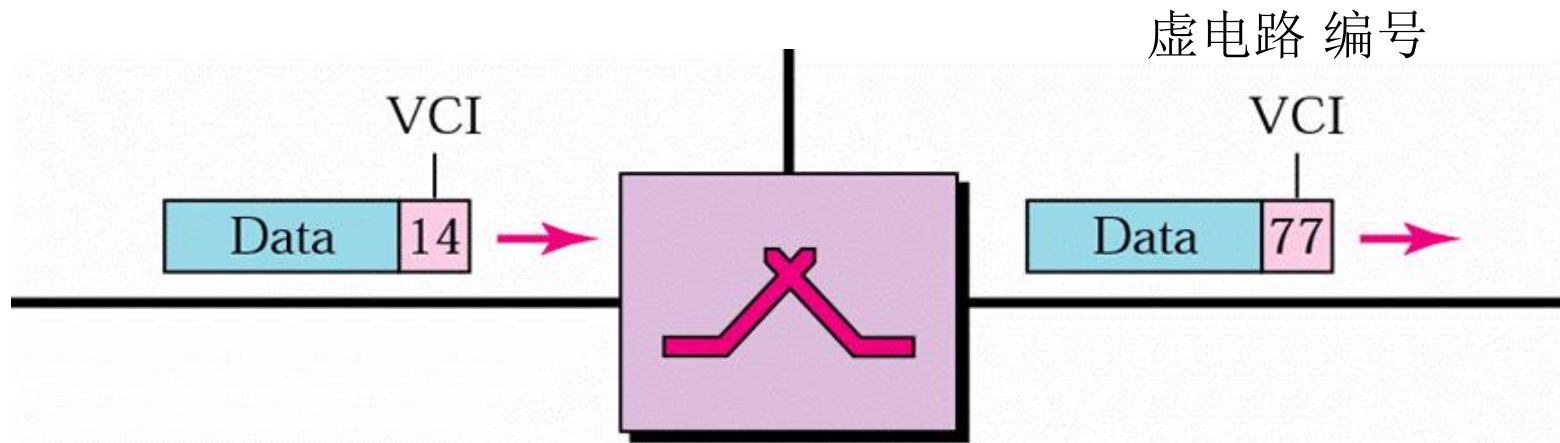
Routing within a Datagram Network



包中需要携带完整地址；包到达可能失序

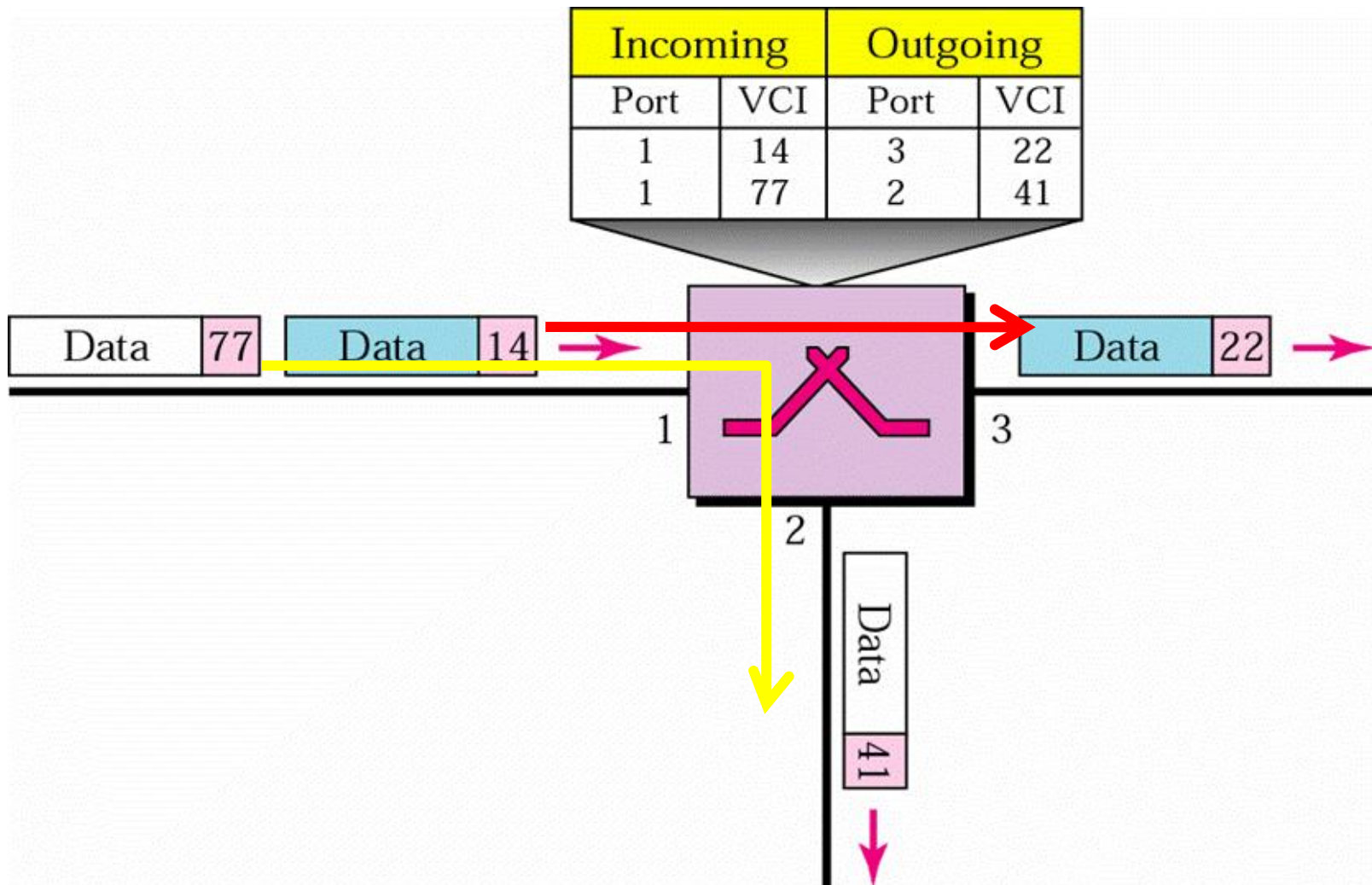
5.1.4 Implementation of Connection-oriented Service: Virtual Circuit(虚电路) Network

- Virtual circuit is established and end-to-end path is decided before data transmission
- Connection-oriented service
- Virtual Circuit Identifier(VCI): local significant
 - ◆ Much shorter than address



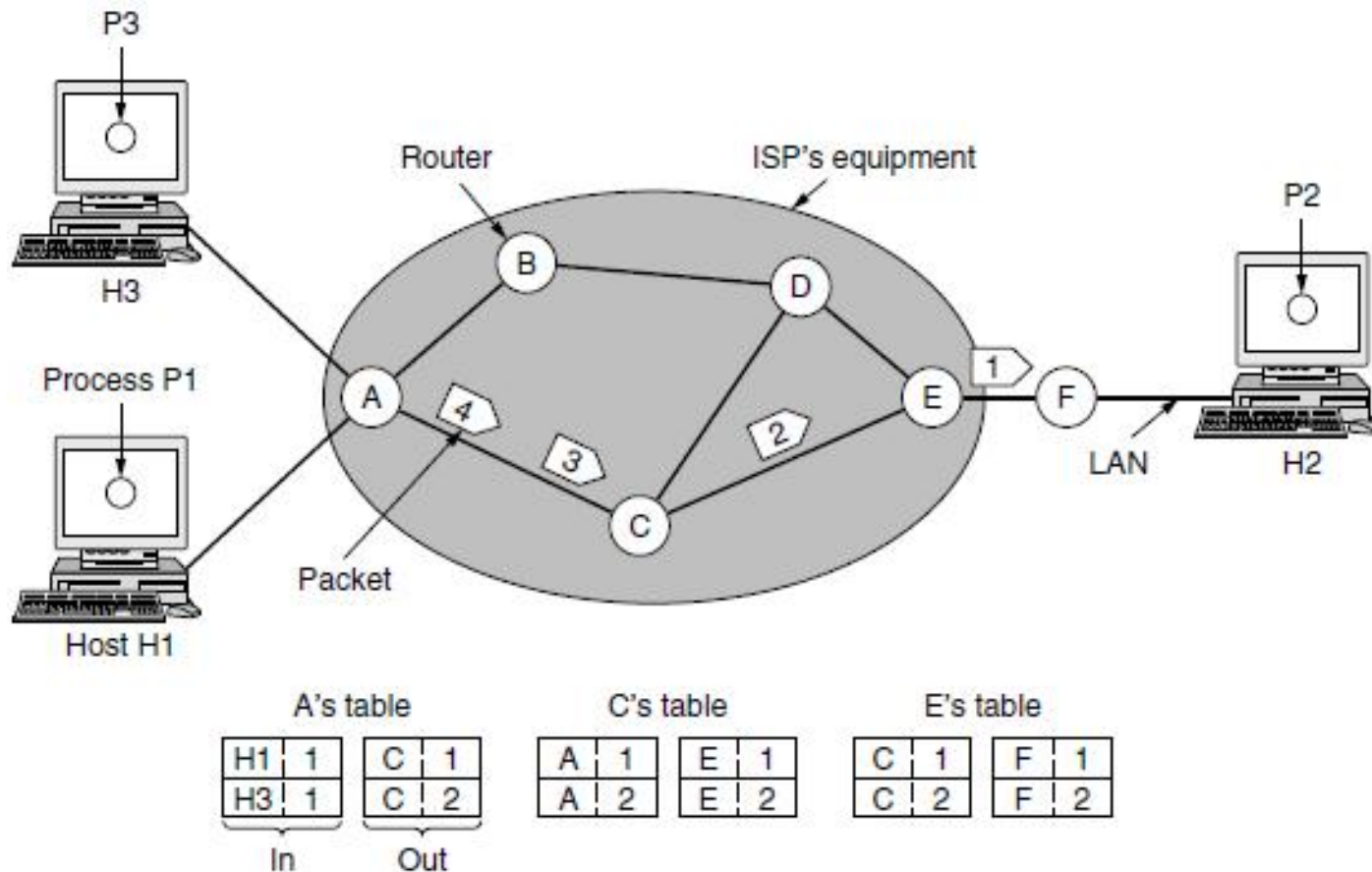
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Virtual Circuit Approach: Routing Table



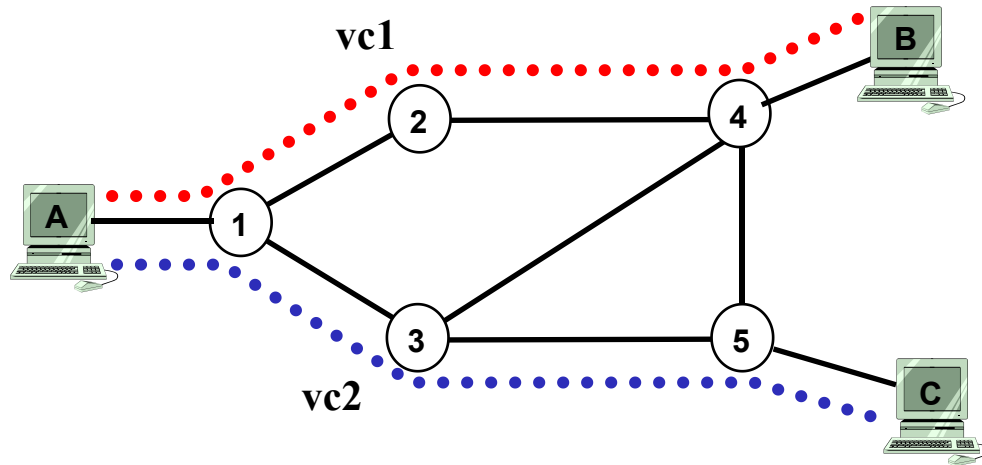
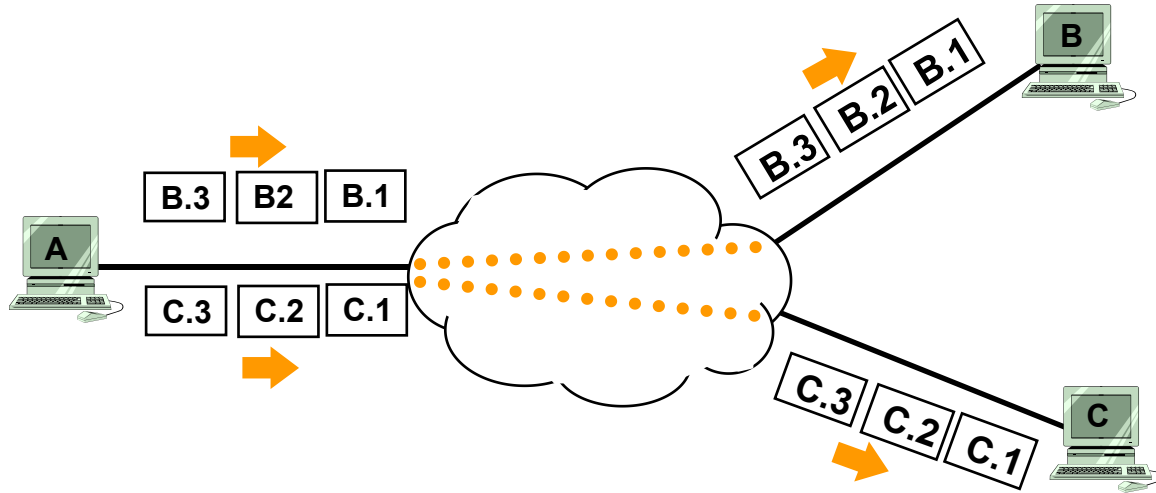
[Forouzan]

Routing within a Virtual-circuit Subnet



VC1: H1-A-C-E-F-H2
VC2: H3-A-C-E-F-H2

Routing within a Virtual-circuit Subnet(2)



● **vc1:** A--1--2--4--B

● **vc2:** A--1--3--5--C

5.1.5 Virtual Circuit vs. Datagram

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service 服务质量	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control 拥塞控制	Difficult	Easy if enough resources can be allocated in advance for each VC

Outline

- 5.1 Introduction to Network Layer
- 5.2 Routing Algorithms
 - ◆ (補) *Concept of Routing and Routing Table*
 - ◆ 5.2.1 Optimality principle
 - ◆ 5.2.2 Dijkstra's "Shortest Path" Algorithm
 - ◆ 5.2.3 Flooding
 - ◆ 5.2.4 Distance Vector Routing (DVR)
 - ◆ 5.2.5 Link State Routing (LSR)
 - ◆ 5.2.6 Hierarchical Routing
- 5.3 Congestion Control
- 5.4 Quality of Service
- 5.5 Internetworking
- 5.6 The network layer in Internet

(1)Routing: Overview

- An internet(互联网) is a combination of networks connected by **routers**. When a packet goes from a source to a destination, it may pass through many routers.
- A router consults a **routing table** when a packet is ready to be forwarded. The routing table specifies the **optimum path** (最佳路径) for the packet.

(1)Routing: Overview(cont.)

- **Routing Algorithms:** algorithms used to construct and update routing tables
- **Routing Protocols:** a combination of rules/procedures that lets routers inform one another when changes occur; mostly based on sharing/combining information between routers at different networks.
 - ◆ RIP, OSPF, BGP

路由表 = 路由算法 + 路由协议

(2) Responsibility of Routing Algorithms

- Responsible for deciding which output line an incoming packet should be transmitted on
 - ◆ If the subnet uses **datagrams** internally, this decision must be made for **every** arriving data packet 数据报网络对每个包选路
 - ◆ If the subnet uses **virtual circuits** internally, routing decisions are made only when a new virtual circuit is being **set up**. Thereafter, data packets just follow the previously-established route 虚电路网络，只在建立虚电路时选路
- Routing(选路) & Forwarding(转发)
 - ◆ **Routing**: **Generate** and **update** routing tables using routing algorithms and protocols
 - ◆ **Forwarding**: **Research** the **outbound route** of the packet according to the routing table

(3)How to evaluate routing algorithms?

■ Goals

- ◆ Correctness
- ◆ Simplicity
- ◆ Robustness: 健壮
- ◆ Stability: 算法可快速收敛达到稳定平衡状态
- ◆ Fairness
- ◆ Optimality
 - Minimizing mean packet delay
 - Maximizing total network throughput

■ Performance Criteria

- ◆ Least cost, such as minimum hops

(4) Classification of Routing Algorithms

■ *Nonadaptive algorithms (Static Routing)*

- ◆ The choice of the route to use *is computed in advance*, off-line, and downloaded to the routers when the network is booted
- ◆ The routing decisions are *not based on* measurements or estimates of *the current traffic and topology*

■ *Adaptive(自适应) algorithms*

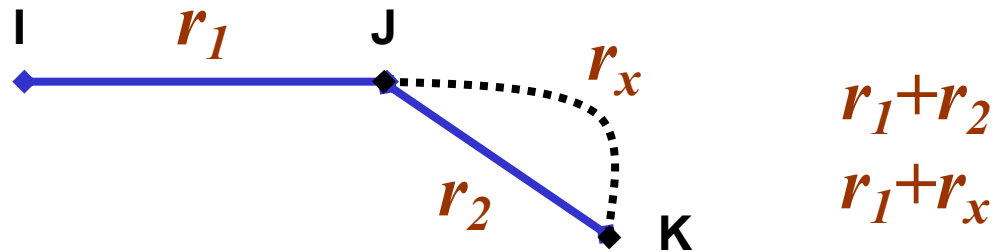
- ◆ Changing their routing decisions to reflect changes in the topology, and usually the traffic as well *适应网络拓扑和业务量的变化*

Routing Algorithms

- (補)Concept of Routing and Routing Table
- *5.2.1 Optimality principle*
- 5.2.2 Dijkstra's "Shortest Path" Algorithm
- 5.2.3 Flooding
- 5.2.4 Distance Vector Routing (DVR)
- 5.2.5 Link State Routing (LSR)
- 5.2.6 Hierarchical Routing

(1)The Optimality Principle(最优化原则)

- If router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route
 - ◆ If a route r_x better than r_2 existed from J to K, it could be concatenated with r_1 to improve the route from I to K
 - ◆ Contradicting our statement that r_1r_2 is optimal

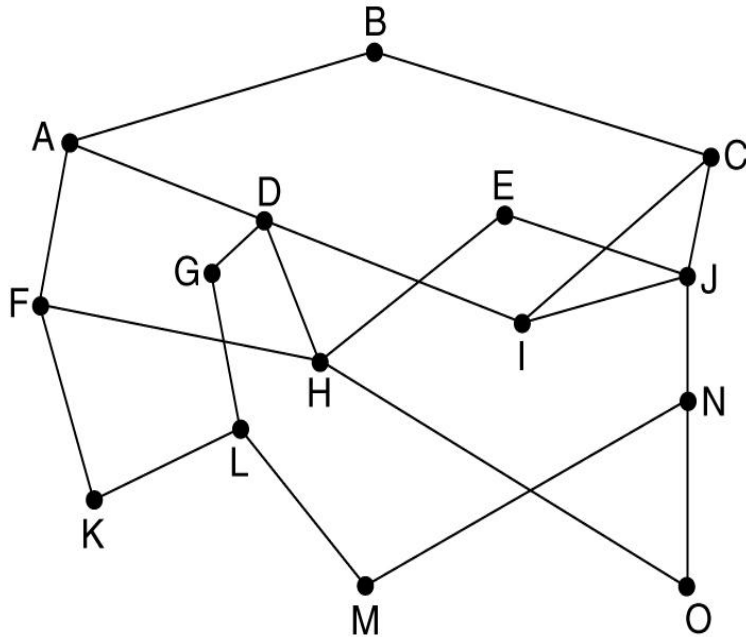


最优路径上任意两点之间的路径也是最优的

(2) Sink Tree(宿树/汇集树)

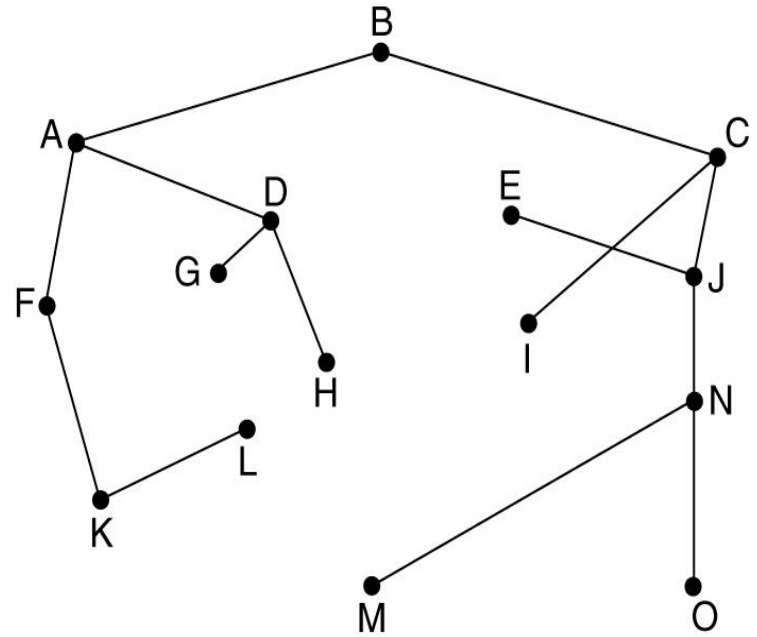
- **Optimal routes** from all sources to a destination, form a tree rooted at the destination 以目的节点为根，由所有源节点到目的节点的最优路由构成
- Does **not contain any loops**, so each packet will be delivered within a **finite number of hops**
- The goal of routing algorithms
 - ◆ Discover and use the **sink trees** for all routers
 - ◆ Different routers may have different ideas about the current topology sometime

Example of Sink Tree



(a)

(a) A subnet



(b)

(b) A sink tree for router B

Routing Algorithms

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(1) Shortest Path Routing(最短路径选路)

- Build a weighted and directed graph of subnet(有向加权图)
 - ◆ Nodes represent **routers**
 - ◆ Arcs represents communication lines
- Weight
 - ◆ **Hops**
 - ◆ Geographic **distance** in kilometers
 - ◆ Mean queuing and transmission **delay**
 - ◆ **A function of** the distance, bandwidth, average traffic, communication cost, mean queue length, measured delay, and other factors

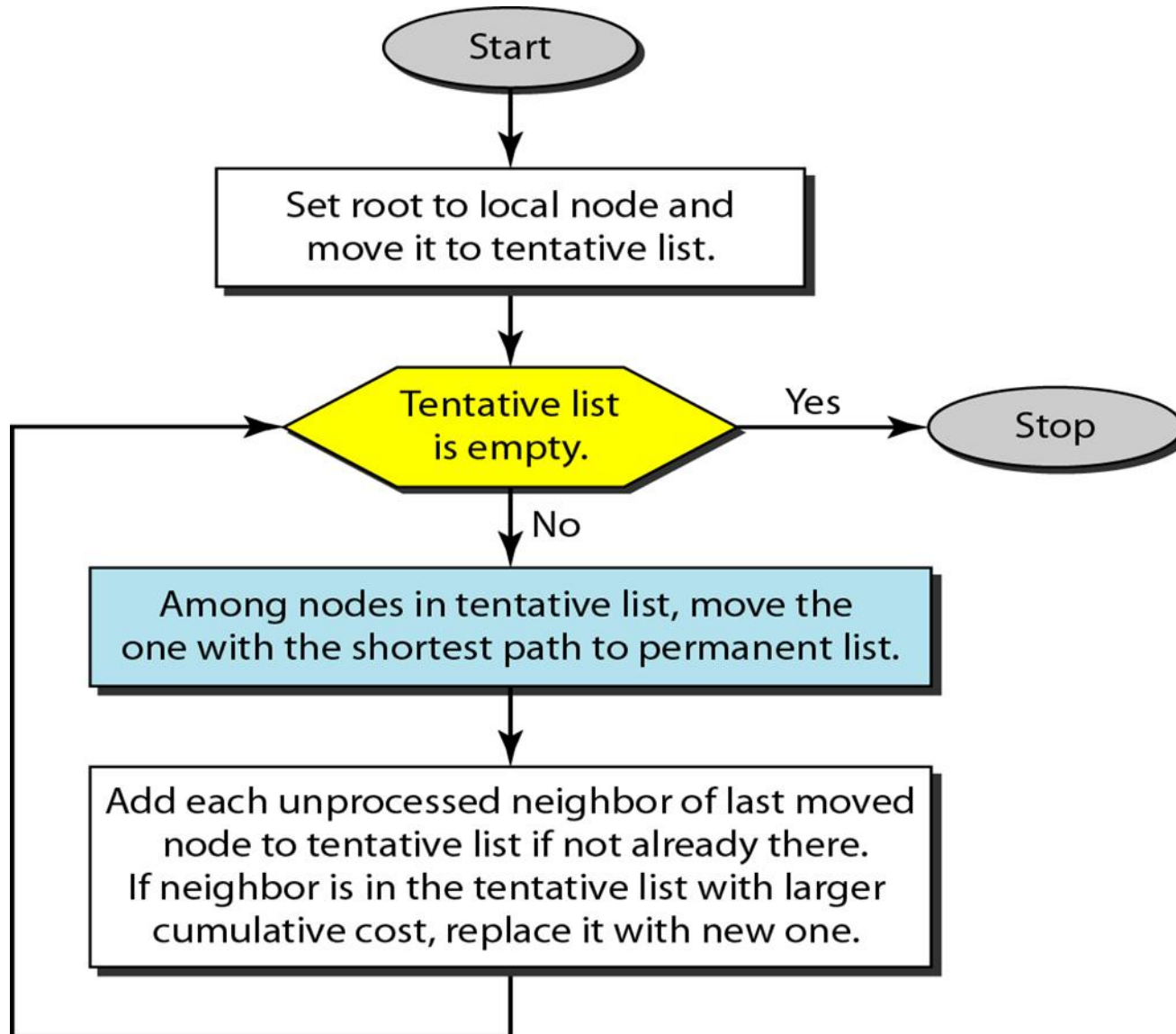
(2) Dijkstra

- Find **shortest paths** from given source node to all other nodes

- Routing Algorithm

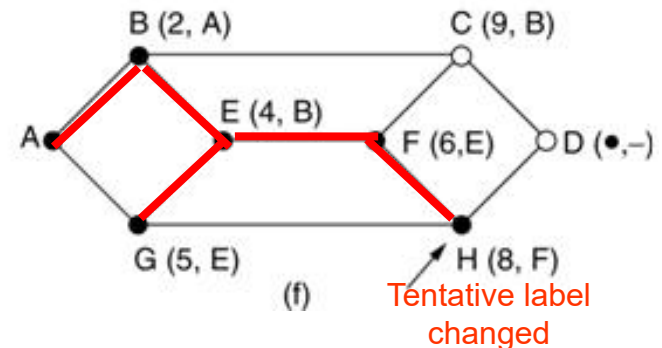
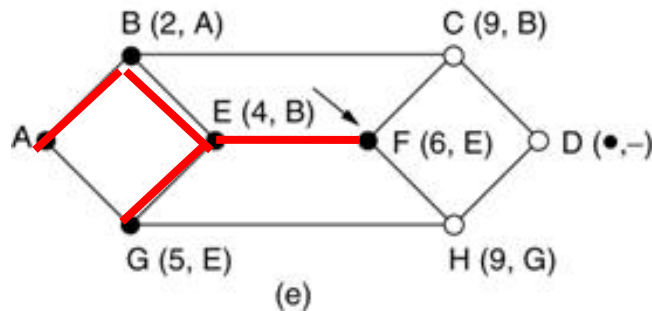
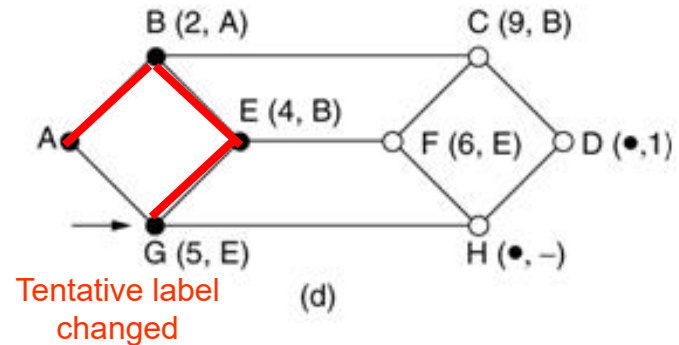
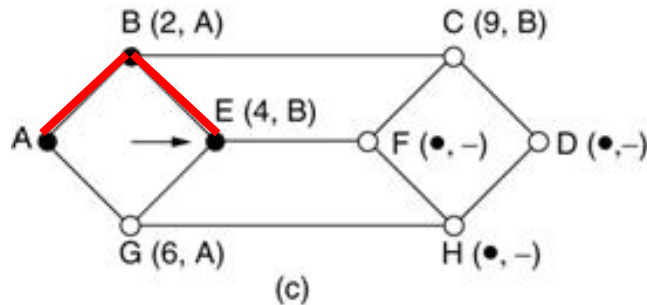
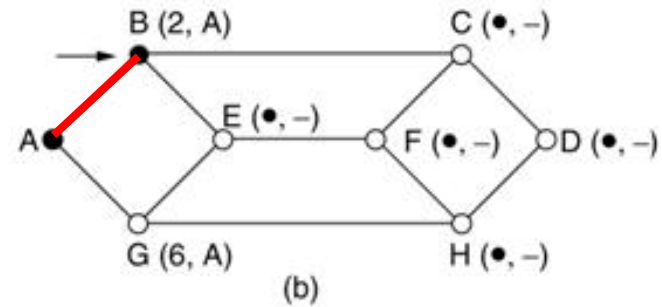
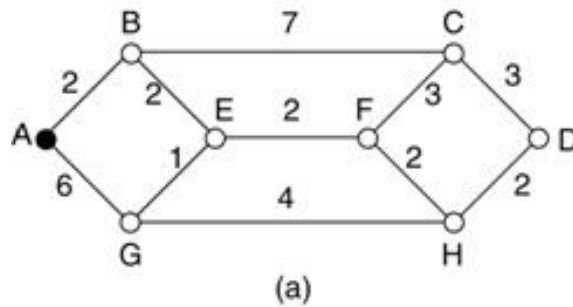
- ◆ 1. Starting with the **local node** **s** (router): the **root** of the tree.
- ◆ 2. Assign a cost of 0 to this node and make it the first **permanent node**(永久节点).
- ◆ 3. Examine each neighbor node of the node that was the last permanent node **P**.
- ◆ 4. Assign a **cumulative cost** (累计开销) to each node and make it **tentative** (临时节点) .
- ◆ 5. Among the list of tentative nodes
 - Find the node with the **smallest** cumulative cost and make it permanent.
 - If a node can be reached from more than one direction, Select the direction with the shortest cumulative cost.
- ◆ 6. Repeat steps 3 to 5 until **every** node becomes permanent.

Dijkstra算法的流程图

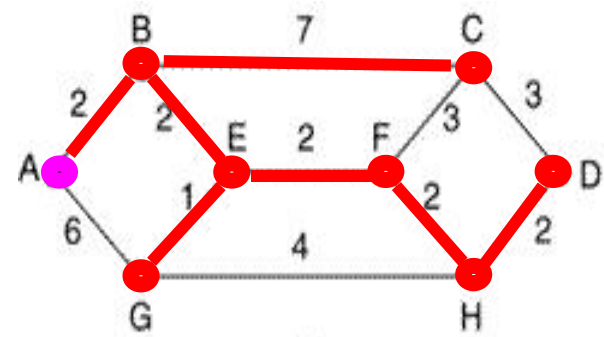


[Forouzan]

Example of Dijkstra Algorithm



Example of Dijkstra

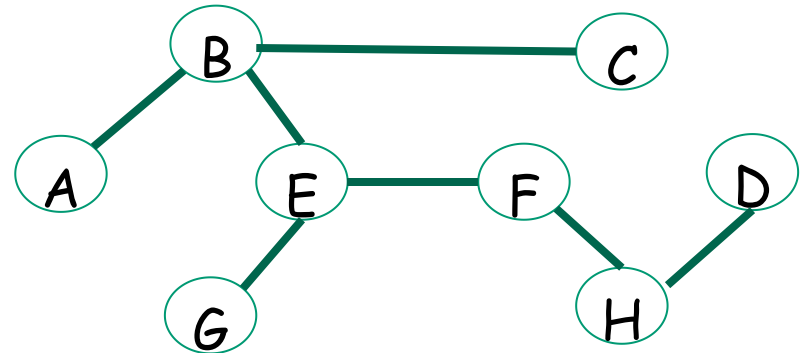
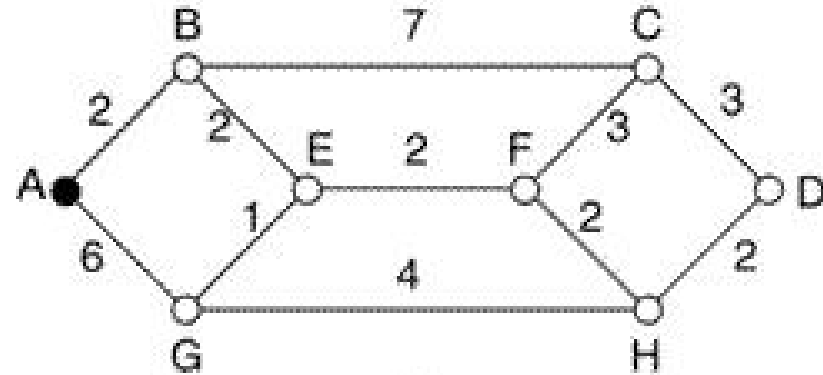


A	B	C	D	E	F	G	H	Permanent Node
∞	∞	∞	∞	∞	∞	∞	∞	
0	∞	∞	∞	∞	∞	∞	∞	A
0	2, A	∞	∞	∞	∞	6, A	∞	B
0	---	9, B	∞	4, B	∞	6, A	∞	E
0	---	9, B	∞	---	6, E	5, E	∞	G
0	---	9, B	∞	---	6, E	---	9, G	F
0	---	9, B	∞	---	---	---	8, F	H
0	---	9, B	10, H	---	---	---	---	C
0	---	---	10, H	---	---	---	---	D

What we get by using Dijkstra?

A's routing table

Dest.	Cost	Next Hop
A	0	--
B	2	B
C	9	B
D	10	B
E	4	B
F	6	B
G	5	B
H	9	B



A's shortest path tree

Routing Algorithms

- (補)Concept of Routing and Routing Table
- 5.2.1 Optimality principle
- 5.2.2 Dijkstra's "Shortest Path" Algorithm
- *5.2.3 Flooding*
- 5.2.4 Distance Vector Routing (DVR)
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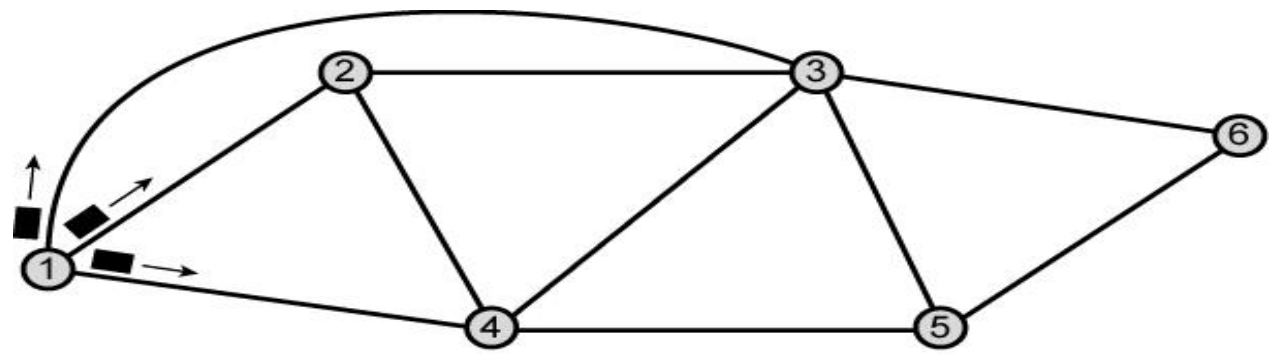
(1) Major Routing Strategies

- Fixed Routing(固定查表选路)
 - ◆ Single permanent route for each source to destination pair
- Flooding(洪泛/泛洪)
 - ◆ Packet sent to every neighbor
 - ◆ Incoming packets forwarded to every link except incoming link
- Random Routing(随机选路)
 - ◆ Selects one outgoing path to forward incoming packet
 - ◆ Selection can be random or round robin
- Adaptive Routing(自适应选路)
 - ◆ Routing decisions change as conditions on the network change
 - ◆ Requires information about network
 - ◆ Used by almost all packet switching networks

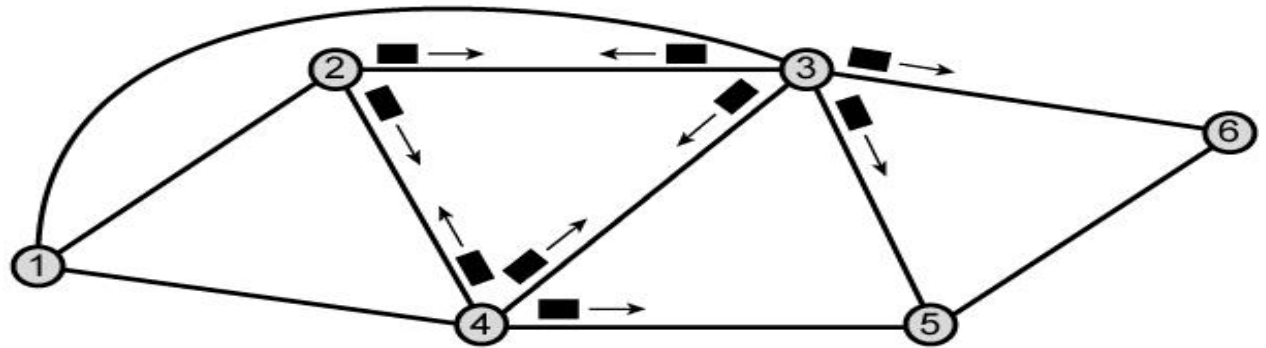
(2) Flooding(洪泛法): Ideas

- 无需路由表
- No network information required
- Incoming packets retransmitted on every link except incoming link
- Eventually a number of copies will arrive at destination

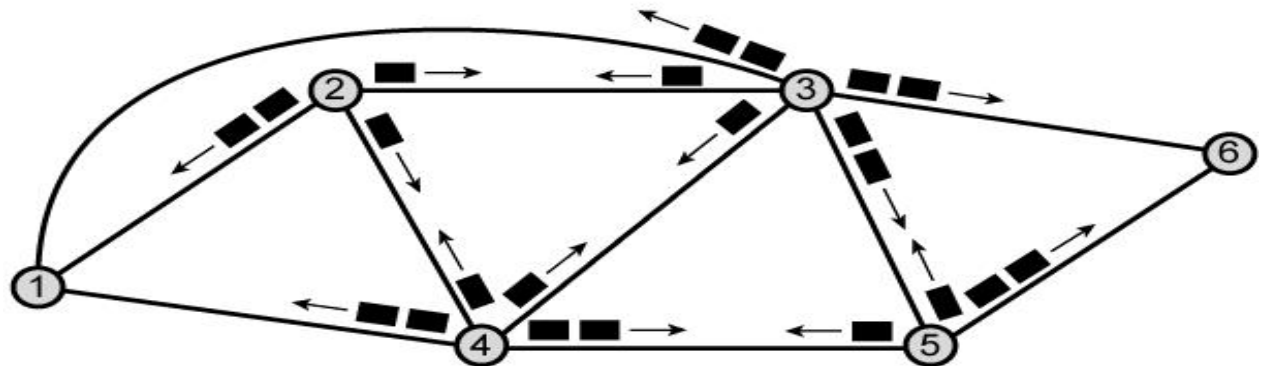
Example of Flooding



(a) First hop



(b) Second hop



(c) Third hop

(3) Flooding: Problems and Solutions

■ Problem

- ◆ Generates vast numbers of duplicate packets

■ Solutions

- ◆ The header of each packet contains a hop counter(跳计数器)
 - Counter is decremented at each hop
 - When the counter reaches zero, discard the packet
 - Sender initializes the hop counter as the length of the path from source to destination, or the subnet diameter
- ◆ Source puts a sequence number in each packet (序列号)
 - Each router records maximal seq per source telling which seq have already been seen

(4) Properties of Flooding

- All possible routes are tried
 - ◆ Very robust 健壯
- At least one packet will have taken minimum hop count route
- All nodes are visited
 - ◆ Useful to **distribute routing information**

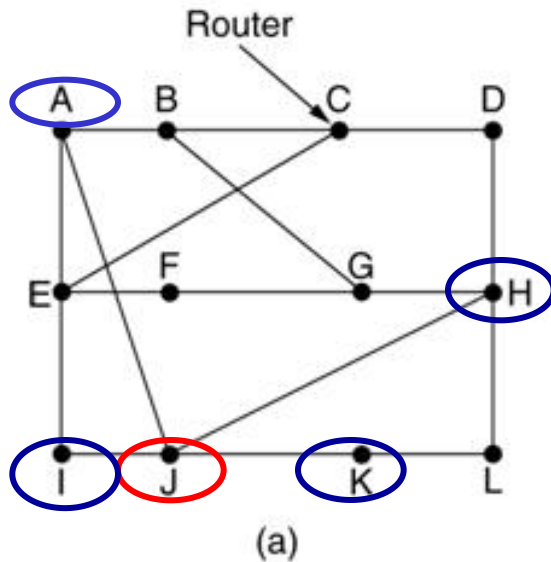
Routing Algorithms

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(1) Distance Vector Routing(距离矢量选路)

- Based on Bellman-Ford algorithm
- Each router maintains a routing table
 - ◆ The best known **distance** to each destination(距离)
 - The distance might be number of hops, time delay in milliseconds, ...
 - ◆ Which line to use (下一跳/接口)(方向)
Vector (distance, direction)
- The table is updated by exchanging information with **neighbors**
 - ◆ Routing information sent: **(Destination, Distance)**--**矢量**
 - Periodically (**RIP: 30s**)
 - ◆ When routing information received
 - Router updates local table if receive a **better** route
 - Refreshes existing routes
 - ◆ Delete routing table items if they time out (**RIP: 180s**)

DVR: Example in textbook



Taking destination B as an example,

$$JB = JA + AB = 8 + 12 = 20$$

$$JB = JI + IB = 10 + 36 = 46$$

$$JB = JH + HB = 12 + 31 = 43$$

$$JB = JK + KB = 6 + 28 = 34$$

Thus, A is selected as next hop.

+

To	A	I	H	K	New estimated delay from J	
					↓	Line
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	-
K	24	22	22	0	6	K
L	29	33	9	9	15	K

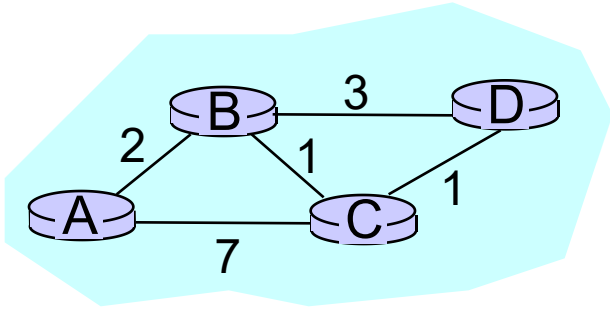
JA delay is 8 JI delay is 10 JH delay is 12 JK delay is 6

Vectors received from J's four neighbors

New routing table for J

(b)

DVR Example 2: Initial Status



Node A

Dest.	Cost	NextHop
B	2	B
C	7	C
D	∞	-

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	3	D

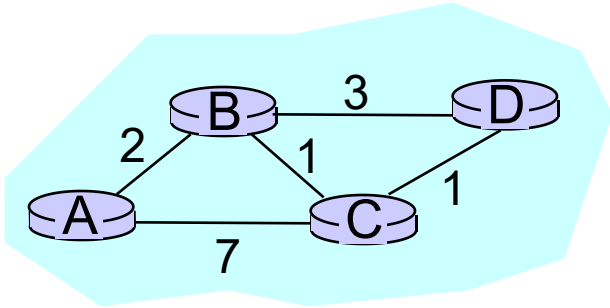
Node C

Dest.	Cost	NextHop
A	7	A
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	∞	-
B	3	B
C	1	C

Example 2: 1st Iteration (C → A)



Node A

Dest.	Cost	NextHop
B	2	B
C	7	C
D	8	C

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	3	D

$$D(A, D) = D(A, C) + D(C, D) = 7 + 1 = 8$$

(D(C,A), D(C,B), D(C,D))

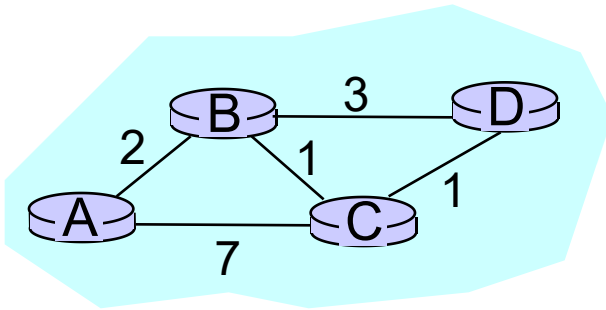
Node C

Dest.	Cost	NextHop
A	7	A
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	∞	-
B	3	B
C	1	C

Example 2: 1st Iteration ($B \rightarrow A$, $C \rightarrow A$)



Node A

Dest.	Cost	NextHop
B	2	B
C	3	B
D	5	B

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	3	D

$$D(A,D) = D(A,B) + D(B,D) = 2 + 3 = 5$$

$$D(A,C) = D(A,B) + D(B,C) = 2 + 1 = 3$$
$$D(A,C) = 7$$

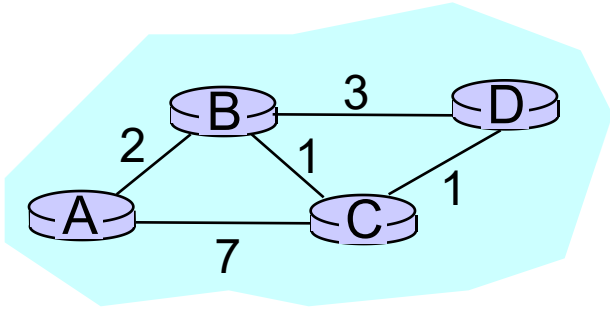
Node C

Dest.	Cost	NextHop
A	7	A
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	∞	-
B	3	B
C	1	C

Example 2: End of 1st Iteration



Node A

Dest.	Cost	NextHop
B	2	B
C	3	B
D	5	B

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	2	C

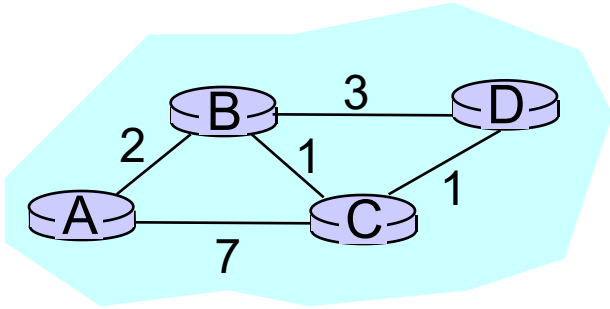
Node C

Dest.	Cost	NextHop
A	3	B
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	5	B
B	2	C
C	1	C

Example 2: End of 2nd Iteration



Node A

Dest.	Cost	NextHop
B	2	B
C	3	B
D	4	B

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	2	C

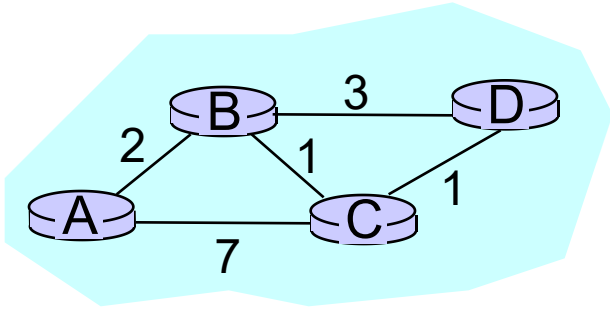
Node C

Dest.	Cost	NextHop
A	3	B
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	4	C
B	2	C
C	1	C

Example 2: End of 3rd Iteration



Node A

Dest.	Cost	NextHop
B	2	B
C	3	B
D	4	B

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	2	C

Node C

Dest.	Cost	NextHop
A	3	B
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	4	C
B	2	C
C	1	C

Nothing changes → algorithm terminates

(2) Features and Problems of DVR

■ 4 key features

- ◆ Sharing **knowledge about the entire network** with neighbours.
- ◆ Sharing only with immediate **neighbours**.
- ◆ Sending knowledge about the entire network through all its interface. (**Broadcast Updates**)
- ◆ Sharing at **regular intervals (Periodic Updates)**.

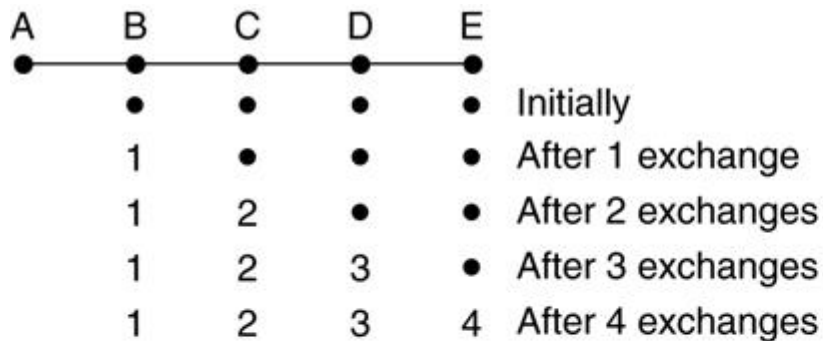
路由器 **R** 周期地 向所有 邻居 广播 整个 路由表

■ Issues

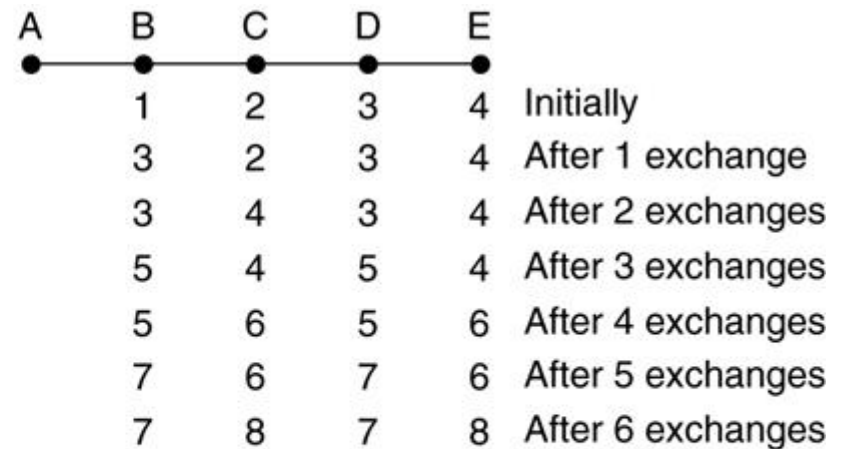
- ◆ 收敛速度可能会比较慢
- ◆ react rapidly to good news, but leisurely to bad news
(好消息反应迅速, 坏消息反应慢)

(3) Count-to-Infinity(无穷计数) Problem

- Problems: Tedious comparing/updating process, slow response to **infinite loop (count-to-infinity, 无穷计数)** problem, huge list to be maintained!!



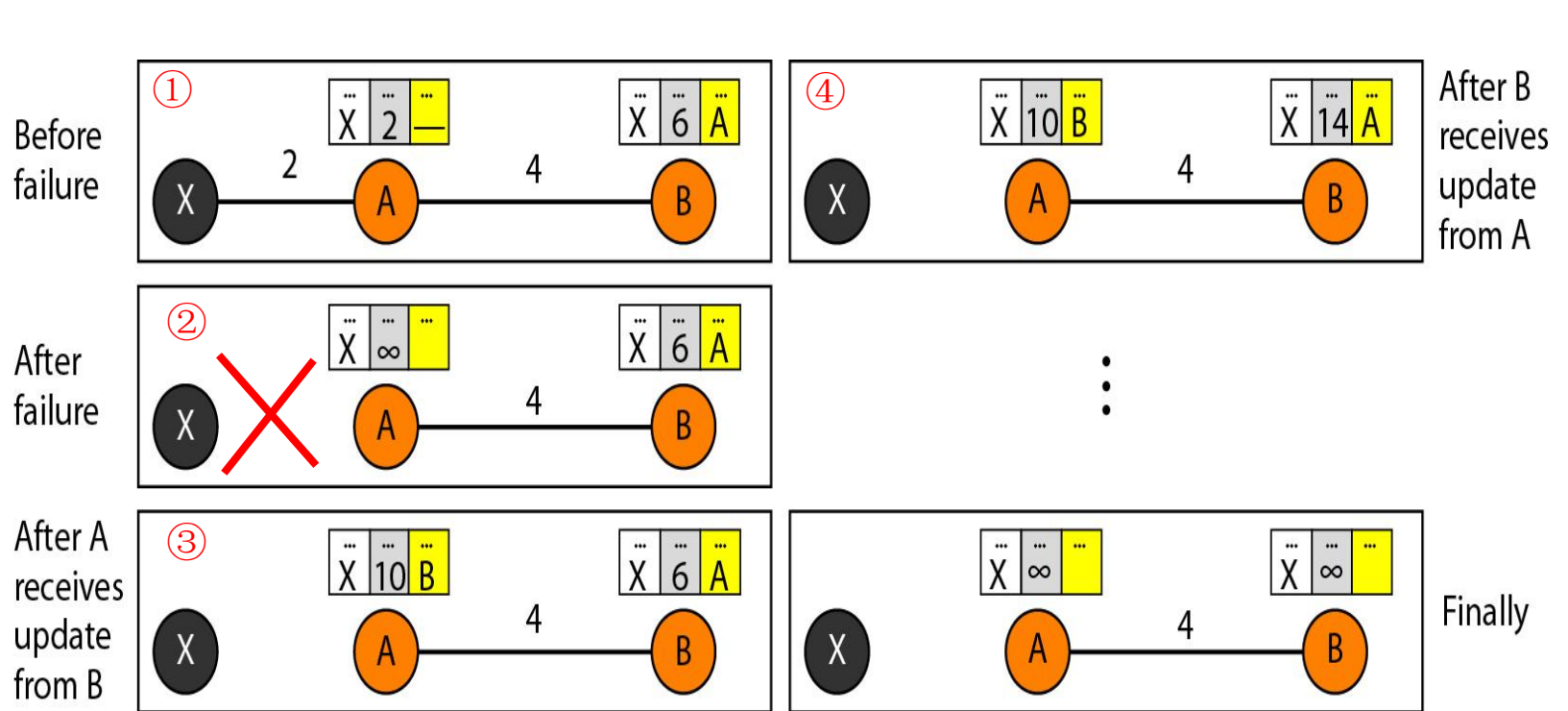
(a)



(b)

Bad news travels slowly

DVR: Infinite Loop Example



For a destination, router only knows distance, no idea of the path
(只知距离，不知路径)

DVR: Summary

■ Used in

- ◆ RIP (Routing Information Protocol)
- ◆ Cisco EIGRP (Enhanced Interior Gateway Routing Protocol)

■ Problem of DVR

- ◆ When X tells Y that it has a path somewhere, Y has no way of knowing whether itself is on the path(whether there is a loop)
- ◆ Rumor based routing (依照传闻进行路由)

■ Improvement: Path Vector Routing

- ◆ BGP-4 (Border Gateway Protocol)
- ◆ IDRP (Inter-domain Routing Protocol)

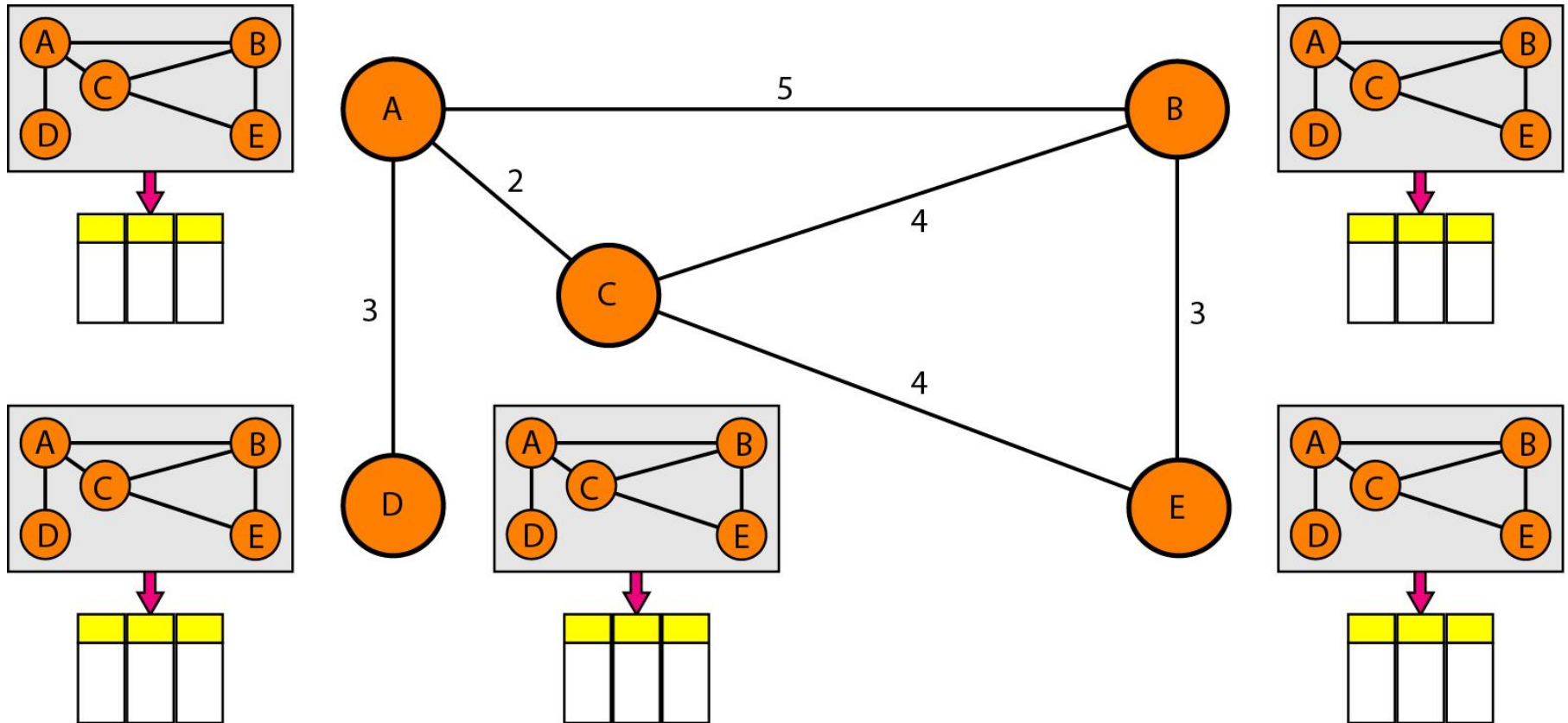
Routing Algorithms

- (補)Concept of Routing and Routing Table
- 5.2.1 Optimality principle
- 5.2.2 Dijkstra's "Shortest Path" Algorithm
- 5.2.3 Flooding
- 5.2.4 Distance Vector Routing (DVR)
- *5.2.5 Link State Routing (LSR)*
- 5.2.6 Hierarchical Routing

(1) Link State Routing(链路状态选路)

- In LSR, each router shares its knowledge about the its neighbourhood with **every other** routers in the network.
- However, in LSR, the Link-State Packet (LSP) defines the **best known network topology** is sent to every routers after it is constructed locally.
(**flooding**)
- 3 keys to understand how LSR works:
 - ◆ Sharing knowledge about **network topology**.
 - ◆ Sharing **with every other routers**.
 - ◆ Sharing **when there is a change**.
- From the received LSPs and knowledge of **entire topology**, a router can then calculate the **shortest path** from itself to every other routers in the area.

LSR: What routers know?



The routers have knowledge of **entire** topology

(2) Link State Routing: Operations

- Each router must do the following
 1. Discover its neighbors, learn their network address
 2. Measure the delay or cost to each of its neighbors
 3. Construct a packet (**LSP**: Link-state Packet, 链路状态包) telling all it has just learned
 4. Send LSP to **all other routers** (not only neighbors) reliably (**flooding**)
 5. Compute **the shortest path** to every other router

Step1: Learning about the Neighbors

■ HELLO packet

- ◆ For a broadcast network, Broadcasting a special HELLO packet
- ◆ For a point-to-point network, Unicast HELLO to neighbors

■ Neighbors send back a reply telling who it is

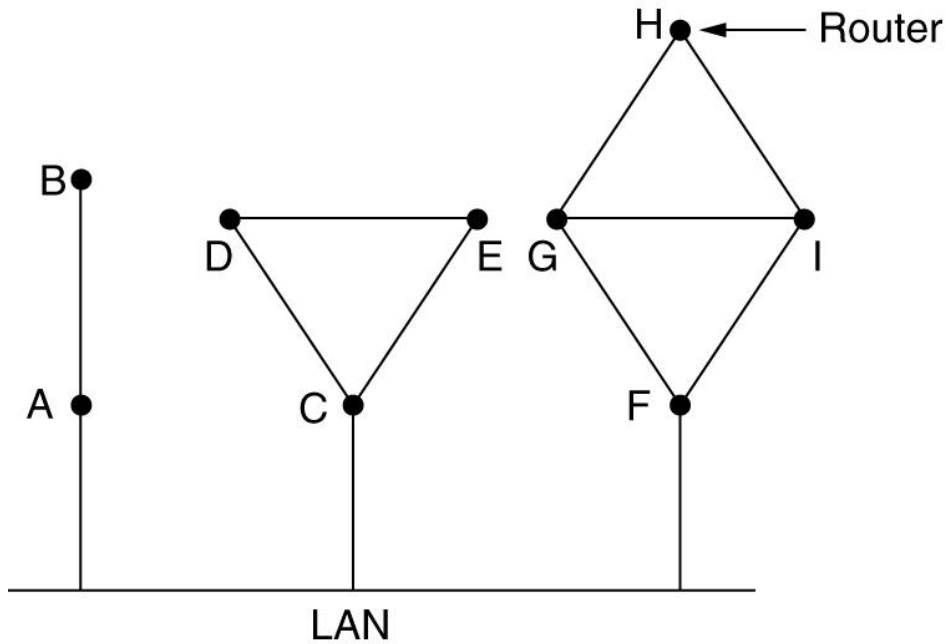
- ◆ Names (Route IDs) must be globally unique

■ Simplify the topology

- ◆ When two or more routers are connected by a LAN or other multi-access network

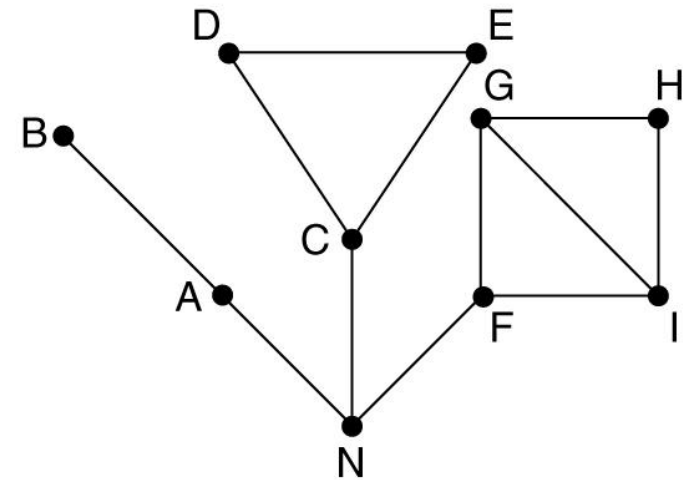
How to model the LAN

A LAN is considered as a Node(虚拟节点)



(a)

(a) Nine routers and a LAN.

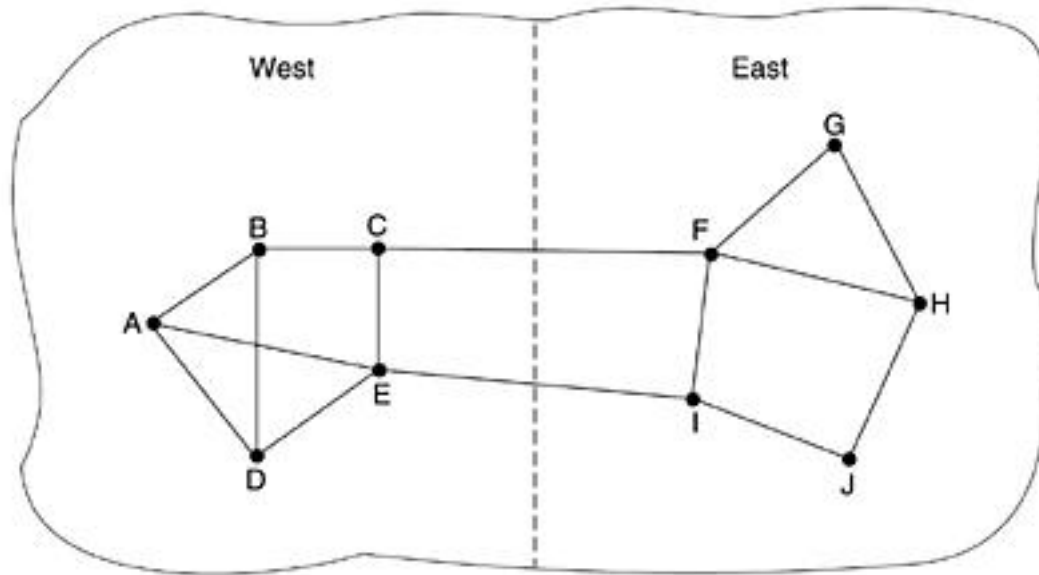


(b)

(b) A graph model of (a)

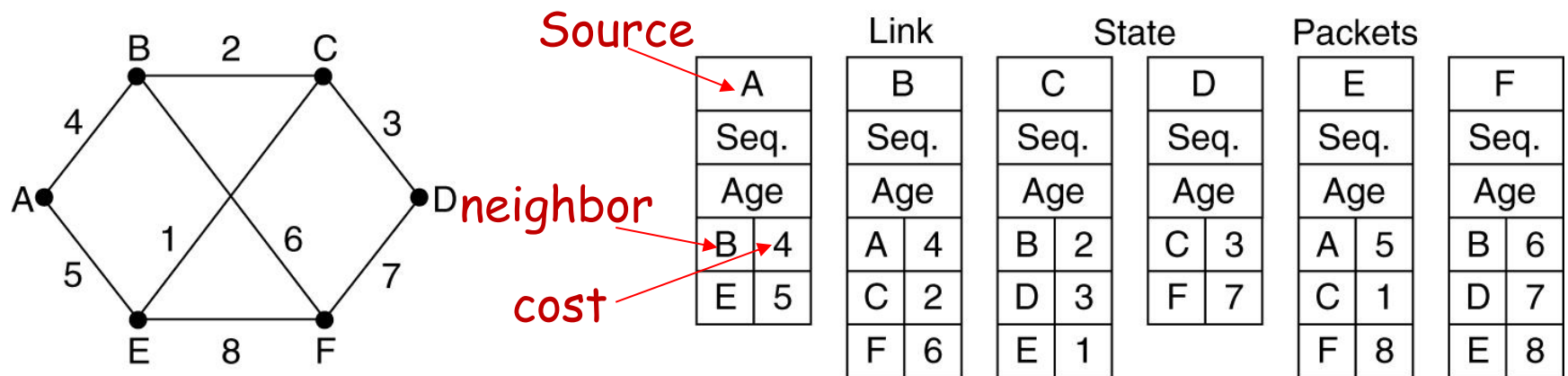
Step 2: Measuring Line Cost

- The delay to each of its neighbors
 - ◆ Measuring the round-trip time (ECHO Packet)
 - ◆ Bandwidth
 - ◆ An issue is whether to take the load into account when measuring the delay? (routing tables may oscillate, leading to erratic routing and many potential problems)



路由振荡，
导致不稳定

Step 3: Building Link State Packets



(a) A subnet. (b) The link state packets for this subnet

■ When to build LSP

- ◆ Build LSP **periodically**
- ◆ Build LSP when some **significant event** occurs
 - such as a line or neighbor going down or coming back up again or changing its properties appreciably

About Link State Packets

- **Sequence number:** used to check whether the received LSP is new
 - ◆ If LSP is new, forward with **flooding**
 - ◆ If LSP is a duplicate, it is **discarded**
 - ◆ If lower than the highest one seen so far ever arrives, it is rejected as being obsolete
 - ◆ If a router is down, and then restart, ...
- **Age:** used to delete looped LSP
 - ◆ If a router is down
 - ◆ If a router changed its routerID
 - ◆ If a router need to delete an LSP

Step 4: Distributing the Link State Packets

■ Responsibility

- ◆ Distributing the LSP *reliably*

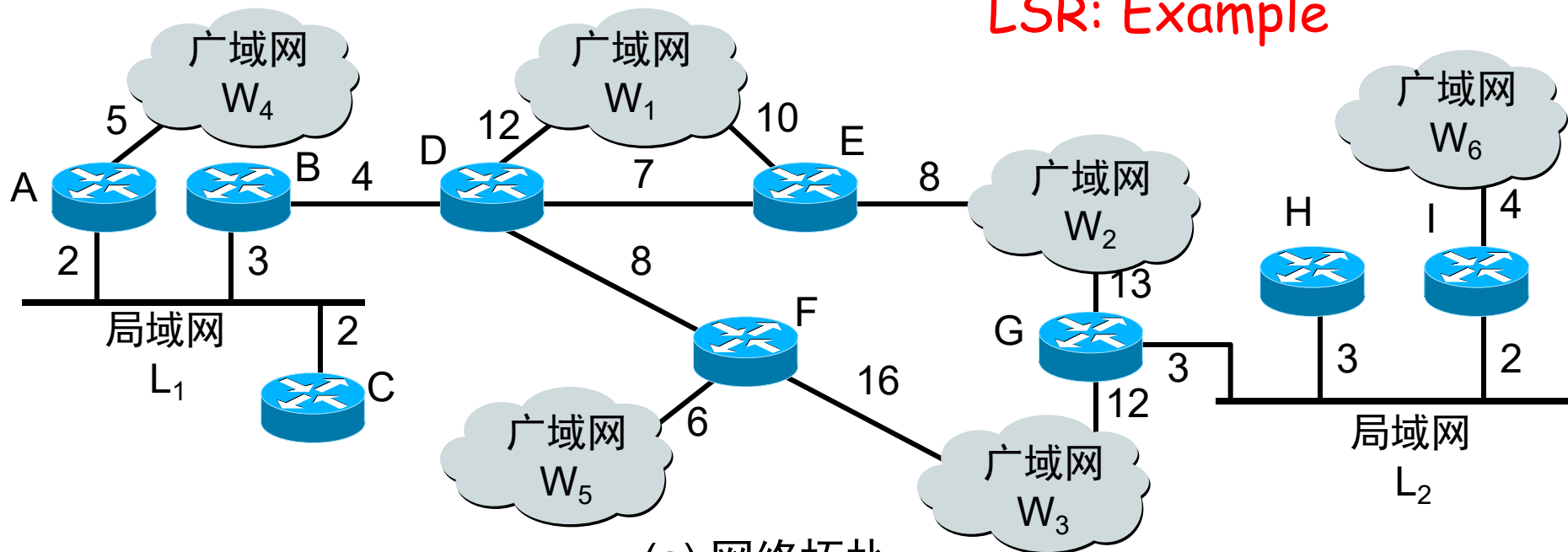
■ The basic distribution algorithm

- ◆ Use *flooding* to distribute LSPs
- ◆ Guard against errors on the router-router lines, all LSP are acknowledged

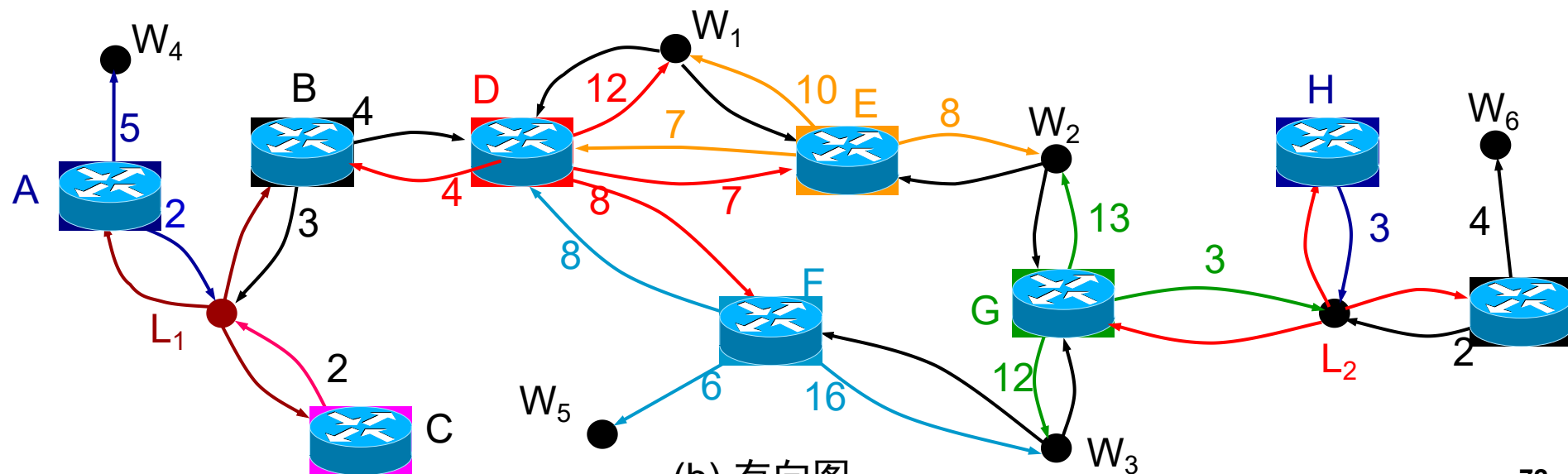
Step 5: Computing the New Route

- Once a router has accumulated a full set of LSPs, it can construct the entire graph
 - ◆ Every link is represented twice, once for each direction. The two values can be averaged or used separately.
- Run **Dijkstra's algorithm** locally to construct the shortest path to all possible destinations
- The results of this algorithm can be installed in the routing tables

LSR: Example

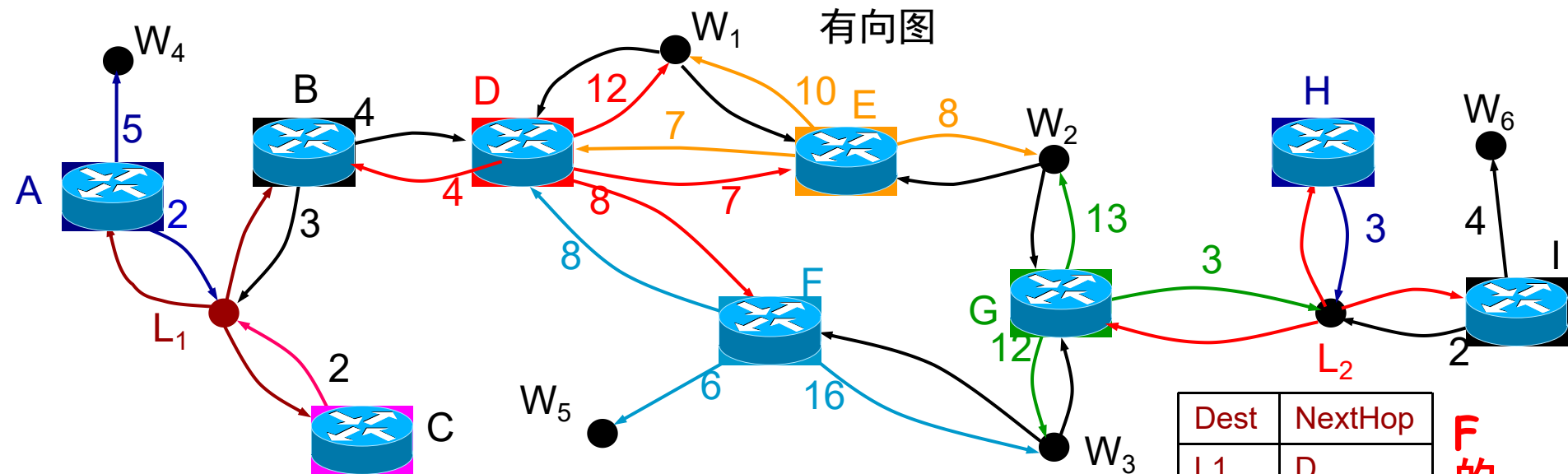


(a) 网络拓扑

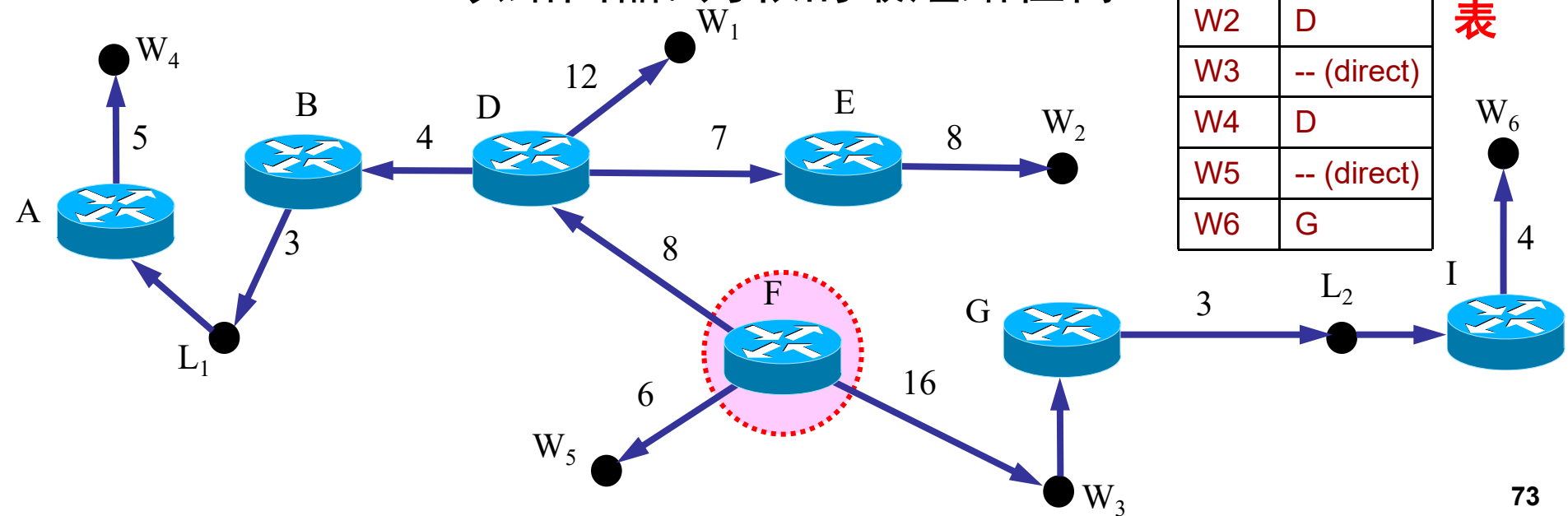


(b) 有向图

有向图



以路由器F为根的最短路径树



Dest	NextHop
L1	D
L2	G
W1	D
W2	D
W3	-- (direct)
W4	D
W5	-- (direct)
W6	G

F 的路由表

(3) Adoptions of Link State Routing

■ IS-IS (Intermediate System-Intermediate System):

- ◆ support multiple network layer protocols at the same time
- ◆ DECnet, ISO CLNP, IP, AppleTalk, Novell NLSP (IPX)

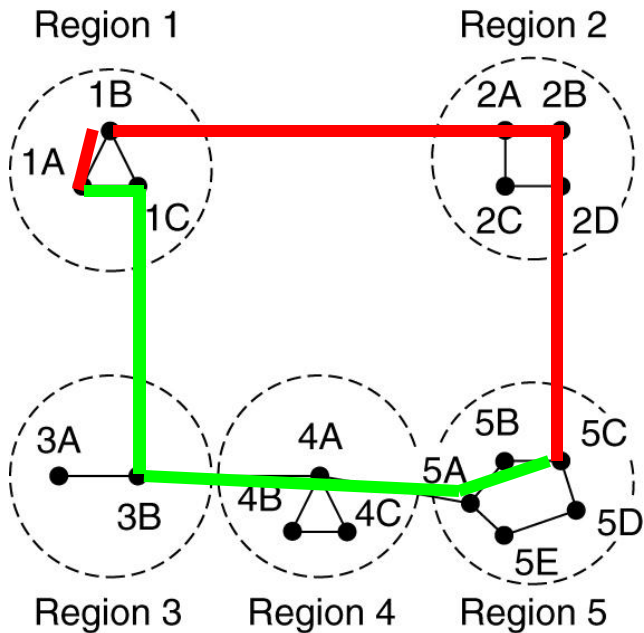
■ OSPF (Open Shortest Path First)

- ◆ designed several years after IS-IS
- ◆ Only for IP

Routing Algorithms

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- 5.2.3 Flooding
- 5.2.4 Distance Vector Routing (DVR)
- 5.2.5 Link State Routing (LSR)
- *5.2.6 Hierarchical Routing*

Hierarchical Routing(层次选路/分级选路) (1)



1A→5C

(a)

Full table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

May take
sub-optimal
route
(次优路由)

(c)

Hierarchical Routing (2)

Levels : **regions**区域 -> **clusters**簇 -> **zones**区 -> **groups**群

Consider a subnet with 720 routers.

- If there is no hierarchy
 - ◆ 720 routing table entries
- If the subnet is partitioned into 24 regions of 30 routers each (Total:53)
 - ◆ 30 local entries
 - ◆ 23 remote entries
- If a three-level hierarchy is chosen, with 8 clusters, each containing 9 regions of 10 routers (Total: 25)
 - ◆ 10 entries for local routers
 - ◆ 8 entries for routing to other regions within its own cluster
 - ◆ 7 entries for distant clusters

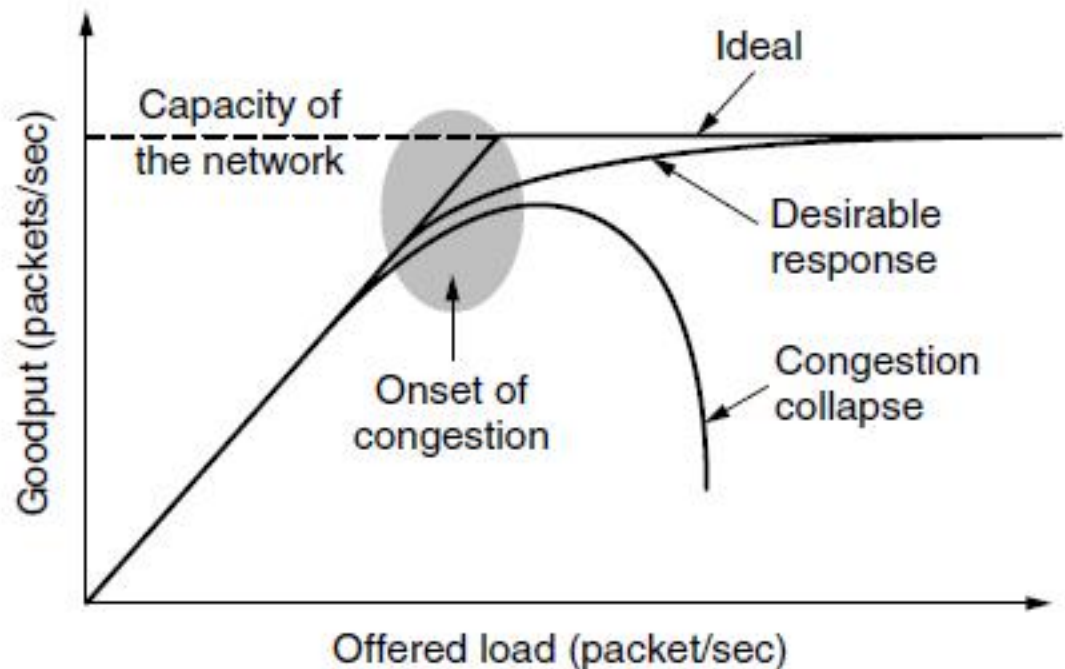
Outline

- 5.1 Introduction to Network Layer
- 5.2 Routing Algorithms
- 5.3 Congestion Control
- 5.4 Quality of Service
- 5.5 Internetworking
- 5.6 The network layer in Internet

(1) What is Congestion(拥塞)?

- Informally, “too many sources sending too much data too fast for **network** to handle” (资源不足)
- Congestion may occur if the **load** on the network (the number of packets sent to the network) is **greater than** the **capacity** of the network (the number of packets a network can handle)

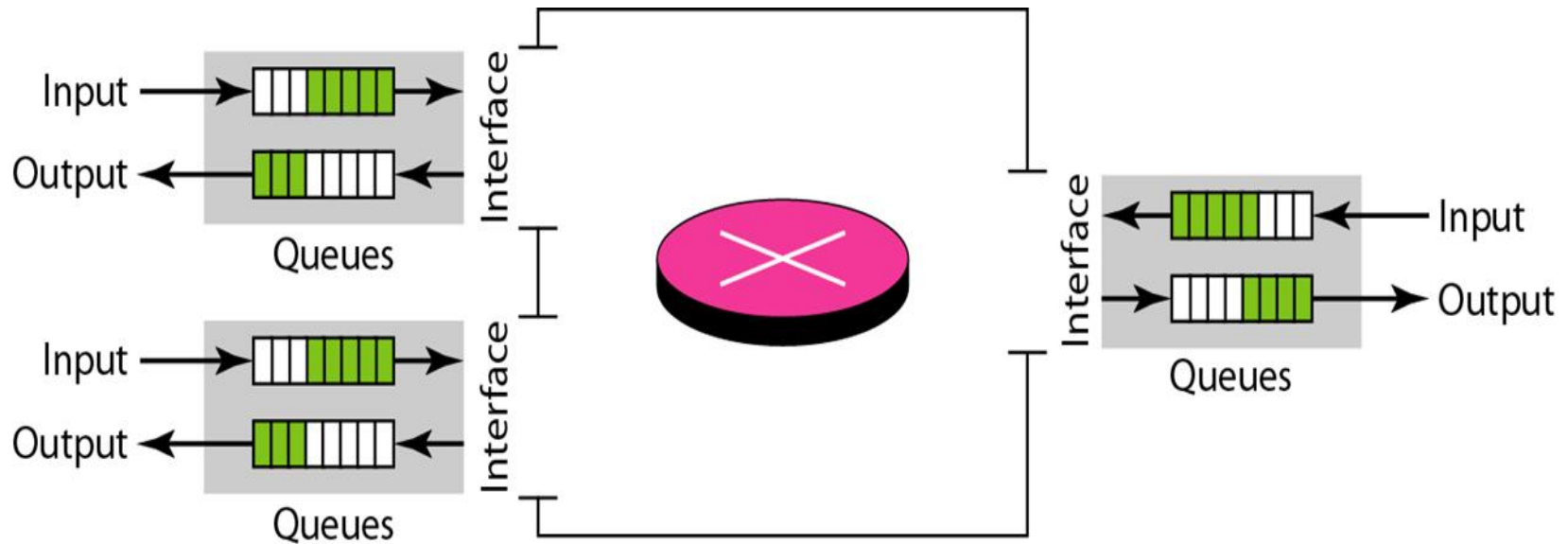
When **too much traffic** is offered, congestion sets in and performance degrades sharply



(2) Symptoms of Congestion

- Congestion happens in any system that involves waiting or **queuing** 
- 导致拥塞的次要因素：
 - ◆ CPU处理速度慢
 - ◆ 链路带宽低
 - ◆ 缓冲队列短
- Symptoms:
 - ◆ **Long delays** (Queuing in router buffers)
 - ◆ **Lost packets** (Buffer overflow at routers)

Queues in a Router

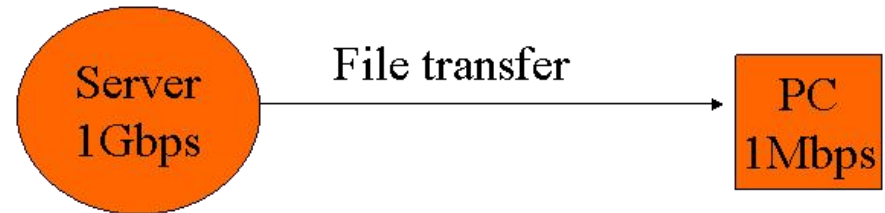


- If the rate of packet arrival is higher than the packet processing rate, the input queues become longer and longer
- If the packet departure rate is lesser than the packet processing, the output queues become longer and longer.

(3)Flow control vs. Congestion control

■ Flow Control(流量控制)

- ◆ To avoid overflowing the receiver
- ◆ Local issue, involves **only 2 endpoints**, the sender and the receiver



■ Congestion Control(拥塞控制)

- ◆ To avoid congesting the network
- ◆ **Global issue**, concerns all nodes/hosts along the path

5.3.1 Approaches to Congestion Control

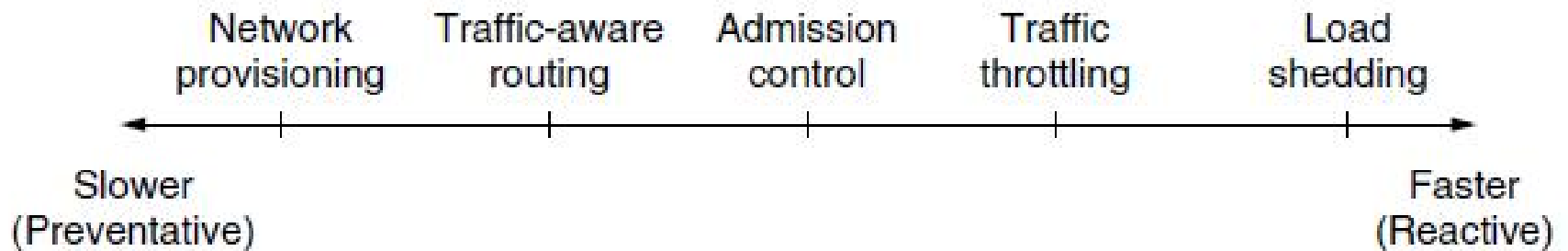
■ Two types of solutions

◆ Increasing the resources

- Network provisioning(网络供给)
- Traffic aware-routing(业务量感知路由)

◆ Decreasing the load

- Admission control(准入控制)
- Traffic throttling(流量限制)
- Load shedding(负载掉落)



5.3.4 Traffic Throttling(业务量减速/流量调节)

■ Feedback based solution(基于反馈的解决方案)

■ 关键问题： 如何确定路由器将要发生拥塞？

■ Step 1: Router--detects congestion, by

- ◆ utilization of the output links(输出链路利用率： 不够精确)
- ◆ buffering of queued packets inside the router (排队分组)
- ◆ number of packets that are lost due to insufficient buffering (分组丢包数量： 太迟)

Estimate of the queuing delay:

EWMA (Exponentially Weighted Moving Average, 指数加权移动均值)

$$d_{\text{new}} = \alpha d_{\text{old}} + (1 - \alpha)s$$

- ◆ d: recent queue length
- ◆ s: sample of the instantaneous queue length
- ◆ $0 < \alpha < 1$ 平衡因子，体现了历史值对当前估计值的影响程度

Traffic Throttling

■ Step 2: Transfer Information about Congestion

(通知发送方：抑制分组、反压、显式/隐式拥塞通知)

- ◆ Sending a packet(**Choke Packet**, 抑制分组) to the source host, giving it the destination found in the packet
- ◆ The original **packet is tagged** so that it will not generate any more choke packets
- ◆ To avoid increasing load on the network, the router may only send **choke packets at a low rate**

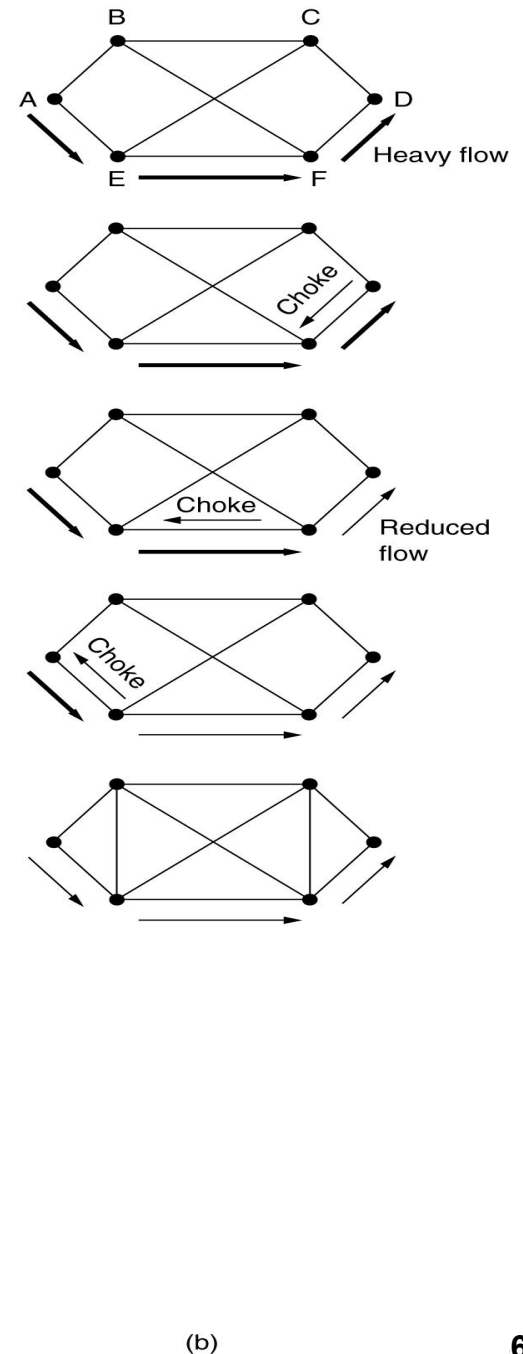
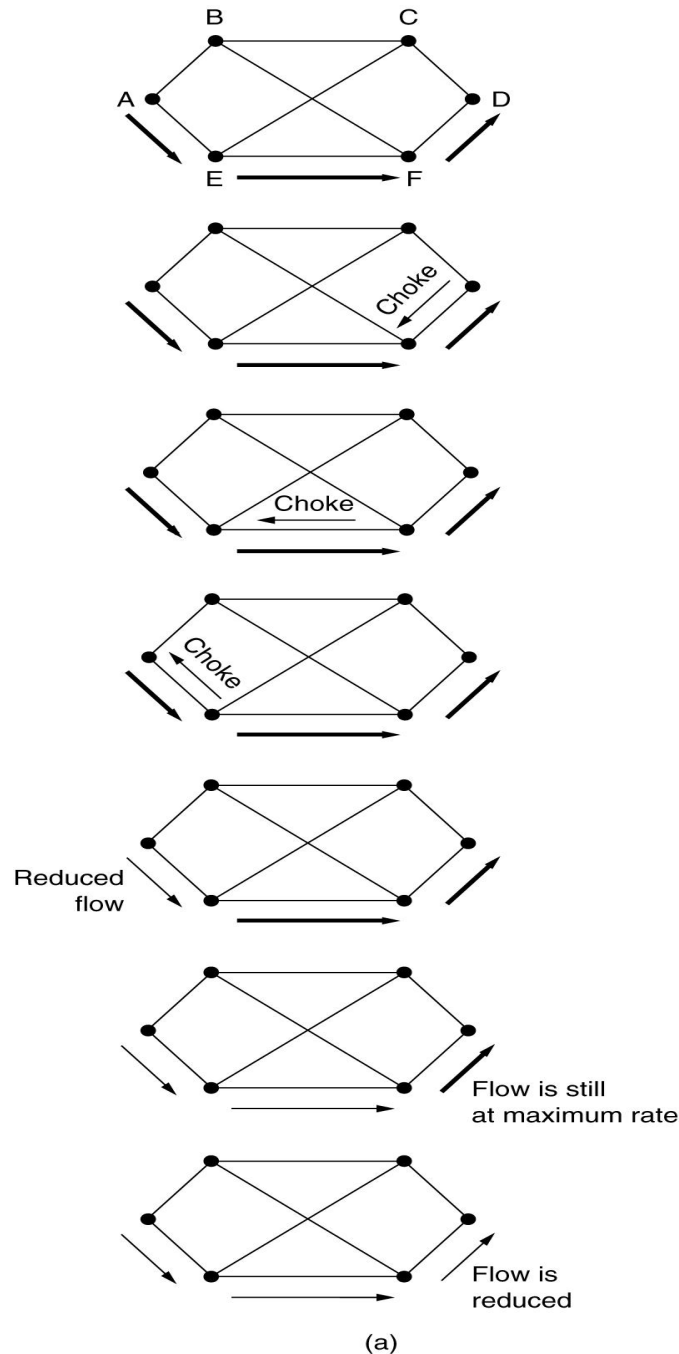
■ Step 3: **Source--adjust System Operation**

- ◆ Receiving choke packet, the source may **reduce the traffic** sent to the specified destination by x percent (e.g. 50%)

Other Feedback Mechanisms: Hop-by-Hop Backpressure

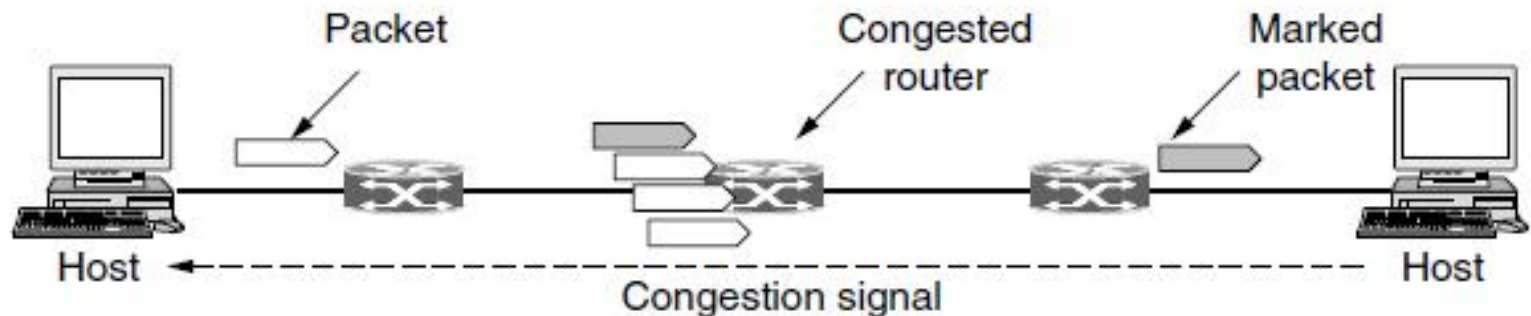
(a) A choke packet that affects only the source.

(b) A choke packet that affects each hop it passes through.



Other Feedback Mechanisms: ECN

- Explicit Congestion Notification, 显式拥塞通知
 - ◆ Used by IP and TCP
 - ◆ Two bits in the IP packet header are used to record whether the packet has experienced congestion.
 - ◆ Router detecting congestion marks the packet as it is forwarded.
 - ◆ The destination will then echo any marks back to the sender as an explicit congestion signal in its next reply packet.



5.3.5 Loading Shedding(负载掉落)

■ 丢包

- ◆ When routers are being inundated by packets, just throw them away

■ 关键问题：选择丢弃哪些数据分组

■ Policies: depend on the type of applications

- ◆ Pick packets at **random** to drop

◆ Wine & Milk

- E.g. file transfer, multimedia

◆ 智能丢弃：Requires cooperation from the senders

- E.g. algorithms for compressing video, ...
- The low-priority packets being cheaper to send than the high-priority ones
- Allow hosts to exceed the limits specified in agreement, but all excess traffic must be marked as low priority

Load shedding in Internet: RED

- Random Early Detection (随机早期检测)
- Drop packets before the situation has become hopeless
 - ◆ When the **average queue length** on some line exceeds a threshold, the line is said to be congested, and a small fraction of the **packets are dropped at random** (随机丢包)
- How should the router tell the source about the problem?
 - ◆ Just **discard** the selected packet
 - ◆ The affected sender will notice the loss when there is no acknowledgement, **implicit notification** (隐式通知)
- Sources respond to lost packets by slowing down their transmission rate
 - ◆ In wireless networks, where most losses are due to noise on the air link, this approach cannot be used.

Review of Congestion Control

- Congestion may occur if the load on the is greater than the capacity of the network
- Symptoms:
 - ◆ Long delays (Queuing in router buffers)
 - ◆ Lost packets (Buffer overflow at routers)
- Congestion control approaches
 - ◆ Increasing the resources: Network provisioning, Traffic aware-routing
 - ◆ Decreasing the load
 - Admission control: only in virtual circuit networks
 - Traffic throttling: 1) Router detects congestion
2) Congestion notification: choke packets, ECN
3) Source reduces traffic
 - Load shedding: RED

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5.4.1 Primary QoS Parameters(服务质量参数)

■ Reliability

- ◆ Packet error rate
- ◆ Packet loss rate

■ **Delay**: the time period from 1st bit of packet sent to last bit received

■ **Jitter**(抖动): the **variation** in delay of packets belonging to the same **flow**

■ **Bandwidth**: maximum data rate

QoS Requirements of Various Applications

Application	Reliability	Delay	Jitter	Bandwidth
E-mail	High	Low	Low	Low
File transfer	High	Low	Low	Medium
Web access	High	Medium	Low	Medium
Remote login	High	Medium	Medium	Low
Audio on demand	Low	Low	High	Medium
Video on demand	Low	Low	High	High
Telephony	Low	High	High	Low
Videoconferencing	Low	High	High	High

ATM Networks: Flow Classification

ATM networks classify flows in four broad categories with respect to their QoS demands:

- Constant bit rate

- ◆ e.g., telephony

- Real-time variable bit rate

- ◆ e.g., compressed videoconferencing

- Non-real-time variable bit rate

- ◆ e.g., watching a movie over the Internet

- Available bit rate

- ◆ e.g., file transfer

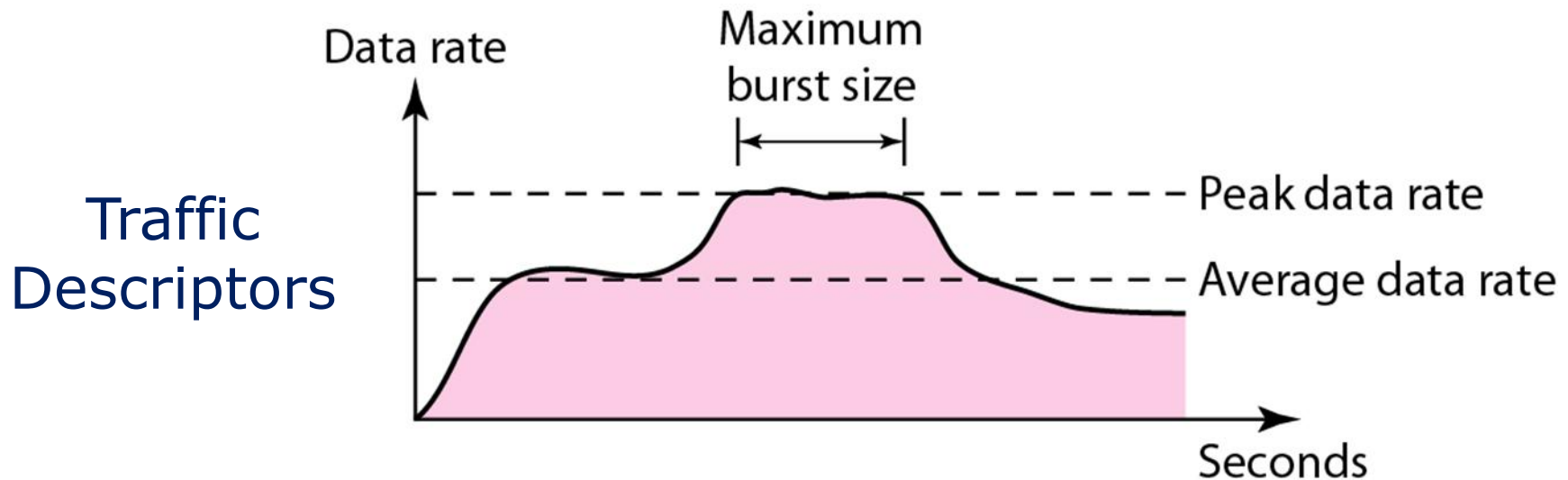
Techniques to improve QoS

- *5.4.2 Traffic shaping*
- 5.4.3 Packet Scheduling
- 5.4.4 Admission control(自阅)
- 5.4.5 Integrated Services(不讲)
- 5.4.6 Differentiated Services(不讲)

(1)Traffic Shaping(业务量/流量整形)

■ Problem

- ◆ **Traffic burst**: Host emits the packets irregularly, which may cause congestion to occur in network



Traffic Shaping

■ Service Level Agreement (SLA)

- ◆ Between the user and the carrier(网络-服务质量提供者)
- ◆ The user: traffic shaping reduces congestion and helps carrier live up to its promise (流量整形)
- ◆ The carrier: traffic policing(流量监管)
(**CAR**: Committed Access Rate) 承诺的接入速度

■ Traffic shaping(流量整形--平滑流量)

- ◆ Regulate the average rate and burstiness of data transmission, **smooth out the traffic**

■ Traffic Policing(流量监管)

- ◆ Monitoring a traffic flow
- ◆ Packets in excess of the agreed pattern might be dropped or marked as having lower priority

(2) Leaky Buckets and Token Buckets

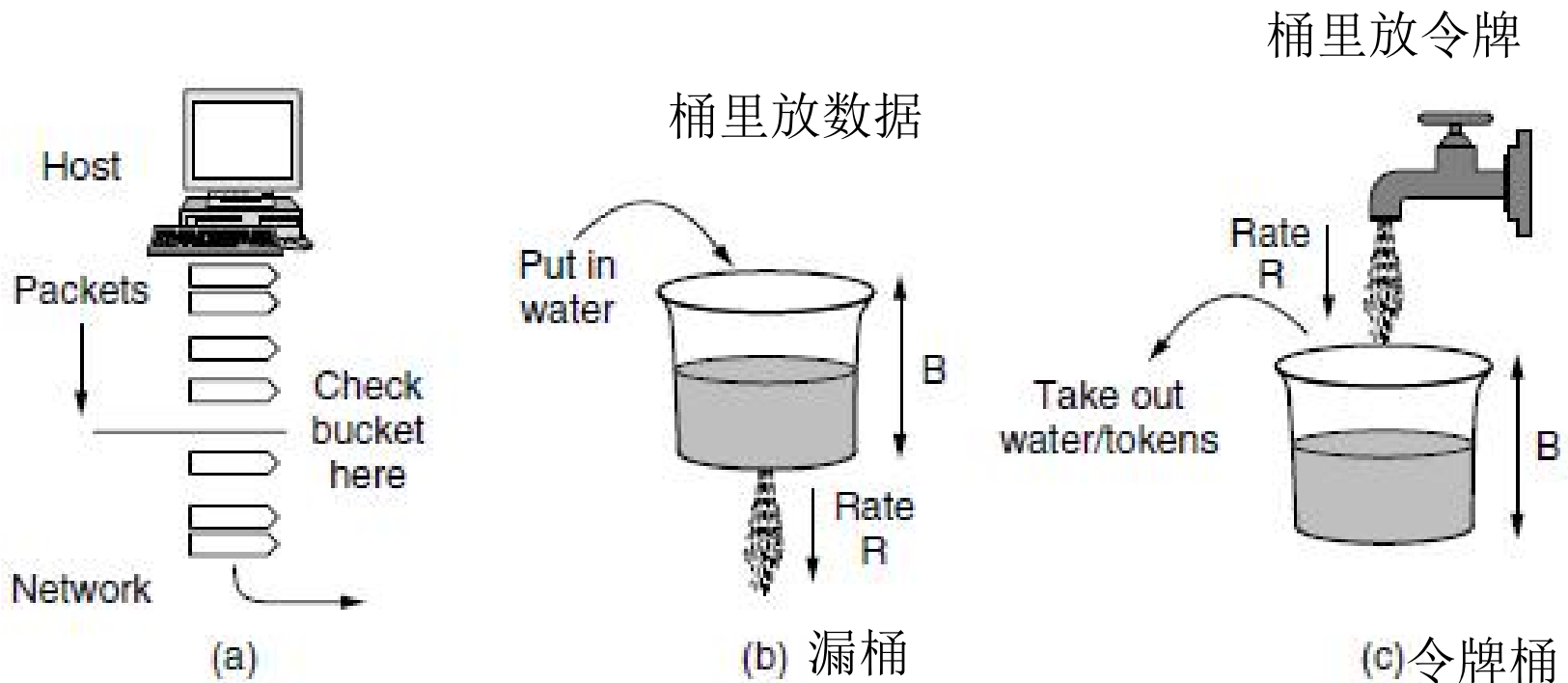
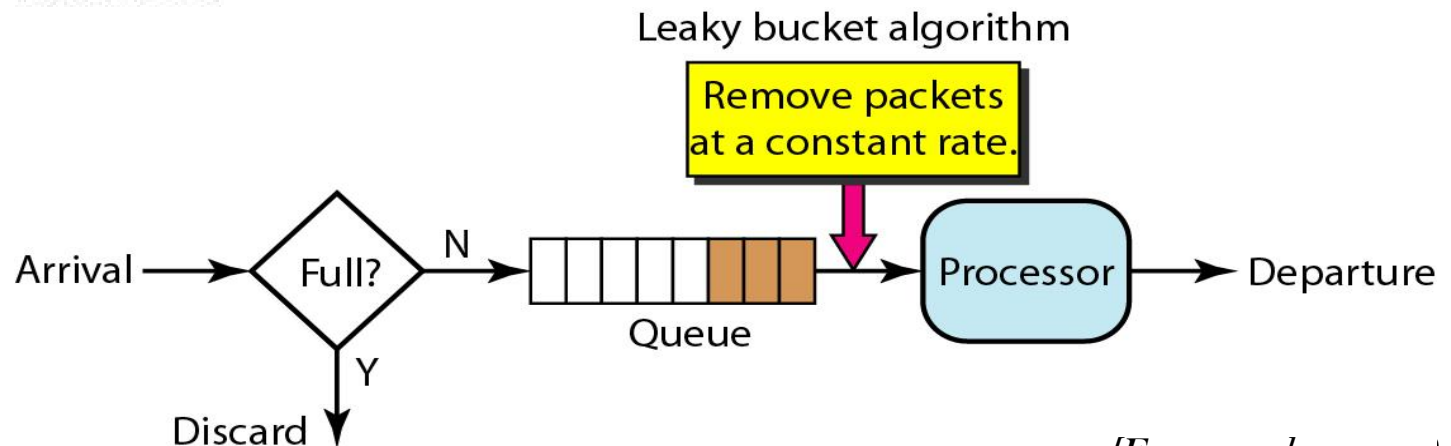
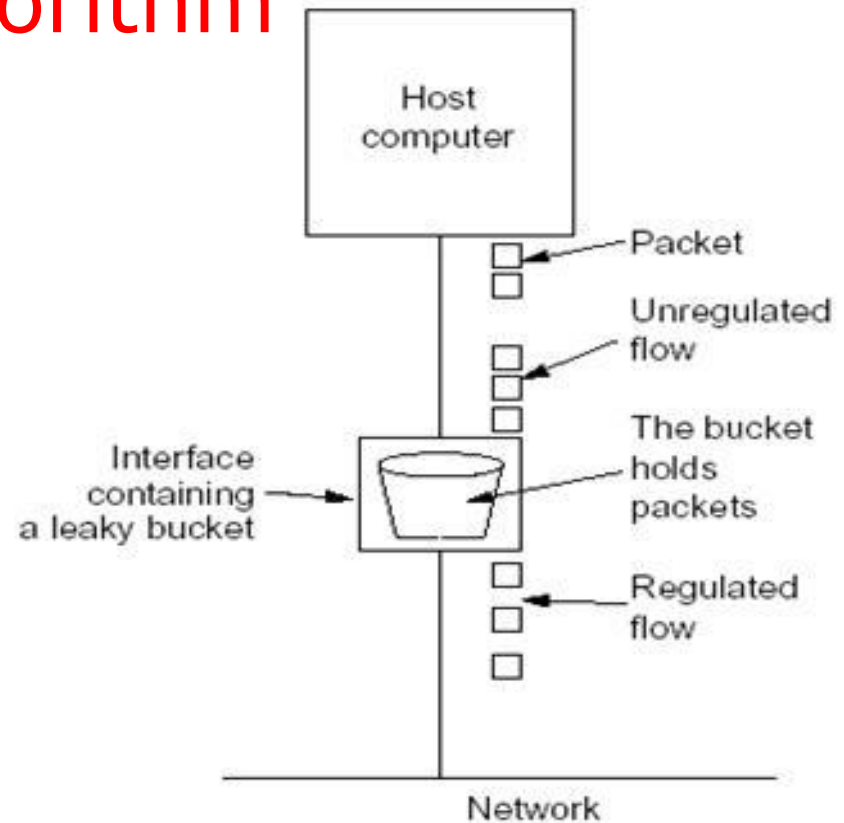
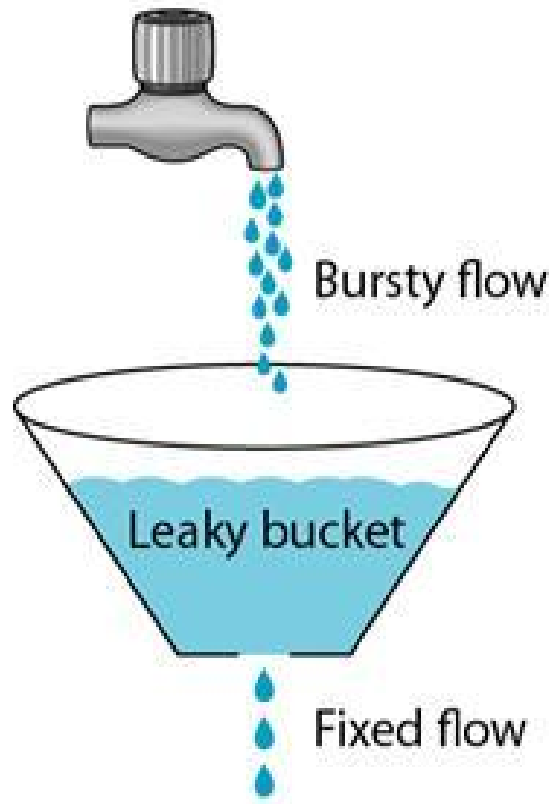


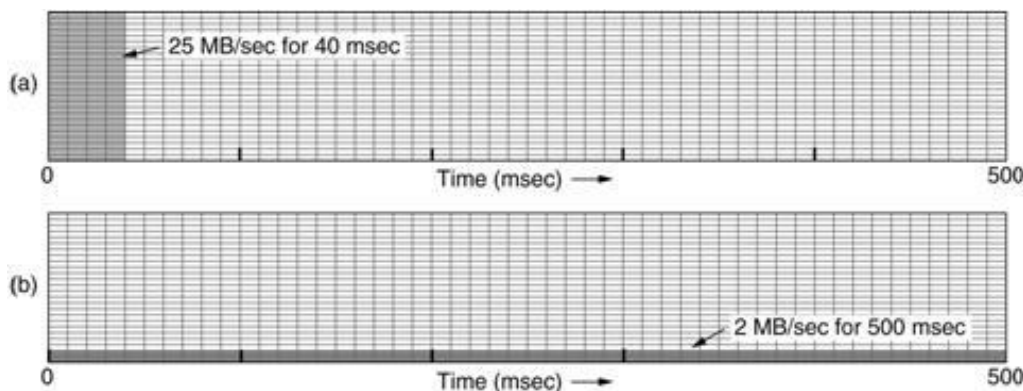
Figure 5-28. (a) Shaping packets. (b) A leaky bucket. (c) A token bucket.

Leaky Bucket (漏桶) Algorithm



Leaky Bucket: Example

- Turns an uneven flow of packets into an even flow
 - ◆ Smoothing out bursts (平滑突发流量)
 - ◆ Reducing the chances of congestion
- The byte-counting leaky bucket (桶满则数据丢失)



Bursting data

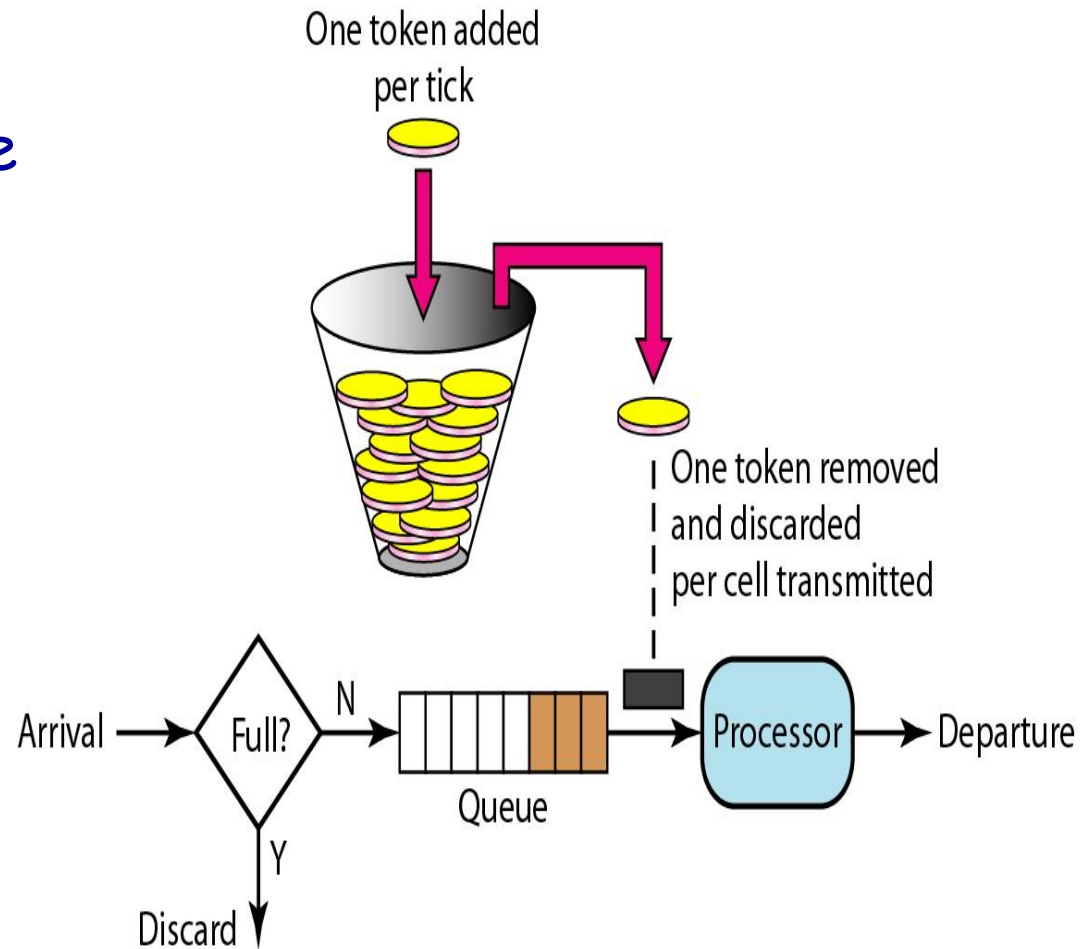
Output from a leaky bucket.

- Bucket can be used to **shape** or **police** packets entering the network

If a packet arrives when the bucket is full, the packet must be queued until enough water leaks out to hold it (**host shapes its traffic**) or be discarded (**network polices traffic**)

(3) Token Bucket (令牌桶) Algorithm

- Token bucket allows idle hosts to **accumulate credit** for the future in the forms of token.
- Token bucket allows **bursts**, but up to a regulated maximum length



Token to be collected based on idle time and can be used later to send bursty packets as long as it is allowed by the bucket.

Leaky Bucket vs. Token Bucket

■ Burstiness

- ◆ **Leaky bucket** algorithm enforces a rigid output pattern at the **average rate**, no matter how bursty the traffic is
- ◆ **Token Bucket** allowing **some burstiness** in the output stream

■ When the bucket fills up

- ◆ Token bucket algorithm **throws away tokens** but **never discards packets**
- ◆ Leaky bucket algorithm **discards packets**

Token Bucket: Example

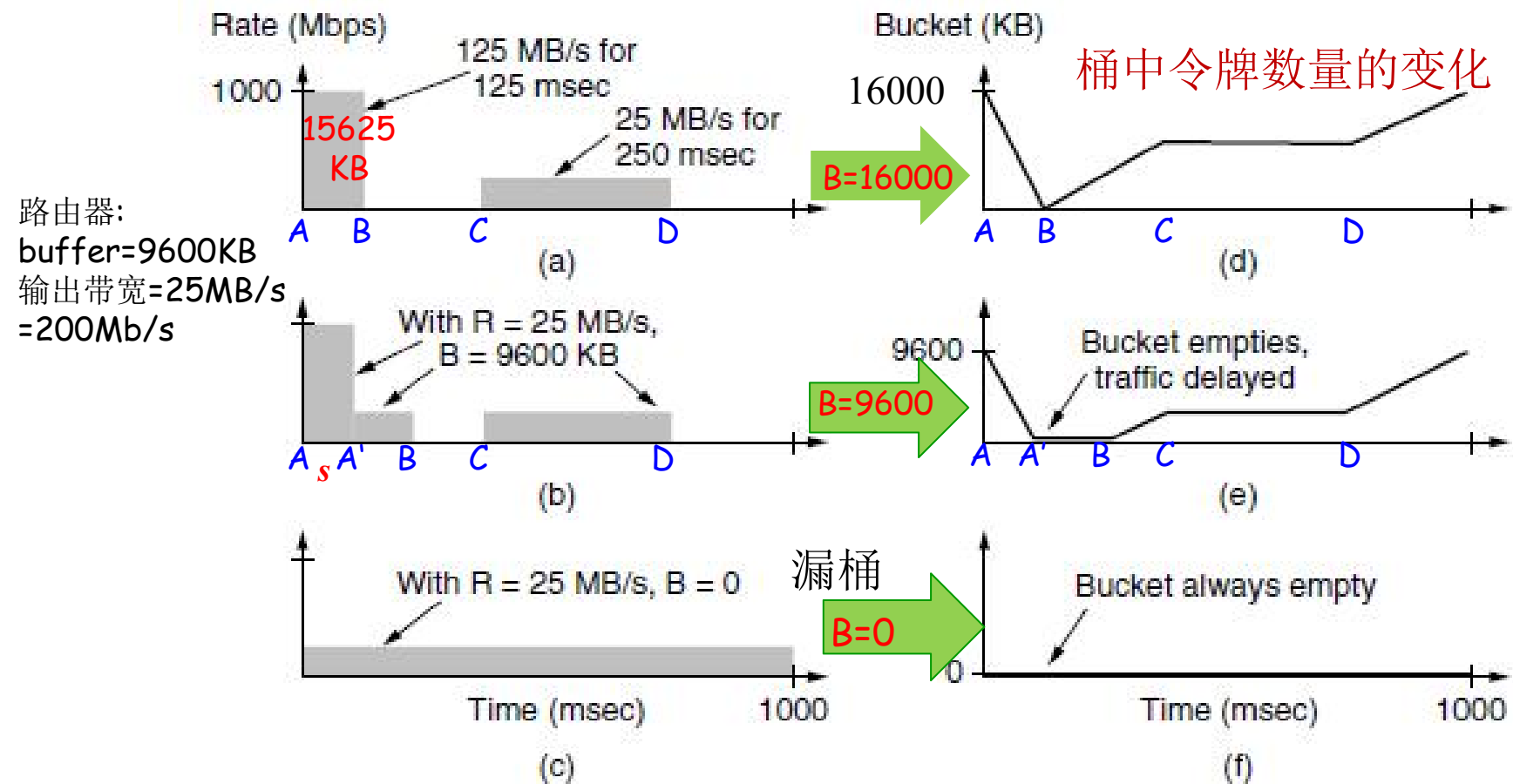


Figure 5-29. (a) Traffic from a host. Output shaped by a token bucket of rate 200 Mbps and capacity (b) 9600 KB and (c) 0 KB. Token bucket level for shaping with rate 200 Mbps and capacity (d) 16,000 KB, (e) 9600 KB, and (f) 0 KB.

The Length of the Maximum Rate Burst

经过令牌桶之后的突发数据持续时间:

burst length **S** sec

token bucket capacity **B** bytes

token arrival rate **R** bytes/sec

maximum output rate **M** bytes/sec

An output burst contains a maximum of **B + RS** bytes

the number of bytes in a maximum-speed burst of length **S** seconds is **MS**

$$B + RS = MS$$

$$S = B / (M - R)$$

If $B = 9600\text{KB}$, $M = 125 \text{ MB/sec}$, and $R = 25 \text{ MB/sec}$
then $S = 96 \text{ ms}$

Techniques to improve QoS

- 5.4.2 Traffic shaping
- *5.4.3 Packet Scheduling*
- 5.4.4 Admission control(自阅)
- 5.4.5 Integrated Services(不讲)
- 5.4.6 Differentiated Services(不讲)

Packet Scheduling

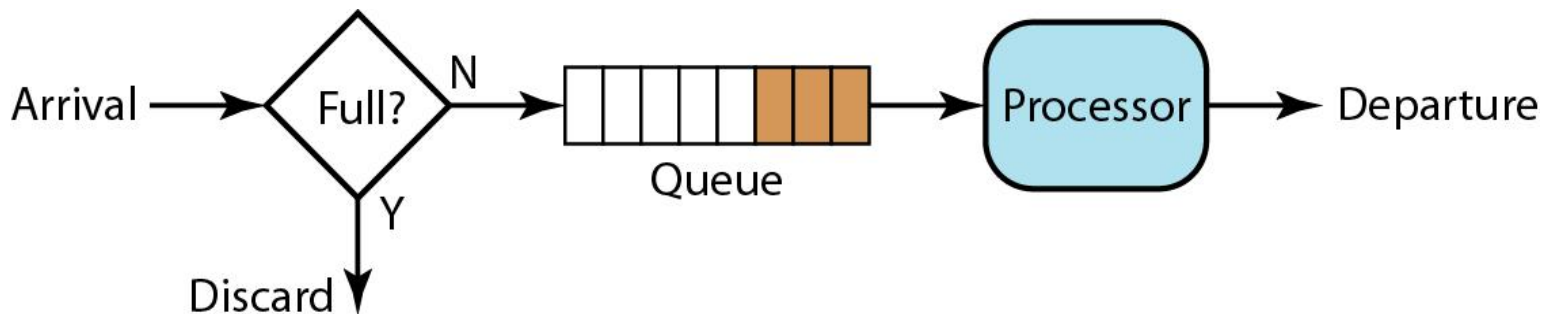
- Resources can be reserved(预留) for different flows
 - ◆ Bandwidth
 - ◆ Buffer space
 - ◆ CPU cycles

- Algorithms
 - ◆ FIFO (FCFS), tail drop
 - ◆ RED Algorithm
 - ◆ Fair queueing algorithm

Packet Scheduling:

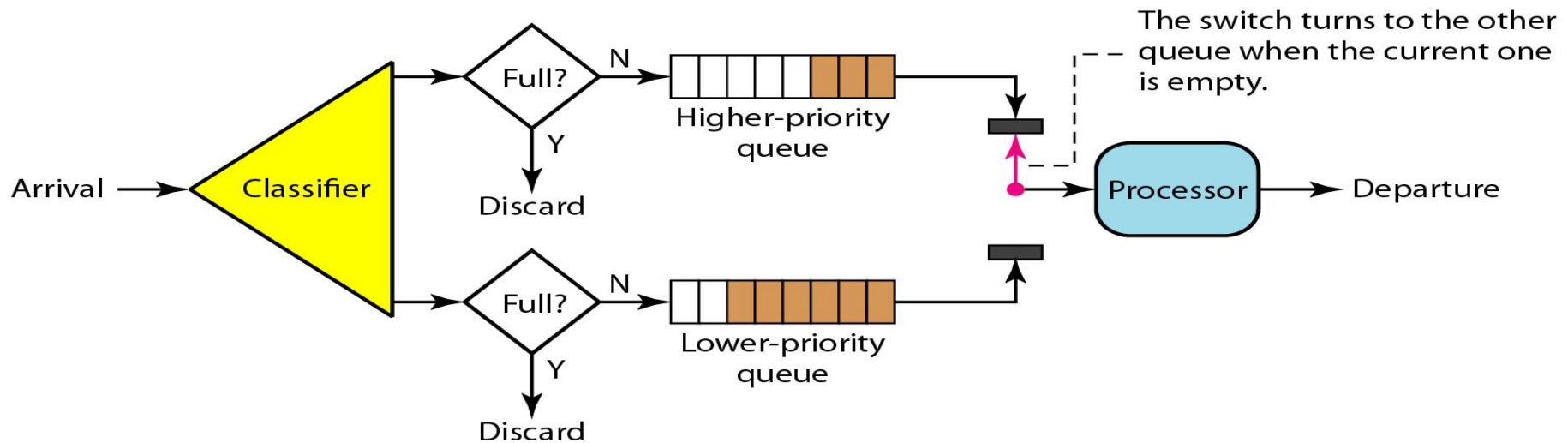
FIFO Queuing(先进先出队列)

- Packets from different flows arrive at a router for processing. A good scheduler treats different flows in a **fair** and **appropriate** manner to improve the QoS.
- **Methods**
 - ◆ FIFO queuing, Priority queuing, Weighted fair queue



FIFO: First In, First Out

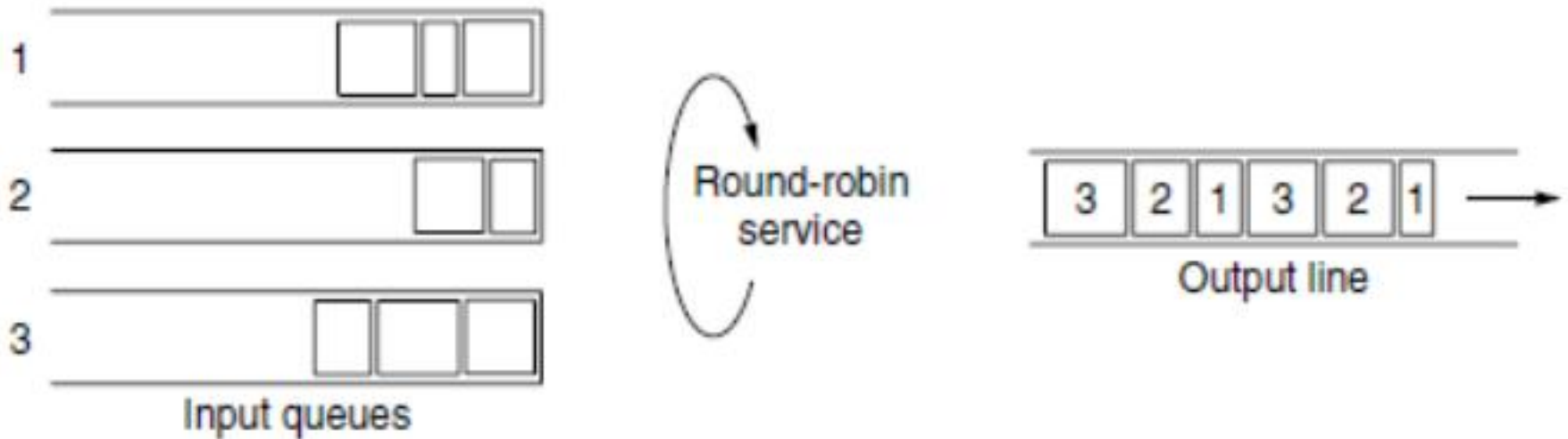
Packet Scheduling: Priority Queuing (优先级队列)



- **Packets** are first assigned a **priority** class. Each priority class has its **own queue**.
- When a line is idle, packets in the highest priority queue are processed first. 高优先级数据先发送
- **Better QoS** because **higher priority traffic** such as multimedia can reach the destination **with less delay**.
- Potential drawback: starvation of lower priority traffic, not fair. 对低优先级数据不公平

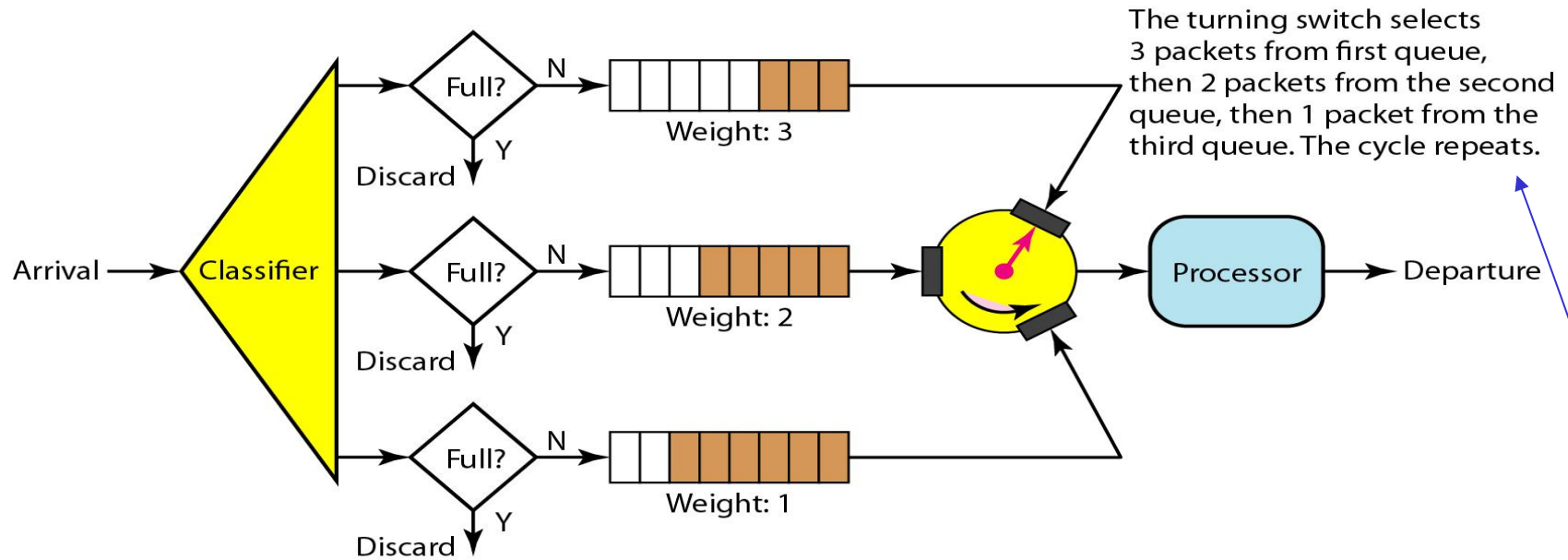
Packet Scheduling: Fair Queue (公平队列)

- Routers have separate **queues** for each flow for a given output line
- Packets are processed in a **round robin**(循环) way.



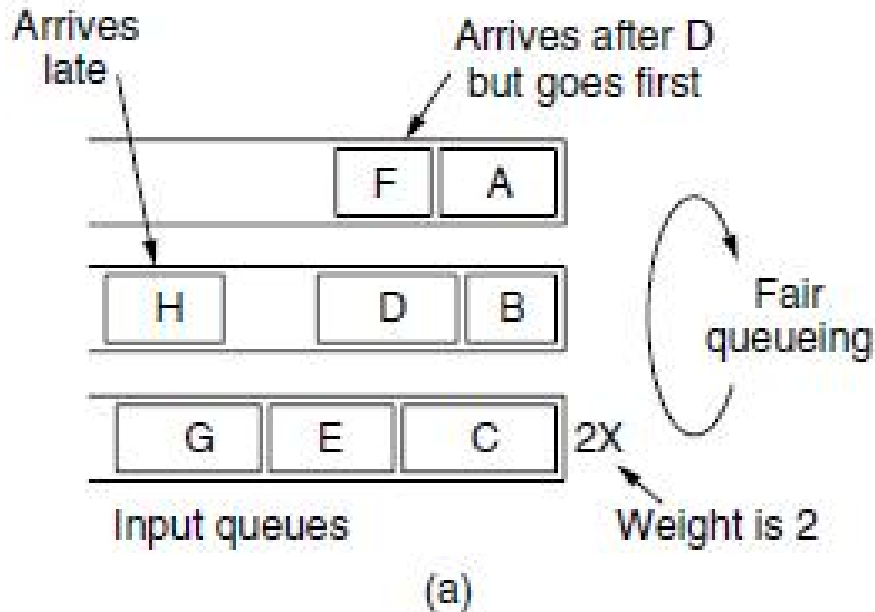
- It gives more bandwidth to hosts that use large packets than to hosts that use small packets

Packet Scheduling: Weighted Fair Queuing(公平加权队列)



- **Priority Queue + Fair Queue** (按权值轮流)
- Queues are weighted based on priority; higher priority means higher weights
- The system processes packets in each queue in a round-robin fashion with the number of packets selected from each queue based on corresponding weight.

WFQ Example



Packet	Arrival time	Length	Finish time	Output order
A	0	8	8	1
B	5	6	11	3
C	5	10	10	2
D	8	9	20	7
E	8	8	14	4
F	10	6	16	5
G	11	10	19	6
H	20	8	28	8

(b)

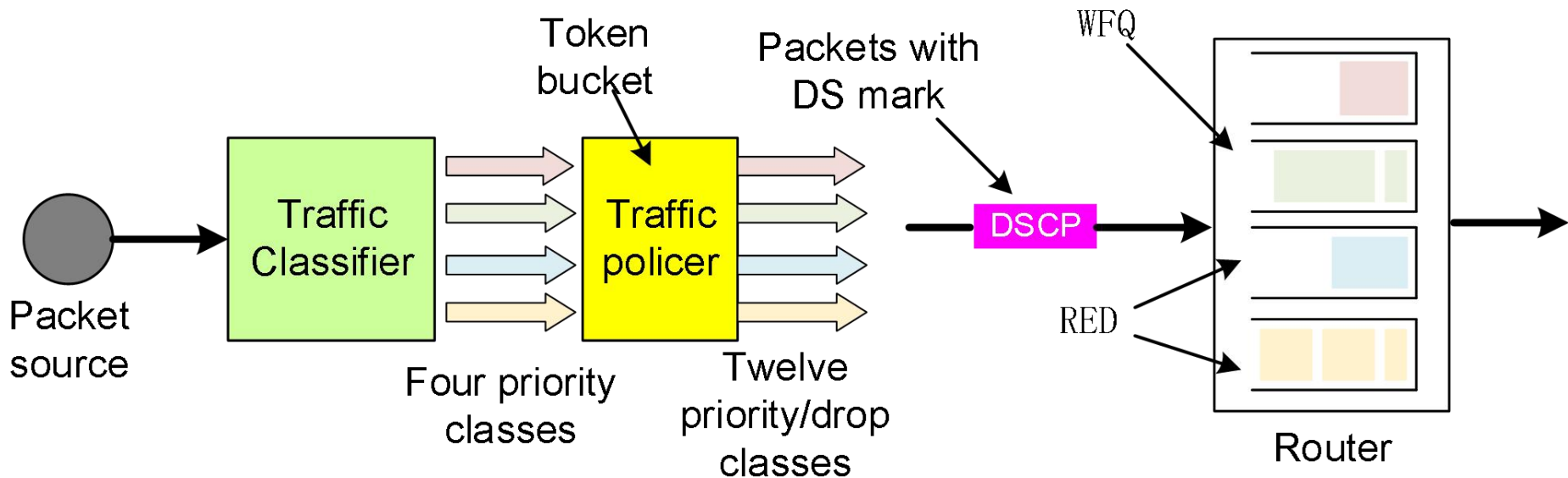
Figure 5-31. (a) Weighted Fair Queueing. (b) Finishing times for the packets.

$$F_i = \max(A_i, F_{i-1}) + \frac{L_i}{W}$$

F : finish time, A : Arrival time, L : length of packet, W : weight

QoS example-Assured Forwarding

- There are **four priority classes** and **three discard classes**(low, medium and high)
- IP Header-Type of Service(**ToS**): **DSCP**(6bit)+ECN(2bit)
- Traffic Classifier
- Traffic Policier



Outline

- 5.1 Introduction to Network Layer
- 5.2 Routing Algorithms
- 5.3 Congestion Control
- 5.4 Quality of Service
- 5.5 Internetworking
- 5.6 The network layer in Internet

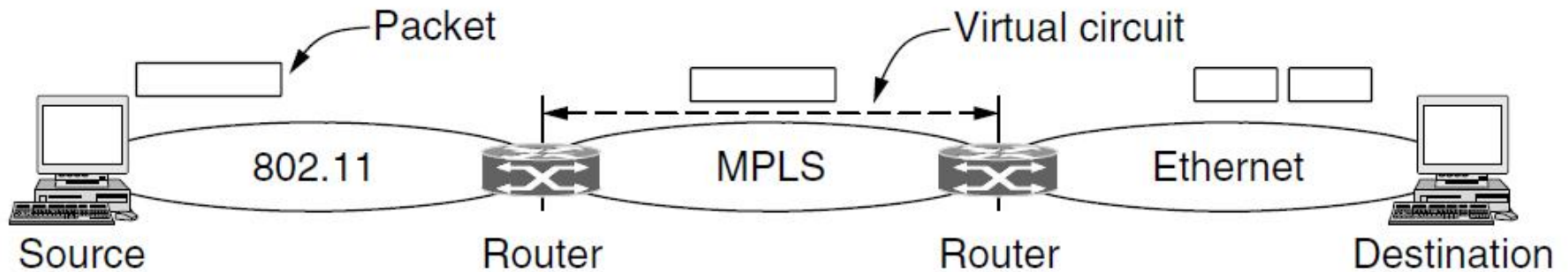
Connecting Networks

- Different networks & different protocols
 - ◆ Ethernet, Internet over cable, Fixed and mobile telephone networks, 802.11, 802.16, ATM, ...
- Purpose of interconnecting
 - ◆ Allow users on any network to communicate with users on all the other ones
- Internetwork(internet, 互联网): two or more networks are connected

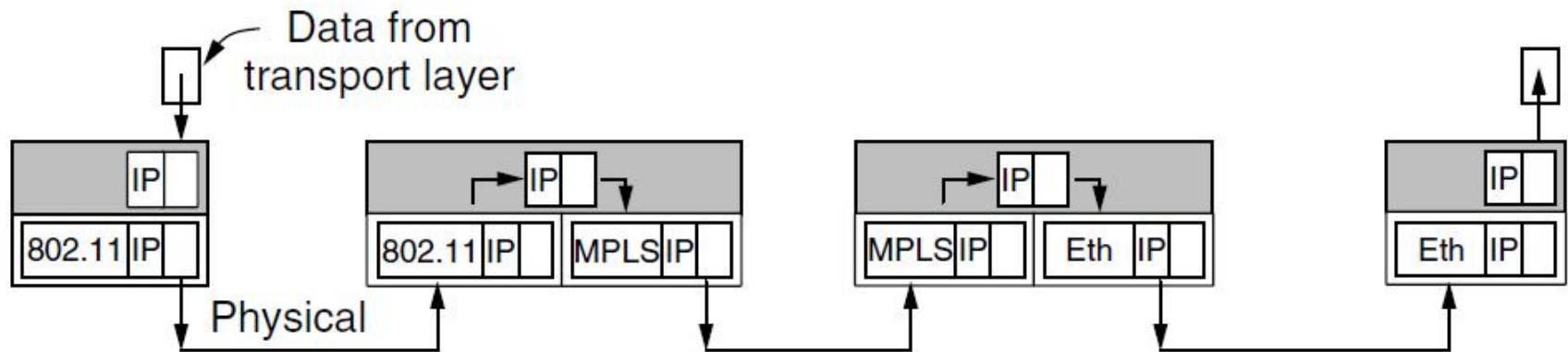
5.5.1 How Networks Differ

Item	Some Possibilities
Service offered	Connectionless versus connection oriented
Addressing	Different sizes, flat or hierarchical
Broadcasting	Present or absent (also multicast)
Packet size	Every network has its own maximum
Ordering	Ordered and unordered delivery
Quality of service	Present or absent; many different kinds
Reliability	Different levels of loss
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, packet, byte, or not at all

5.5.2 How Networks Can Be Connected



A packet crossing different networks

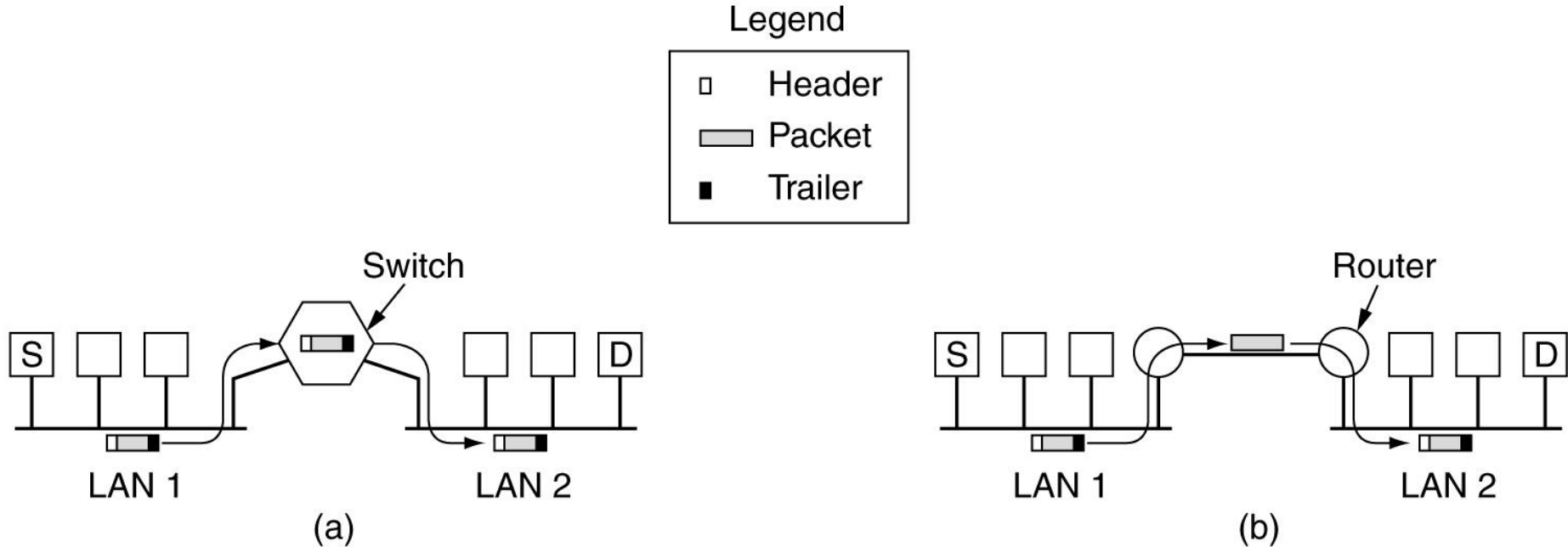


Network and link layer protocol processing

Interconnection at different layers

- Layer 1: Repeater, HUB
- Layer 2: Bridges, Switches
 - ◆ Protocol translation: Ethernets/802.11
- Layer 3: Router (Packet translation)
- Layer 4: Transport gateway
 - ◆ TCP/Wireless TCP
 - ◆ TCP/SNA
- Above layer 4: Application gateway

Bridge vs. Router

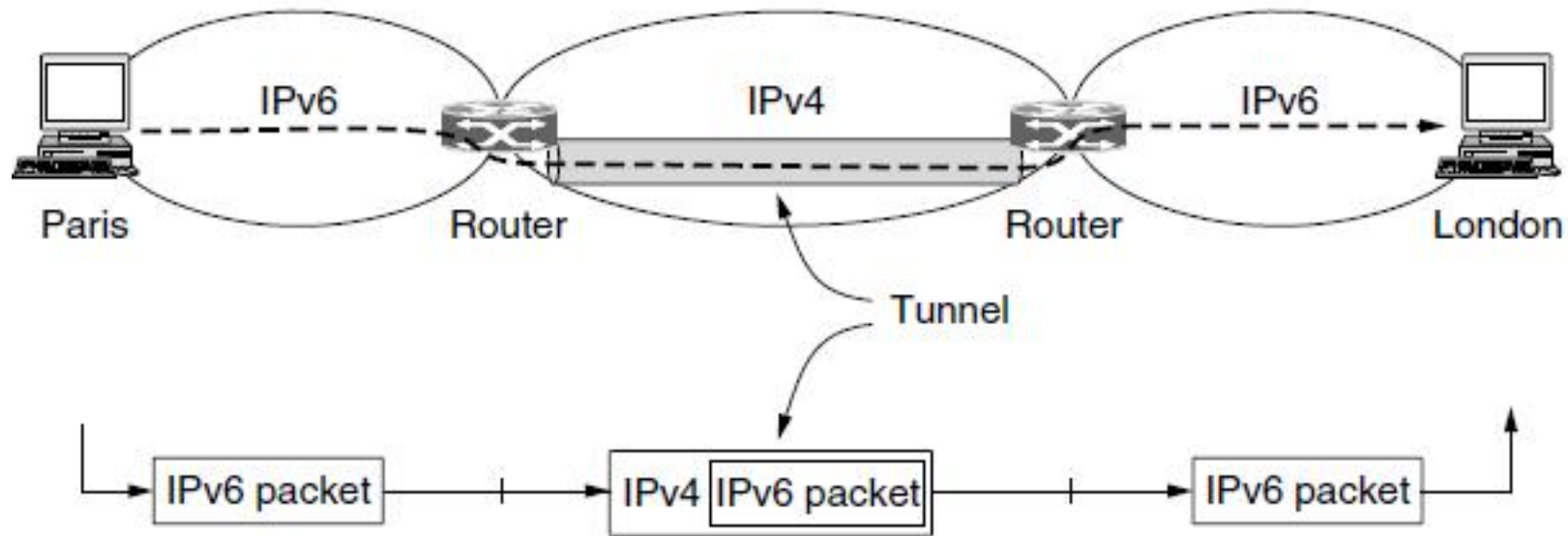


Bridge: the entire frame is transported on the basis of its **MAC address**

Router: packet is extracted from the frame and the **Layer-3 address** is used for deciding where to send it

5.5.3 Tunneling(隧道)

- ◆ Source and destination hosts are on the same type of network, but there is a different network in between
- ◆ Packet from source is **encapsulated** inside packet of the intermediate network



条件：源主机与目的主机网络协议相同，中间网络不同

方法：用中间网络的协议封装源主机发送的数据包

局限：无法到达隧道之下网络的主机

5.5.4 Internetwork Routing

- A two-level routing algorithm is used
 - ◆ Within each network, an intradomain(域间) or interior(域内) gateway protocol is used for routing.
 - ◆ Across the networks that make up the internet, an interdomain or exterior gateway protocol is used.
- See how it is done in Internet

5.5.5 Fragmentation(分段/分片), Why?

Different MTUs

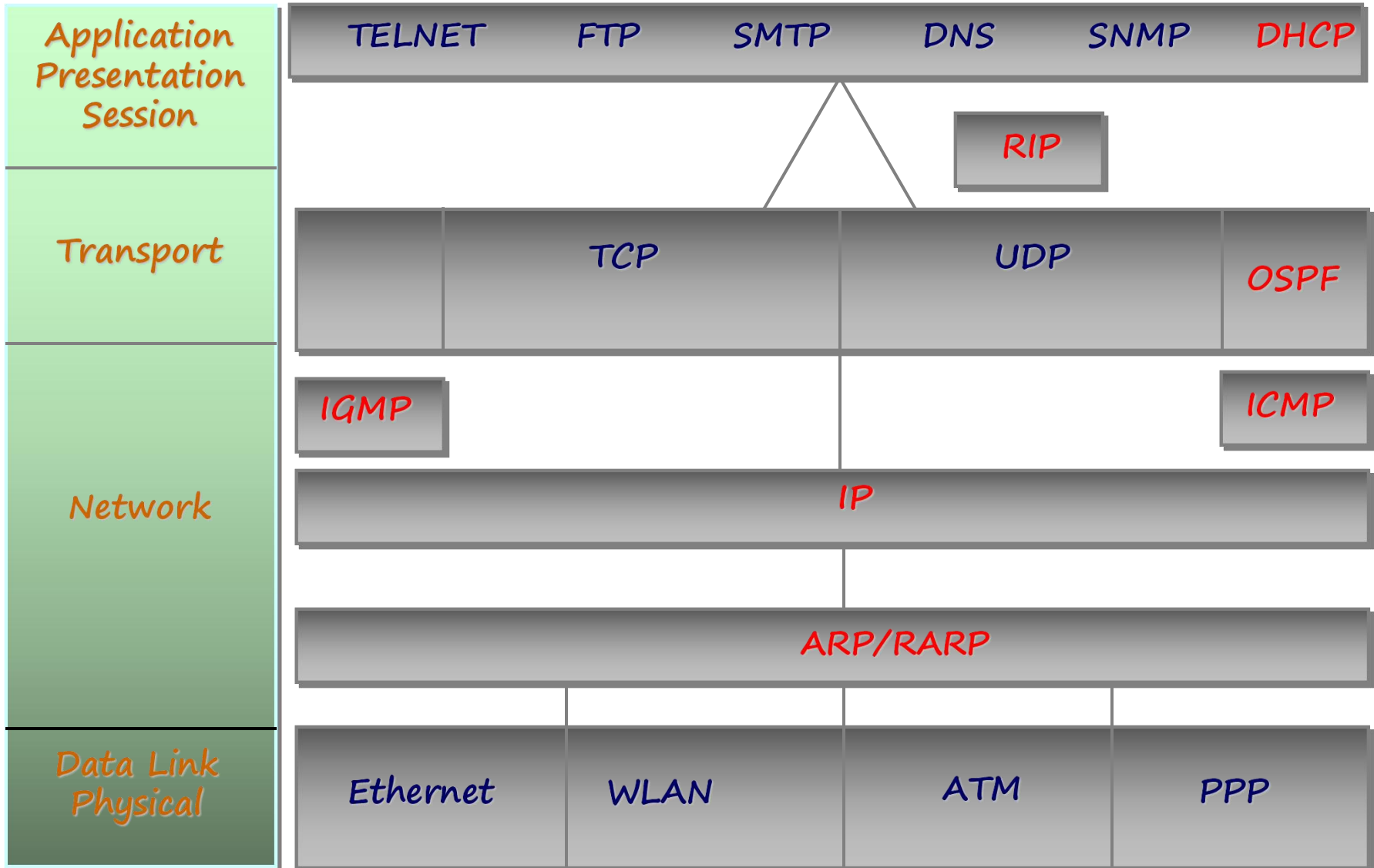
- **MTU**: Maximum Transmission Unit (最大传输单元)
- MTU is determined by
 - ◆ Hardware (e.g., the size of an Ethernet frame)
 - ◆ Operating system (e.g., all buffers are 512 bytes).
 - ◆ Protocols (e.g., the number of bits in the packet length field)
 - ◆ Compliance with some (inter)national standard.
 - ◆ Desire to reduce error-induced retransmissions to some level
 - ◆ Desire to prevent one packet from occupying the channel too long
- How to do fragmentation?
 - ◆ See **how IP does**

Outline

- 5.1 Introduction to Network Layer
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 - ◆ 5.6.1 IPv4 Packet format
 - ◆ 5.6.2 IP Addressing
 - ◆ 5.6.3 IPv6
 - ◆ 5.6.4 ARP, DHCP and ICMP
 - ◆ 5.6.6&5.6.7 Routing Protocols: OSPF and BGP

OSI

TCP/IP

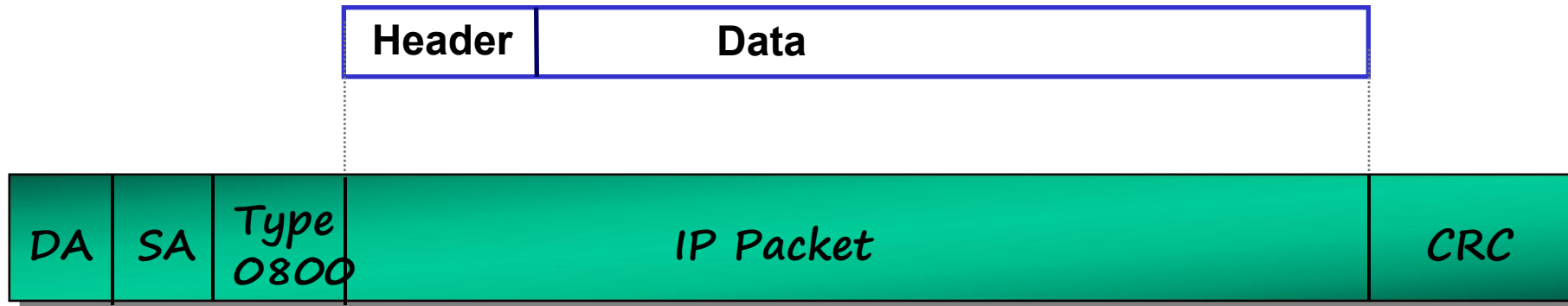


Design Issues of IP

- Design requirements: accommodate differences in constituting networks
 - ◆ Different addressing schemes:
 - using a **global** address—IP地址
 - ◆ Different maximum packet size
 - ◆ Different network access mechanism
 - ◆ Different timeouts
 - ◆ Different transmission modes (connection-oriented, connectionless)
 - ◆ Error control
 - ◆ Flow control

IP Datagram

■ Independent of Hardware



↑
Indicating data in Ethernet
frame is IP packet

Encapsulation of IP Datagram in Ethernet Frame

Design Issues of IP

- Design requirements: accommodate differences in constituting networks:
 - ◆ Different addressing schemes
 - ◆ Different maximum packet size
 - ◆ Different network access mechanism
 - ◆ Different timeouts
 - ◆ Different transmission modes (connection-oriented, connectionless)
 - ◆ Error control
 - ◆ Flow control

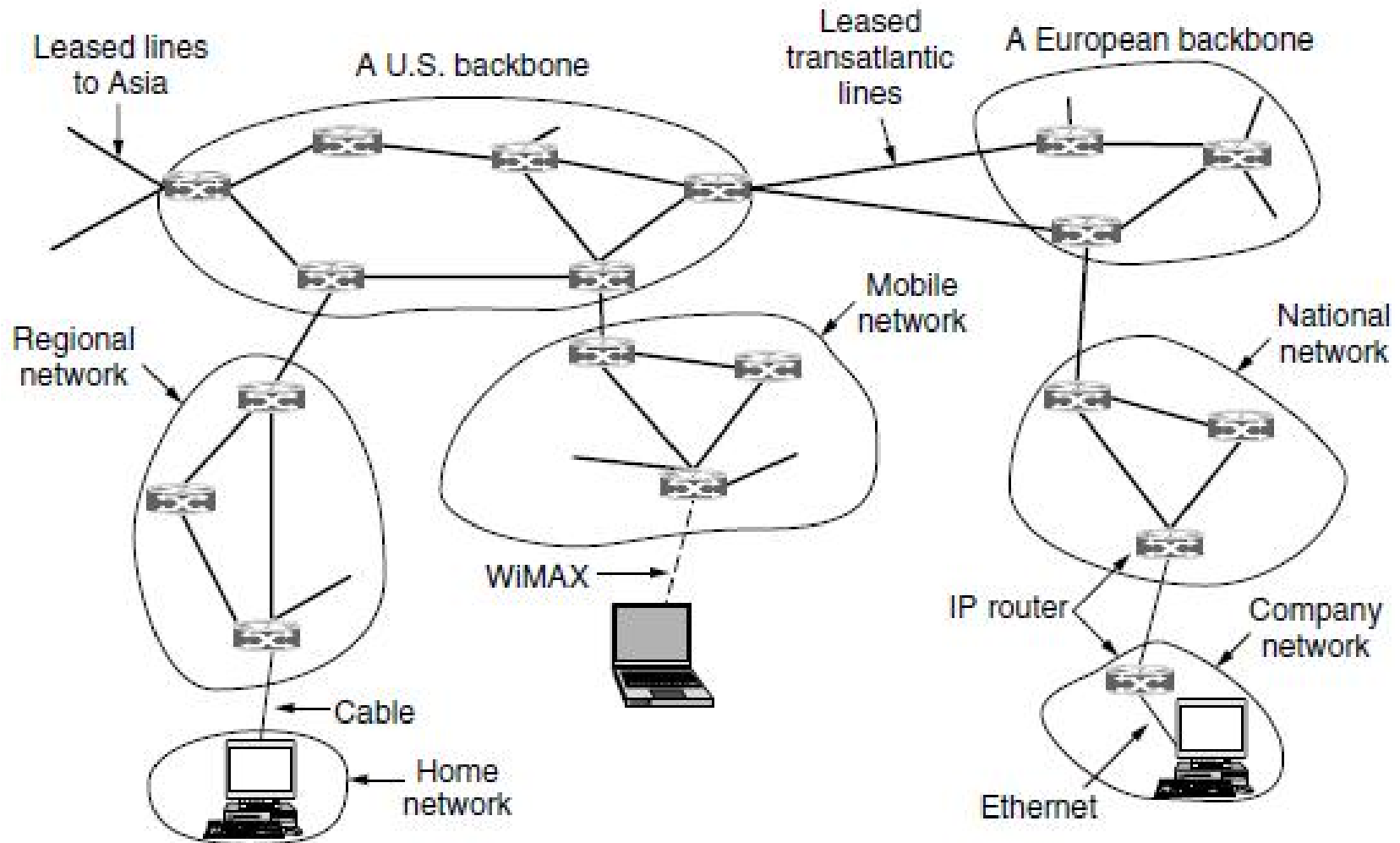
Fragmentation and Re-assembly (分段/分片和重装)

- Problem: A datagram can travel through different networks. different networks have different maximum frame sizes.
 - ◆ **MTU**: maximum transmission unit(最大传输单元)
 - Ethernet: MTU = 1500 bytes.
 - WiFi(IEEE802.11): MTU = 2312 bytes.
 - ◆ MTU may be as small as 128 bytes or less.
- Consequence: IP should adapt to different MTUs.
 - ◆ Choose convenient initial datagram size.
 - ◆ Divide datagrams when they traverse network with smaller MTU.
- IP datagram is **fragmented** by the source host or any router in the path.
- IP datagram is **re-assembles** at destination only
- Size of fragments determined by network with minimum MTU

Design Principles of Internet's network layer

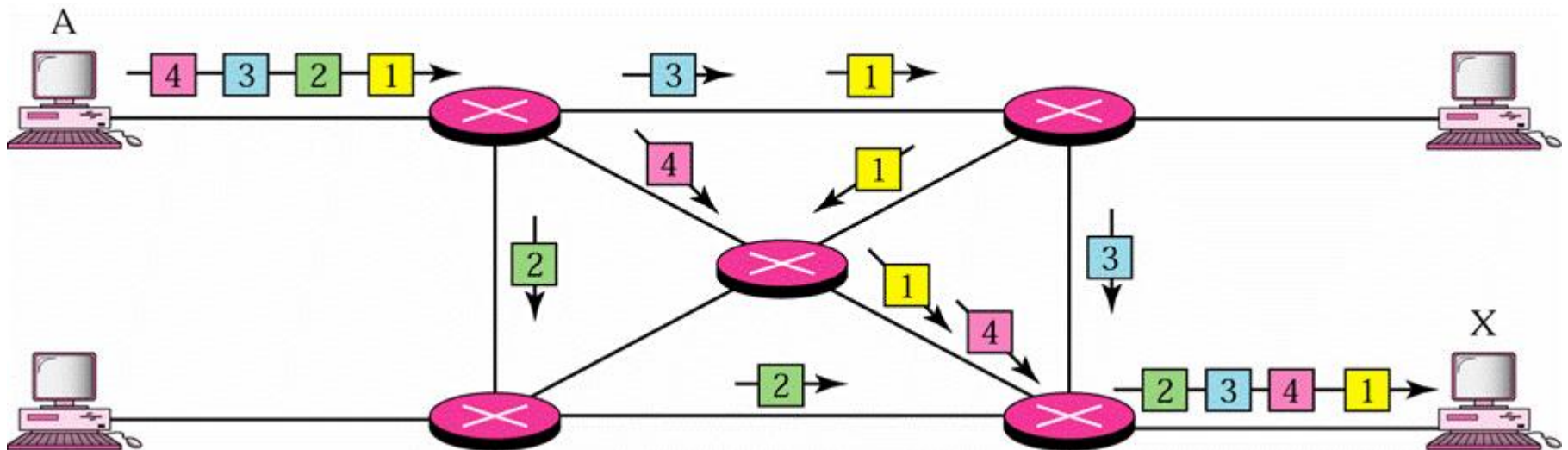
- Make sure it works
- Keep it simple
- Make clear choices
- Exploit modularity (模块化)
- Expect heterogeneity (异构性)
- Avoid static options and parameters
- Look for a good design; it need not be perfect
- Be strict when sending and tolerant when receiving
- Think about scalability (可扩展性)
- Consider performance and cost

Quasihierarchical Organization of Internet (准体系结构)



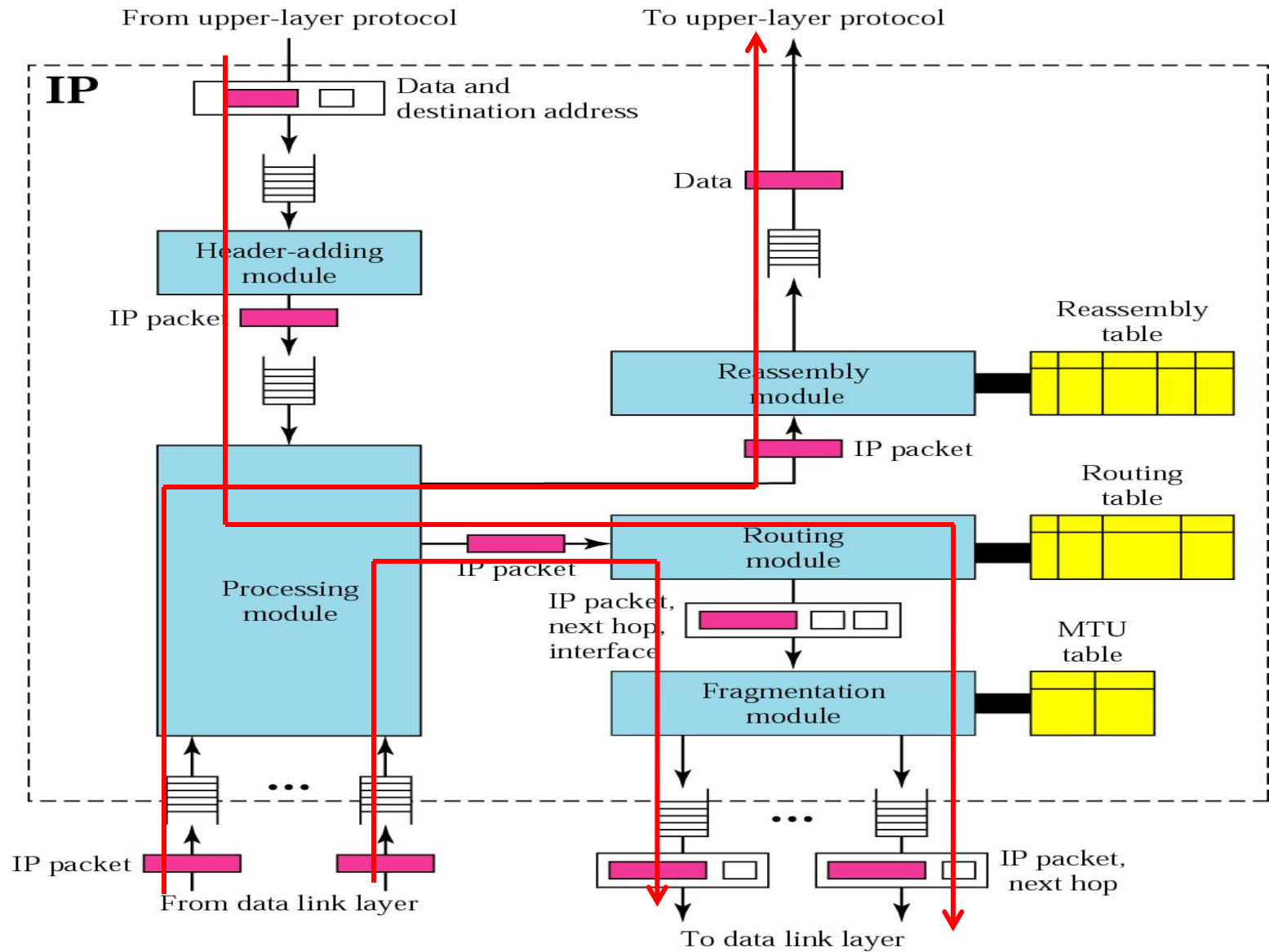
Internet Protocol (IP)

- The **most-widely used** Internetworking protocol
- A **connectionless** network-layer protocol.
- Based on **datagram** routing.
- IP provides **best-effort** service, unreliable !!
- Routing in IP is a mixture of **hierarchical**, **distance vector** and **link state** routing algorithms.



[Forouzan]

IP的组件



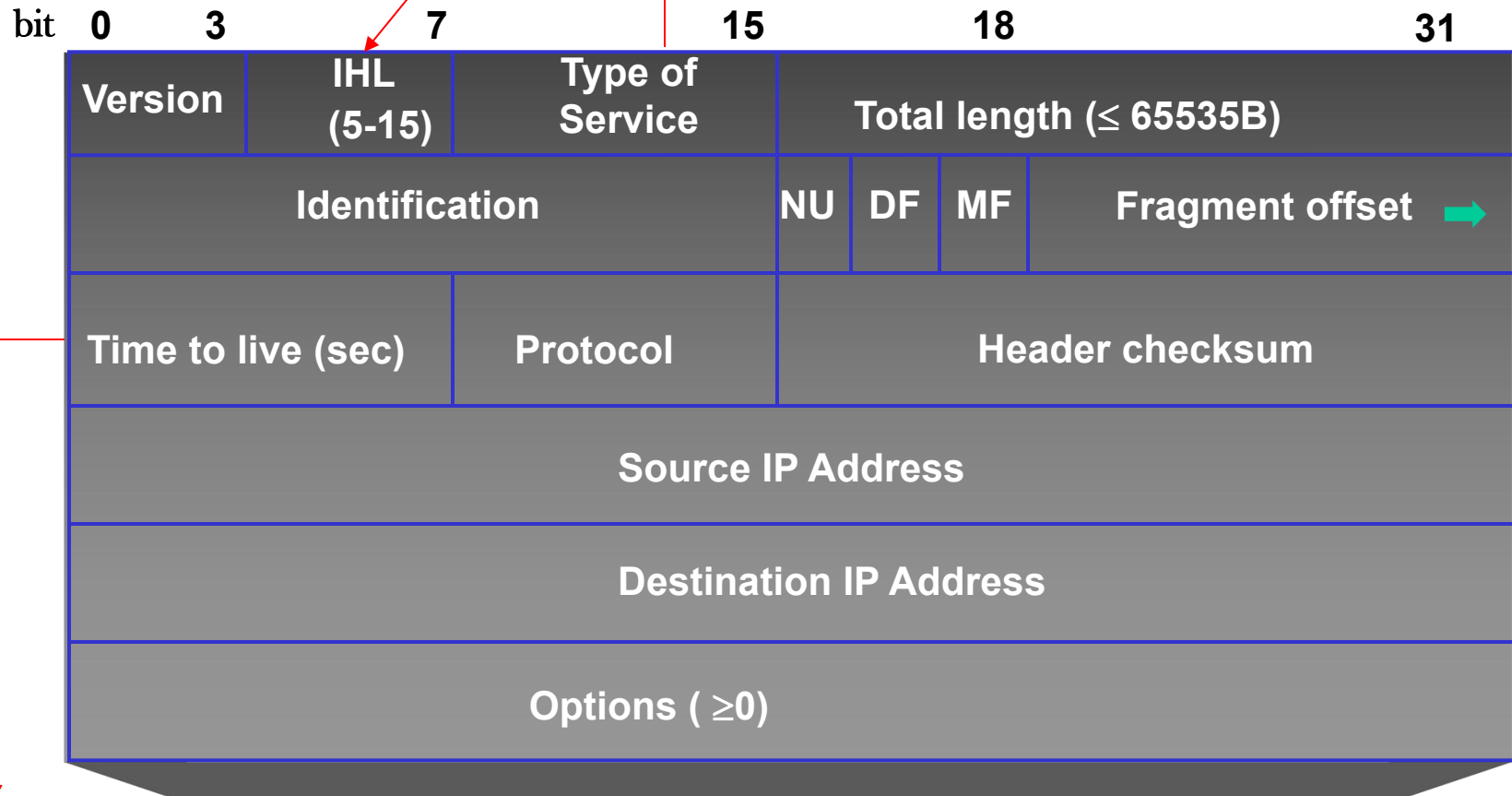
The network layer in the Internet

- 5.6.1 IPv4 Packet format
- 5.6.2 IP Addressing
- 5.6.3 IPv6
- 5.6.4 ARP, DHCP and ICMP
- 5.6.6&5.6.7 Routing Protocols: OSPF and BGP

IP Header

IP Header Length in terms of 32-bit words

defines the class of datagram for QoS purpose



to control max number of hops visited by a datagram by count decrement.

Header length + Data length of the IP packet in terms of number of bytes

The IPv4 Header (1)

■ Version (4 bits)

■ IHL(4 bits)

◆ How long the header is, in 32-bit words, Val:5~15, 20~60 bytes

■ The Differentiated services field

◆ Originally, it was called the Type of service field (TOS)

➤ ~~a 3-bit Precedence field: (Priority 0~7)~~

➤ ~~3 flags: Delay, Throughput, Reliability~~

◆ **Now**, the top 6 bits are used to mark the packet with its **service class** the bottom 2 bits are used to carry explicit congestion notification (**ECN**) information

■ Total Length(16 bits)

◆ Includes **both header and data**, in bytes

◆ The maximum length is 65,535 bytes

45 00 00 30 -> 0000 0000 0011 0000 -> $2^5 + 2^4 = 48$

The IPv4 Header (2)

■ Identification (16 bits)

- ◆ Allow the destination host to determine which datagram a newly arrived **fragment** belongs to
- ◆ All the fragments of a datagram contain the **same Identification** value

■ DF (1 bit)

- ◆ **Don't Fragment**. It is an order to the routers not to fragment the datagram (It is used as part of the process to discover the path MTU)

■ MF (1 bit)

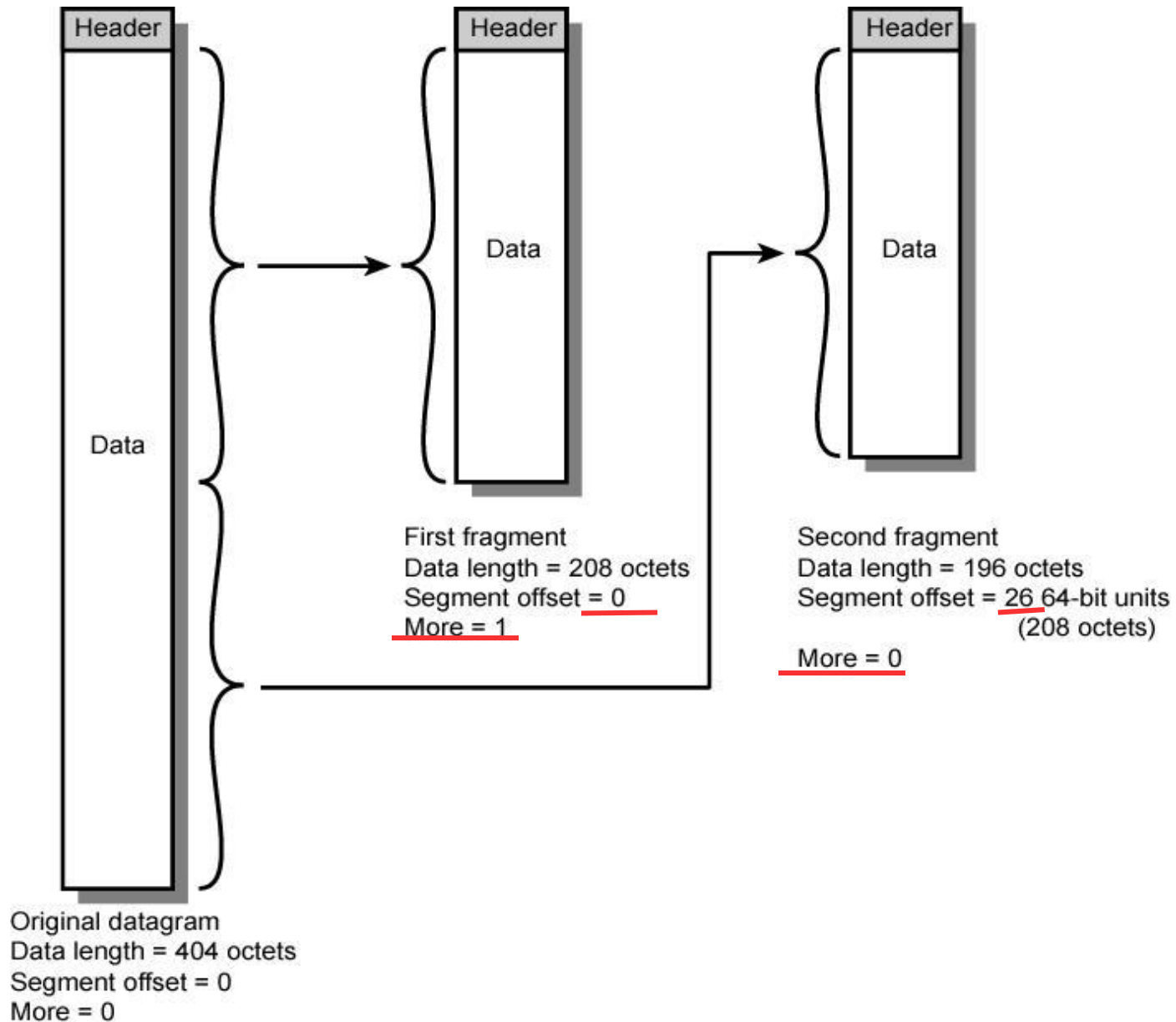
- ◆ **More** Fragments

■ Fragment offset (13 bits)

- ◆ Tell where in the current datagram this fragment belongs, **in 8 bytes** (0~8191)

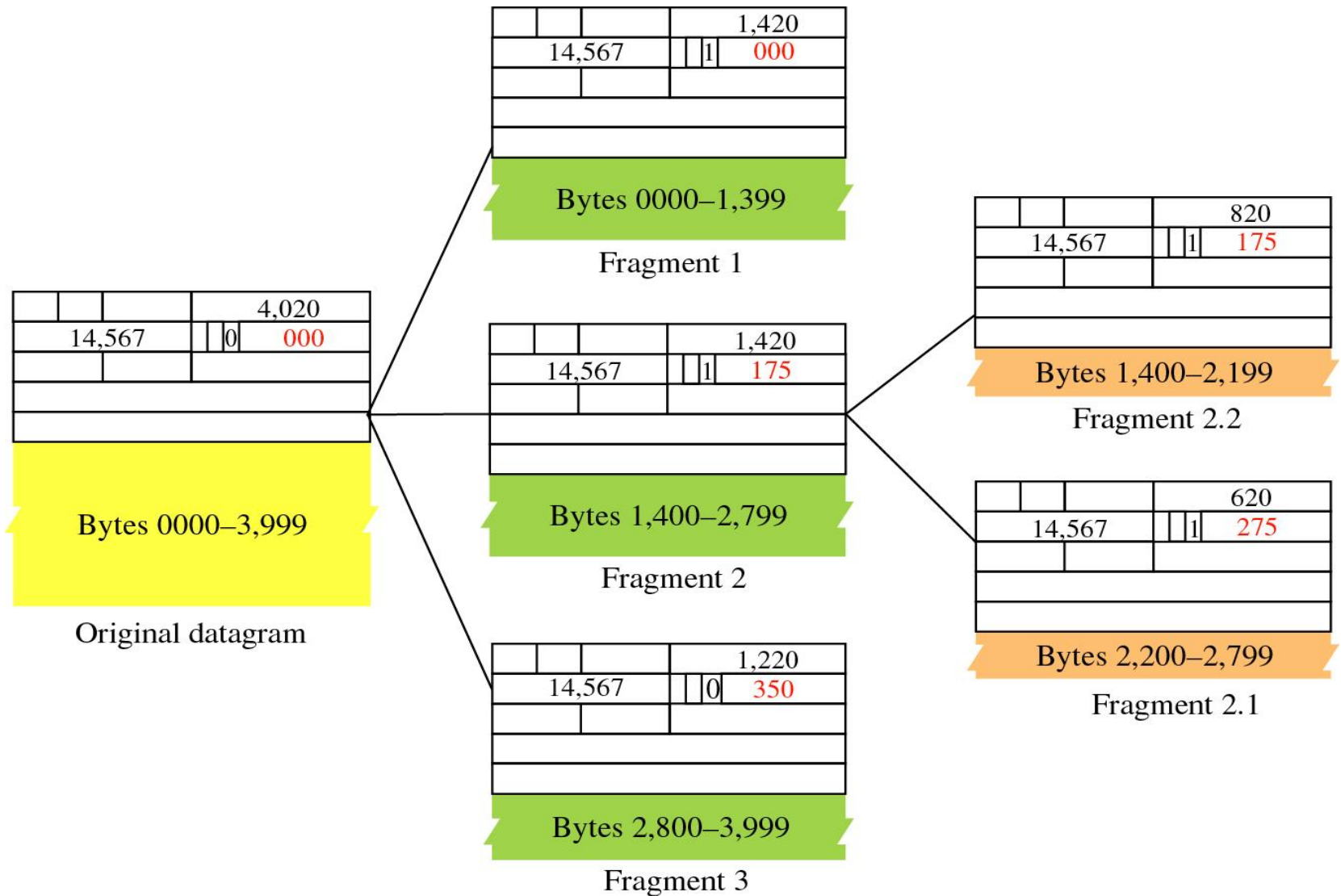
82 fc **40 00**

Fragmentation Example (1)



[Stallings]

Fragmentation Example (2)



The IPv4 Header (3)

■ TTL: Time to live (8 bits)

- ◆ A counter used to limit packet **lifetimes**. It is supposed to count time in **seconds**
- ◆ It must be **decremented on each hop**
- ◆ When it hits **zero**, the packet is discarded and a warning packet is sent back to the source host

■ Protocol field (8 bits)

- ◆ Which transport process to give it to (TCP/UDP,...)
1-ICMP, 6-TCP, 17-UDP, 89-OSPF, ...

■ The Header checksum (16 bits)

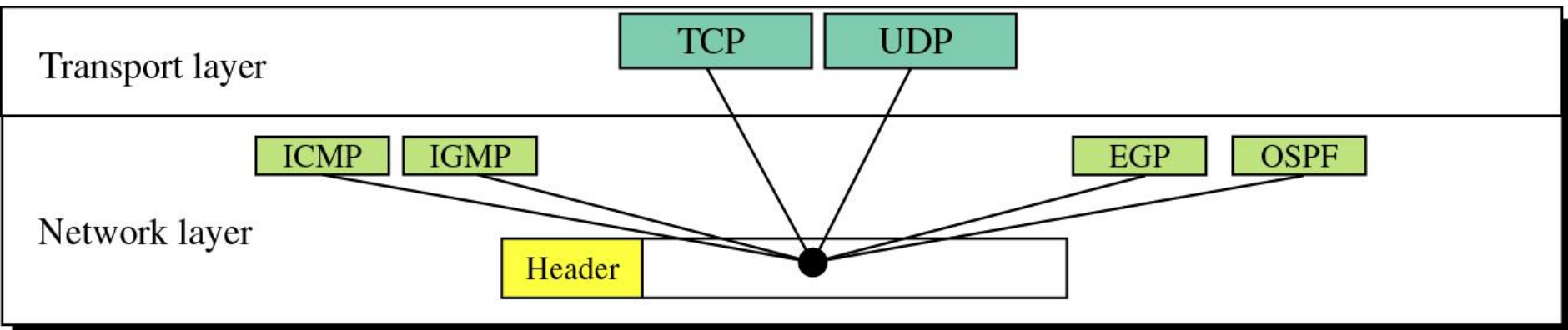
- ◆ Verifies the header only
- ◆ Must be **recomputed** at each hop

80 06 f5 a5

■ Source & Destination address(32+32 bits)

80 a8 b1 99 3b 40 6e d5

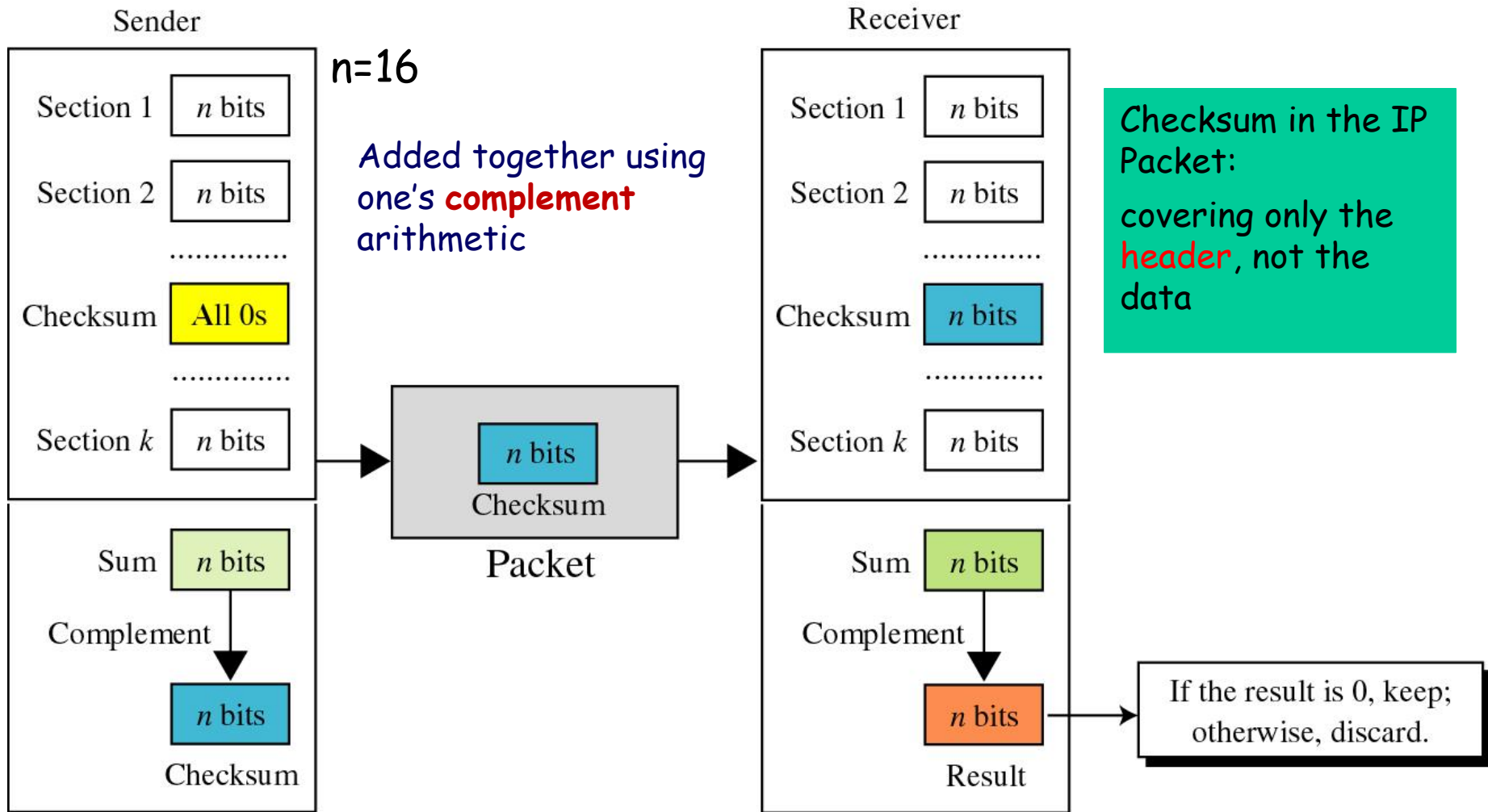
Protocol Field



- Defines the higher level protocol (transport-layer) that uses the services of IP-layer.
- An IP datagram can encapsulate data from several higher-level protocols such as: TCP=6, UDP=17, ICMP=1, IGMP=2, OSPF=89
- Specifies final destination protocol to which IP datagram should be delivered

[Forouzan]

Checksum(校验和/检验和) Concept(RFC 1071)



Example of Checksum Calculation

4	5	0	28	
1			0	0
4	17		0	
10.12.14.5				
12.6.7.9				



4, 5, and 0	→	01000101	00000000
28	→	00000000	00011100
1	→	00000000	00000001
0 and 0	→	00000000	00000000
4 and 17	→	00000100	00010001
0	→	00000000	00000000
10.12	→	00001010	00001100
14.5	→	00001110	00000101
12.6	→	00001100	00000110
7.9	→	00000111	00001001
Sum	→	01110100	01001110
Checksum	→	10001011	10110001

11111111 11111111



Checksum

- How well does the checksum work?
 - ◆ What is the distance of the code?
 - ◆ How many errors will it detect/correct?

Some of the IP options

- Security option
- Strict source routing option 严格源选路
 - ◆ Gives the **complete path from source to destination** as a sequence of IP addresses
- Loose source routing option 宽松源选路
 - ◆ Requires the packet to **traverse the list of routers specified**, and in the order specified, but it is **allowed to pass through other routers** on the way
- Record route option 记录路由
 - ◆ Tells the **routers along the path to append their IP address to the option field**
- The Timestamp option 时间戳
 - ◆ Like the Record route option, except that in addition to **recording its 32-bit IP address**, each router **also records a 32-bit timestamp**

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

The network layer in the Internet

- 5.6.1 IPv4 Packet format
- 5.6.2 IP Addressing
- 5.6.3 IPv6
- 5.6.4 ARP, DHCP and ICMP
- 5.6.6&5.6.7 Routing Protocols: OSPF and BGP

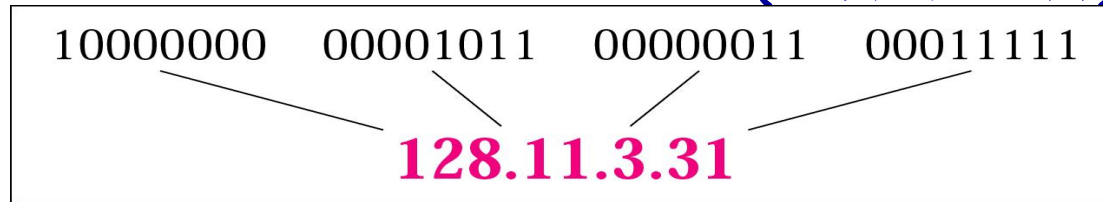
IPv4 Addressing

- **Classful Addressing**(分类地址): 1981
- Subnetting(子网): 1985
- Supernetting(超网): 1993
- **Classless Addressing**(无类别地址): 1995

(1) IPv4 Address

- Identifier used in network layer to identify each device connected to the Internet is called the **Internet address** or **IP address**.

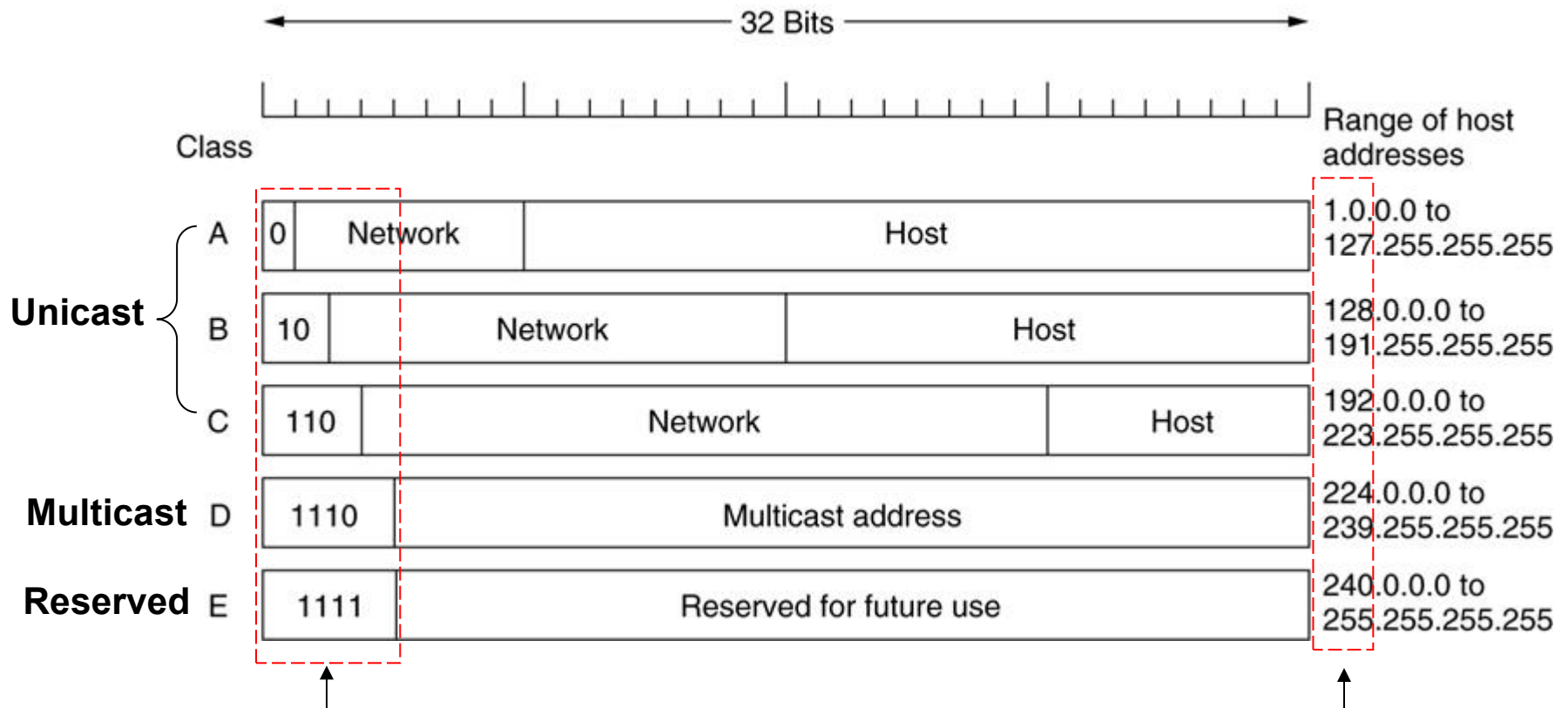
- 32-bit, Dotted-decimal notation(点分十进制)



- **Unified addressing**
- Unique for each **network interface**, irrelative with its physical address, which universally defines **1 connection** to Internet.
- If a host/router has 2 interfaces connecting to Internet, then it has 2 IP addresses
- Two types of IP addressing: **Classful** vs. **Classless**

(2) Classful IP Addressing(分类地址)

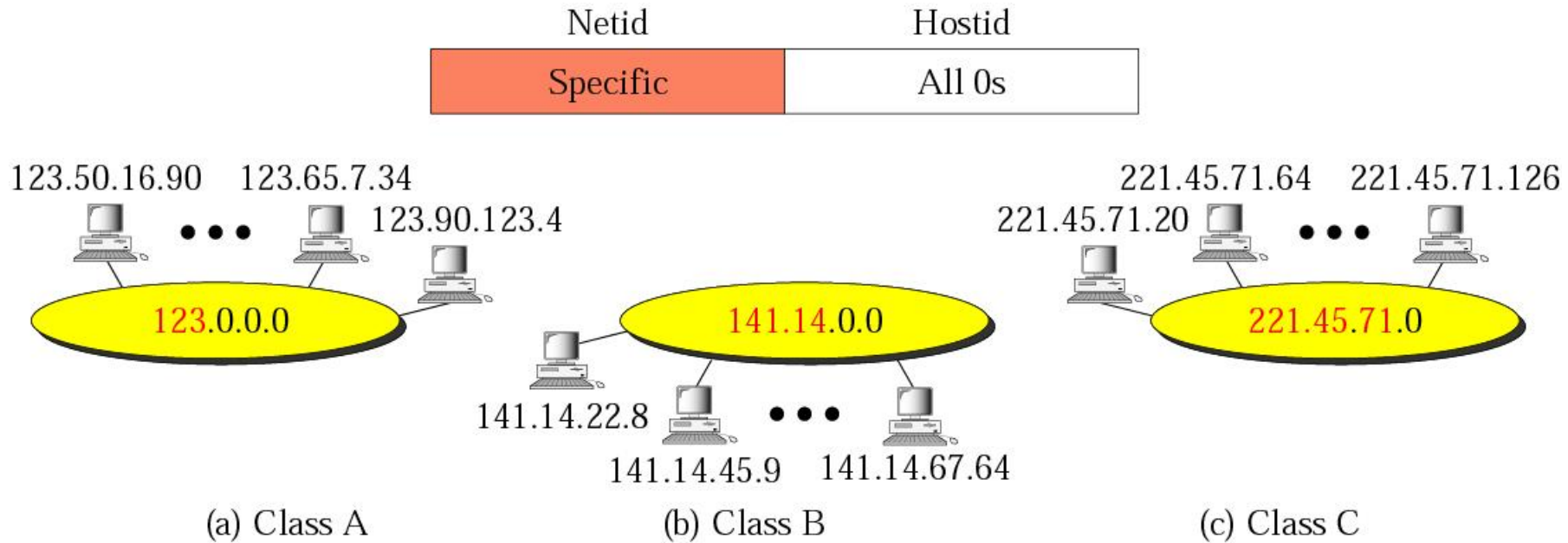
- IP has a two-level (at least, traditionally) hierarchical addressing mechanism in class A, B and C: **Network ID(net-id) + Host ID (host-id)**



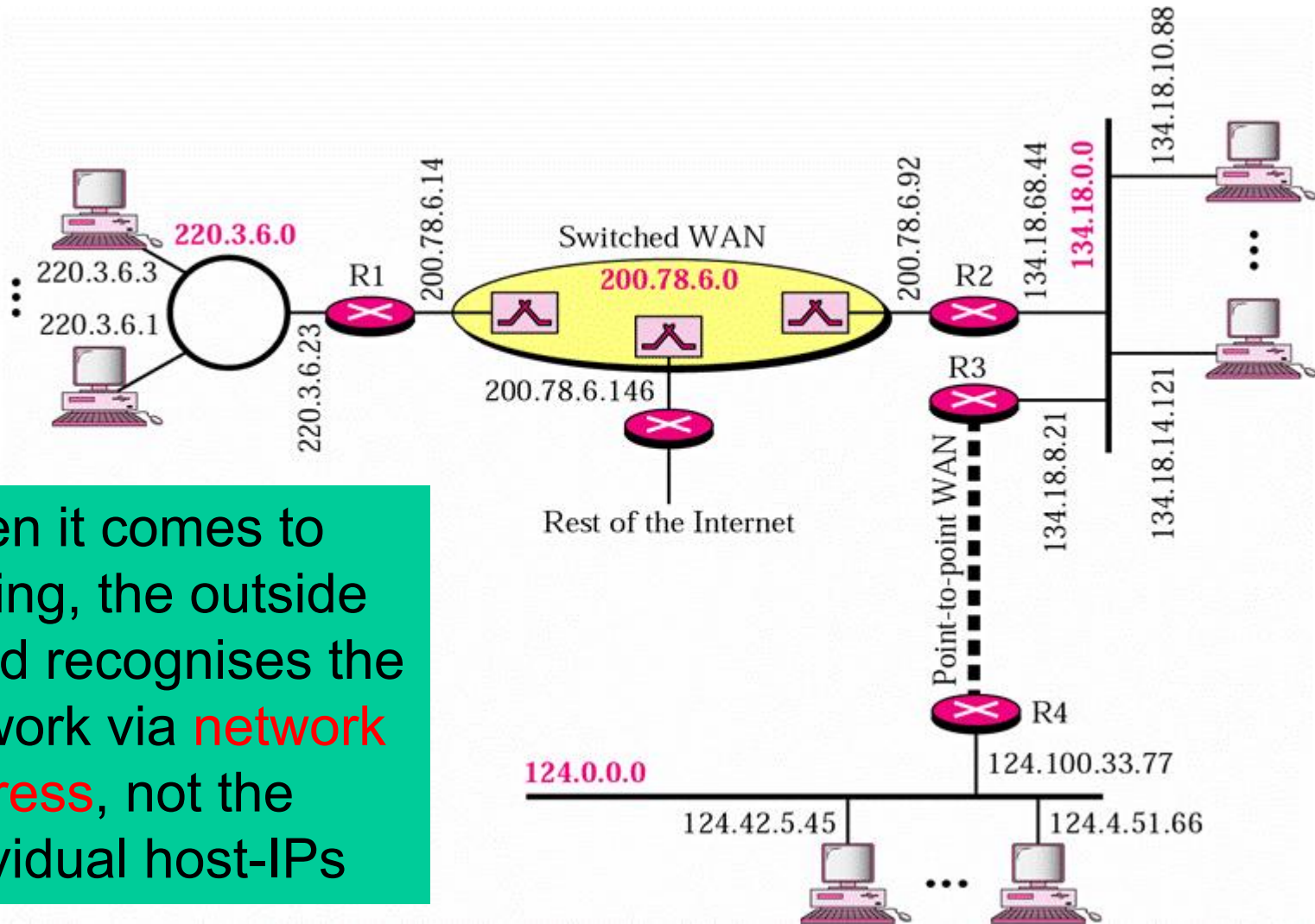
(3)Network address

- Network address is the **first address** in the block, is the one that is assigned to the organization.
- 网络号部分保留，主机号部分为0 
- It can be found by applying the **default mask**(默认/缺省掩码) to any of the IP addresses in the network
 - ◆ IP address bit AND with default mask
- Default masks
 - ◆ Class A: 255.0.0.0
 - ◆ Class B: 255.255.0.0
 - ◆ Class C: 255.255.255.0

Network address: example



Network Addresses: Usage



When it comes to routing, the outside world recognises the network via **network address**, not the individual host-IPs

路由选择使用网络地址



[Forouzan]

(4) Special IP addresses

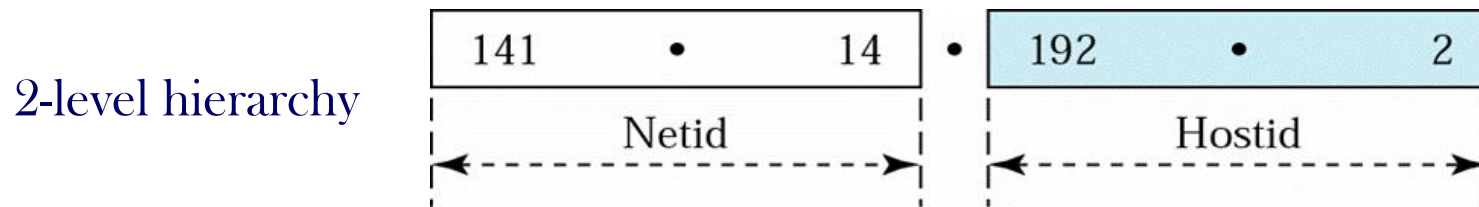
0 0																														This host																				
0 0										...										0 0										Host										A host on this network										
1 1																														Broadcast on the local network																				
Network															1 1 1 1															...										1 1 1 1										Broadcast on a distant network
127										(Anything)																				Loopback 环回地址																				

网内的地址数与可分配(可用)地址

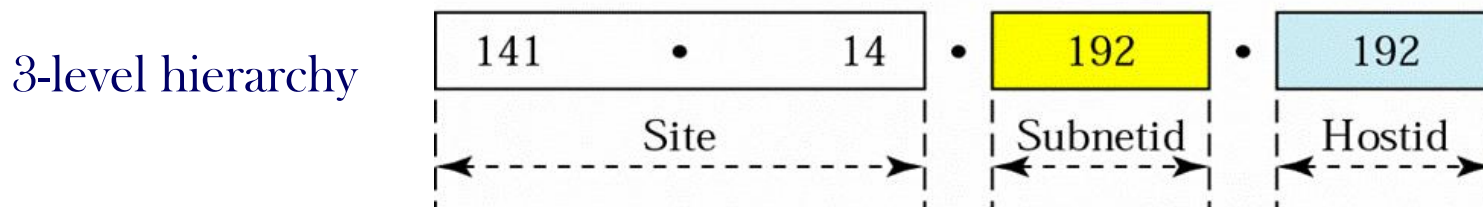
- 网络地址: $x.x.x.x/m$
- 网络号 m 位, 主机号 $=32-m=n$ 位
- 该网络内共有 2^n 个地址
- 其中第一个地址 (主机号为全0) 为网络地址
最后一个地址 (主机号为全1) 为广播地址
- 可分配 (可用) 地址数: 2^n-2

(5) Subnetting(子网划分)

- To manage IP addresses and routing packets more efficiently
- A network is split into several parts for internal use, but still act like a single network to outside world identified by network address
 - ◆ Subnetting is invisible to outside world 子网划分对于外部不可见
- Subnet is identified by using part of **host portion** to identify physical network



a. Without subnetting

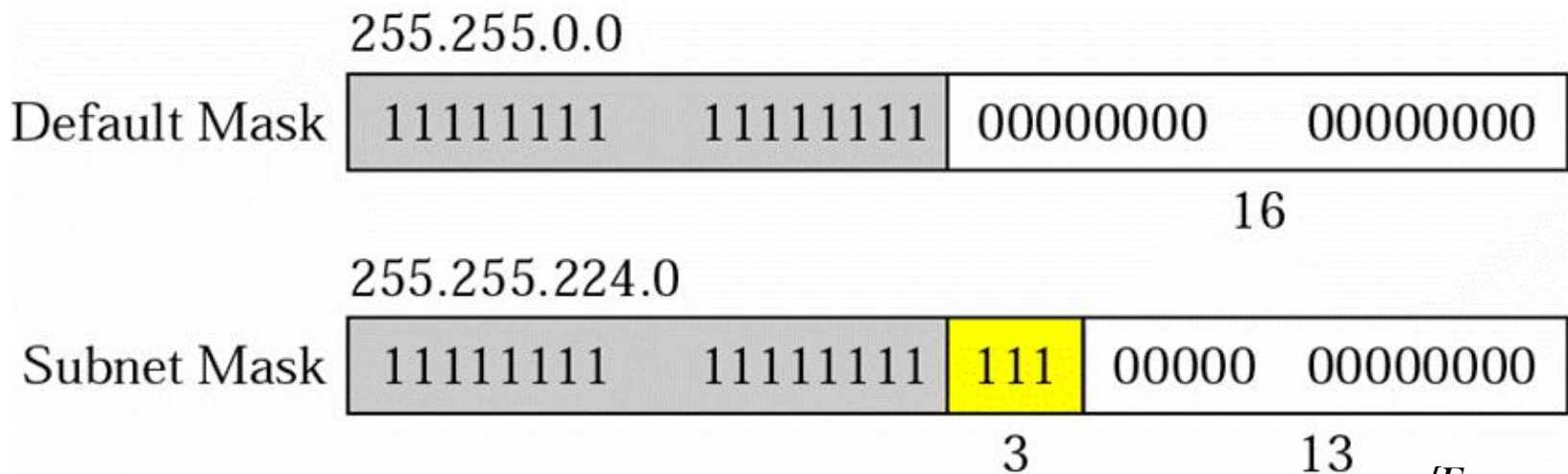


b. With subnetting

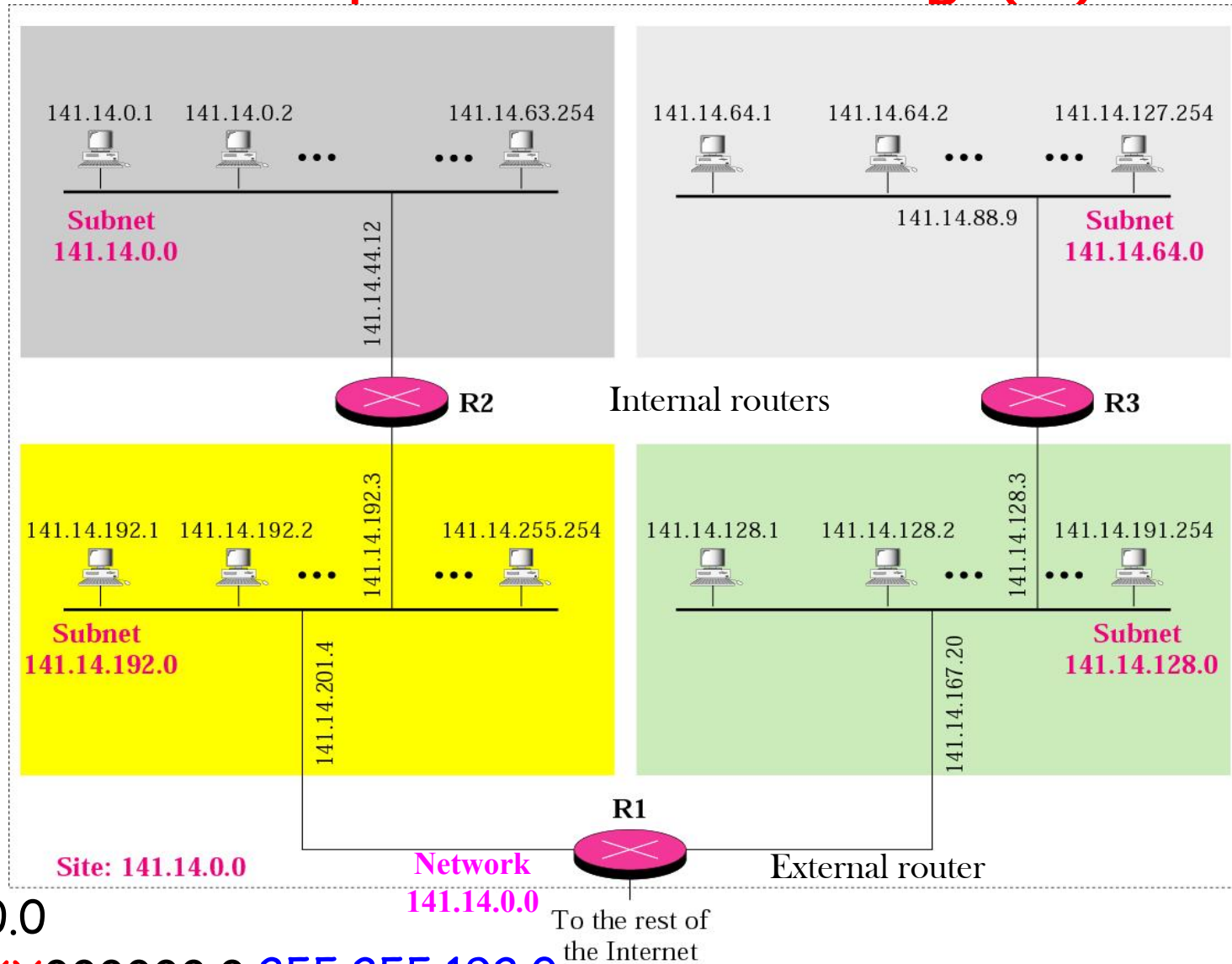
[Forouzan]

Subnet Mask(子网掩码)

- How a router knows the network address?
 - ◆ Using the **subnet mask**.
- **Boolean ANDed** (按位“与”) with destination address when routing
- If IP addresses of source and destination, and subnet mask are known, then we can find the network address
 - ◆ IP packet is sent to a host in **same subnet**, or
 - ◆ A host in **another network**



Example of Subnetting (1)



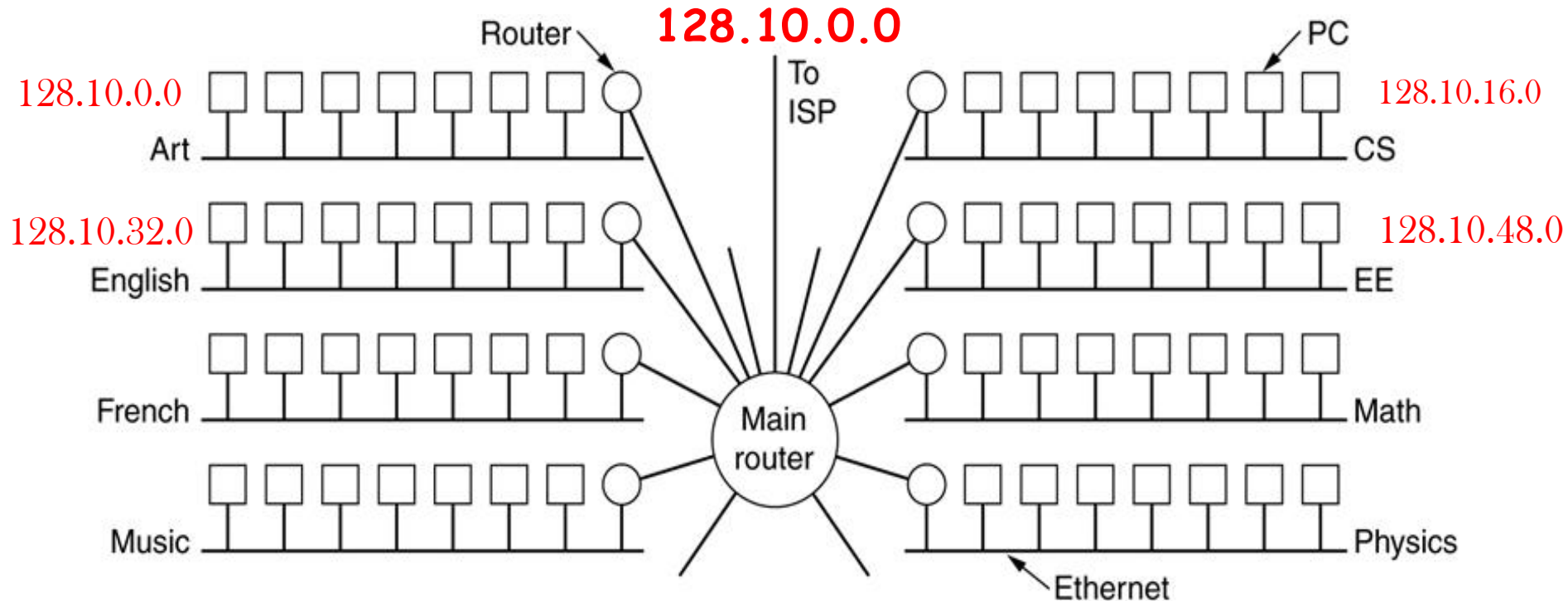
141.14.0.0
 141.14.XX000000.0 255.255.192.0

To the rest of
the Internet

[Forouzan]

Example of Subnetting (2)

A campus network consisting of LANs for various departments



Subnet mask: 255.255.240(11110000).0

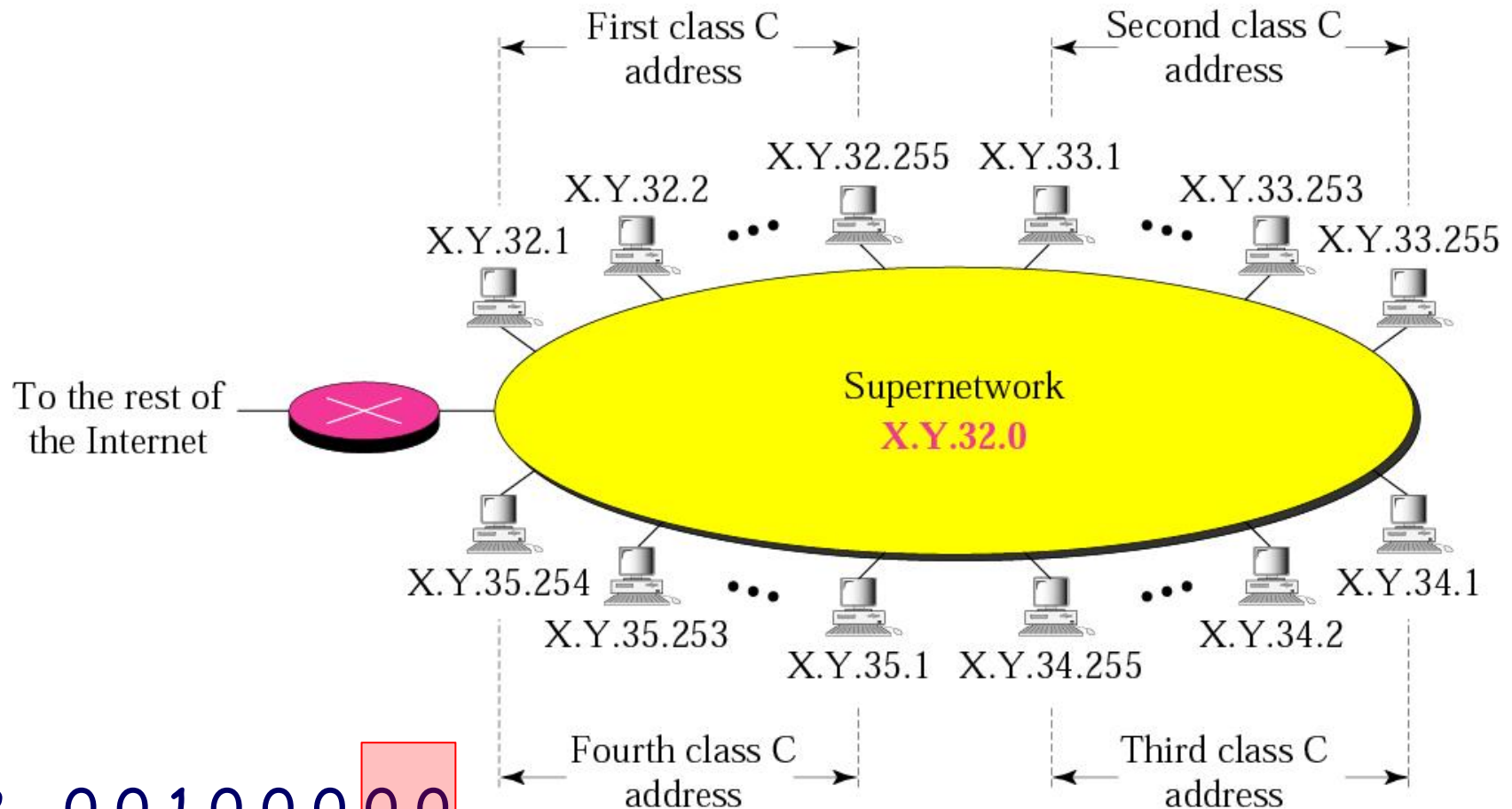
What can we get from subnet mask?

- Number of IP addresses in the subnet.

(6) Supernetting(超网划分)

- Problem of Class C addresses: the size of a block is too small to satisfy the needs of an organisation
- Solution: supernetting
 - ◆ To combine several continuous class C blocks to create a large range of addresses
 - ◆ i.e. several networks are combined to create a supernetwork (超网)
- The number of class C blocks must be a power of 2
- The blocks must be contiguous in the address space (no gaps between the blocks).

超网划分示例



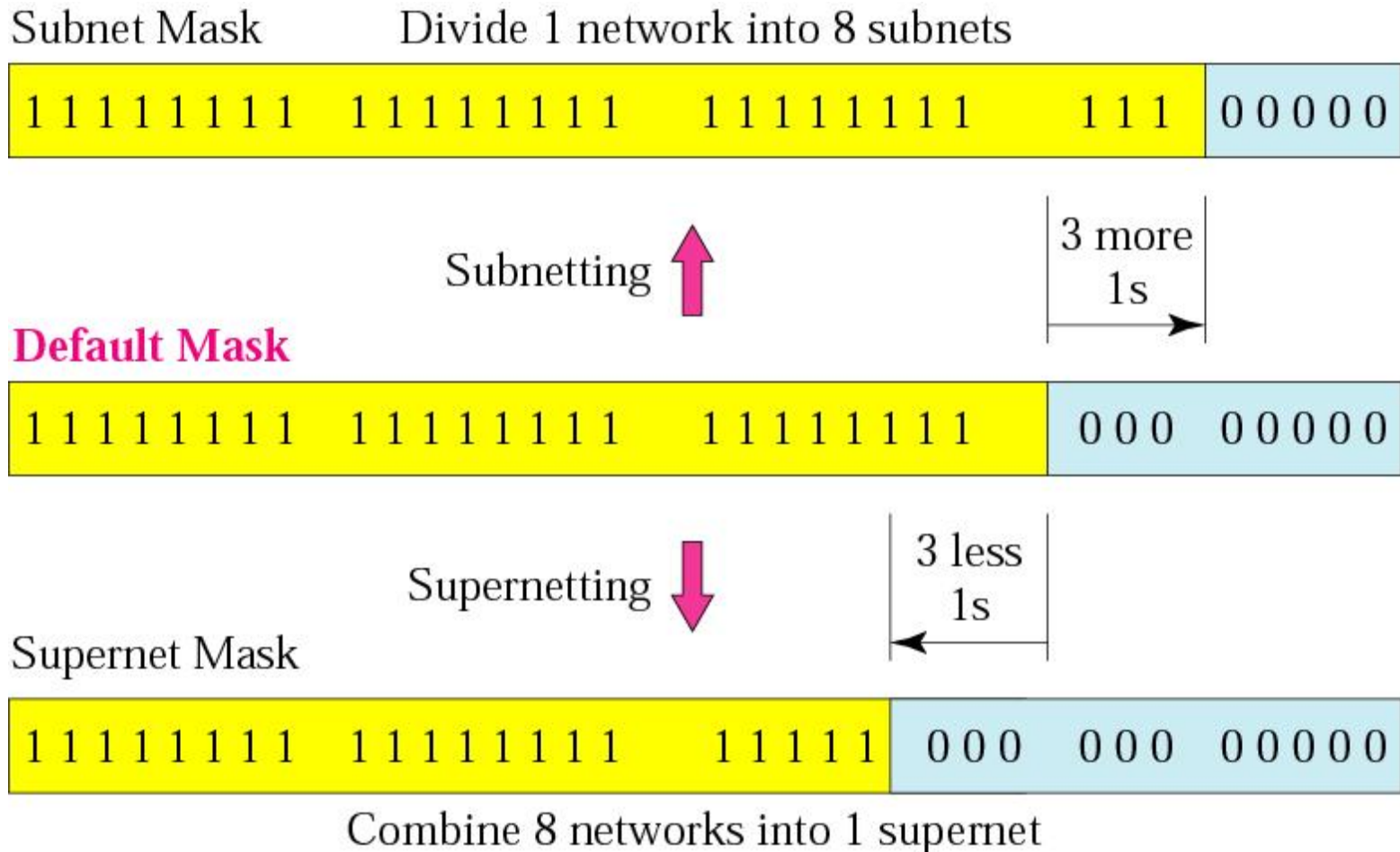
32:	0	0	1	0	0	0
33:	0	0	1	0	0	0
34:	0	0	1	0	0	1
35:	0	0	1	0	0	1

Supernet mask: 255.255.252.0

Network ID: X.Y.32.0/22

[Forouzan]

Comparison of subnet, default, and supernet masks



[Forouzan]

(7) Classless Addressing(无类别地址)

- Until mid-1990s, a range of addresses meant a block of addresses in class A, B or C.
- Problems of Classful addressing
 - ◆ Continued exponential Internet growth causes **address space exhaustion**
 - ◆ **Inefficient use** of address space
 - ◆ **Explosion** in the size of the Internet **routing tables**
- Solution: **Classless addressing** (from 1996)
- The idea is to have **variable-length blocks** that belongs to **no class**

Address Space



Classless InterDomain Routing (CIDR, 无类别域间选路)

■ CIDR Address Format: slash notation

$a.b.c.d/x$, where x is # bits in network portion of address



■ CIDR编址中，前缀指明了网络，长度1-32之间

■ Rules of addressing

- ◆ The number of addresses in a network(subnet) **must be a power of 2**

$N=2^{32-n}$, 其中, n 是前缀长度, N 是地址块的地址数

- ◆ 首地址=(任意地址) AND (网络掩码) (即网络号)

- ◆ 末地址=(任意地址) OR [NOT(网络掩码)]

=首地址+地址块中地址数 N

CIDR Prefixes: Dotted Decimal Equivalents

CIDR Notation	Dotted Decimal	CIDR Notation	Dotted Decimal
/1	128.0.0.0	/17	255.255.128.0
/2	192.0.0.0	/18	255.255.192.0
/3	224.0.0.0	/19	255.255.224.0
/4	240.0.0.0	/20	255.255.240.0
/5	248.0.0.0	/21	255.255.248.0
/6	252.0.0.0	/22	255.255.252.0
/7	254.0.0.0	/23	255.255.254.0
/8	255.0.0.0	/24	255.255.255.0
/9	255.128.0.0	/25	255.255.255.128
/10	255.192.0.0	/26	255.255.255.192
/11	255.224.0.0	/27	255.255.255.224
/12	255.240.0.0	/28	255.255.255.240
/13	255.248.0.0	/29	255.255.255.248
/14	255.252.0.0	/30	255.255.255.252
/15	255.254.0.0	/31	255.255.255.254
/16	255.255.0.0	/32	255.255.255.255

Figure 10.11 Dotted decimal mask values for all possible CIDR prefixes.

随堂练习题解析

- 求三个网络：200.2.48.0/23、200.2.52.0/22和200.2.56.0/22聚合后的网络地址。

A. 200.2.48.0/20

B. 200.2.48.0/21

C. 200.2.32.0/20

D. 200.2.32.0/21

48: 0 0 1 1 0 0 0 0

52: 0 0 1 1 0 1 0 0

56: 0 0 1 1 1 0 0 0

200.2.00110000.0/20

■ 某公司（网络地址为**114.23.5.0/24**）有**A、B和C**三个部门，其主机（及路由器）的总数分别为**64、30和50**台，请给这三个部门根据最小够用原则 顺序划分网络地址，写出每个部门的网络地址。

步骤1：确定各地址块大小 **66(64+2); 32(30+2); 52(50+2)**

步骤2：从主机地址最高位开始 **114.23.5. X X X X X X X X**

66: 114.23.5. N X X X X X X X

66: 114.23.5. 0 X X X X X X X / 25 114.23.5. 0 / 25

32: 114.23.5. N N N X X X X X / 27

32: 114.23.5. 1 0 0 X X X X X / 27 114.23.5. 128 / 27

52: 114.23.5. 1 1 X X X X X X / 26 114.23.5. 196 / 26

■ 某公司（网络地址为**114.23.5.0/24**）有**A、B和C**三个部门，其主机（及路由器）的总数分别为以下要求的大小，请给这三个部门根据**最小够用原则**划分网络地址，写出每个部门的网络地址。

1. 31, 10, 50
2. 10, 50, 100
3. 100, 31, 10
4. 31, 100, 3

- 1、114.23.5.0/26; 114.23.5.64/28; 114.23.5.128.0/26;
- 2、114.23.5.0/28; 114.23.5.64/26; 114.23.5.128/25;
- 3、114.23.5.0/25; 114.23.5.128/26; 114.23.5.192/28;
- 4、114.23.5.0/26; 114.23.5.128/25; 114.23.5.64/29

Subnet

将IP地址空间192.168.1.0/24按网络号递增顺序分配给2个局域网LAN1～LAN2

子网	分配地址数	网络地址	子网掩码
LAN1	95		
LAN2	76		

Subnet

将IP地址空间192.168.1.0/24按网络号递增顺序分配给3个局域网LAN1～LAN3

子网	分配地址数	网络地址	子网掩码
LAN1	95		
LAN2	56		
LAN3	60		

CIDR: 路由聚合/汇聚

- CIDR permits route aggregation
- ◆ reduce the Internet routing table

192.24.000	000000.X	/21
192.24.000	100000.X	/20
192.24.000	010000.X	/22

192.24.000000000.X /19

New York



192.24.0.0/19

(1 aggregate prefix)

London



(3 prefixes)

192.24.0.0/21

Cambridge

192.24.16.0/20

Oxford

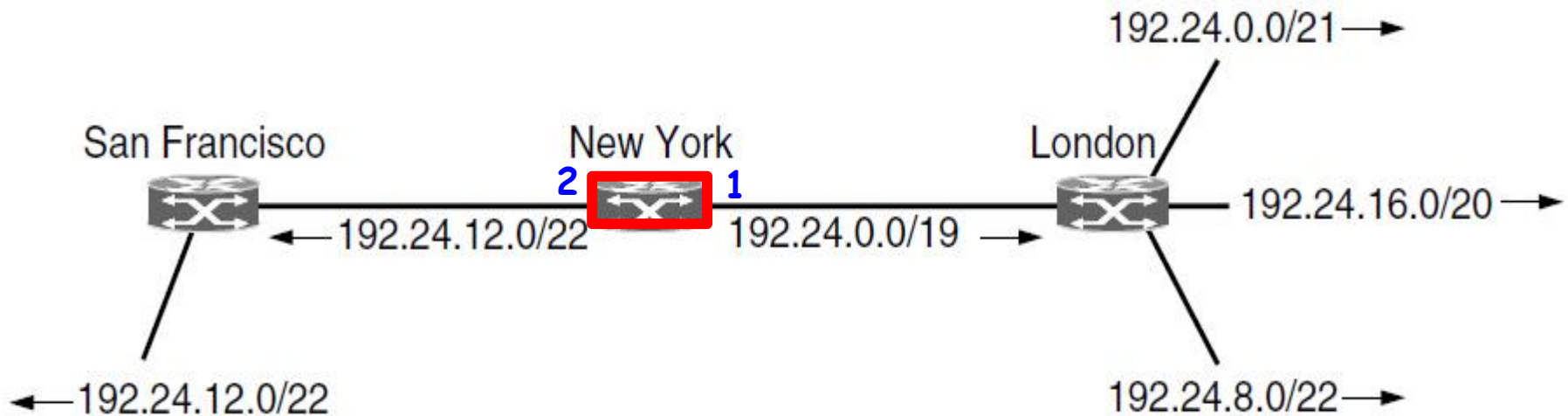
192.24.8.0/22

Edinburgh

192.24.12(00001100).0/22

问题：地址空间被放大

Example of routing aggregation



解决方法：最长匹配原则(路由前缀允许重叠，按最具体路由方向转发)

	IP address	outgoing
192.24.16.3	192.24.0.0/19	Port 1
192.24.12.5	192.24.12.0/22	Port 2

(8)Routing Table

■ Routing table

- ◆ An array of (IP address, netmask, outgoing line, Next hop)

■ Scan the routing table

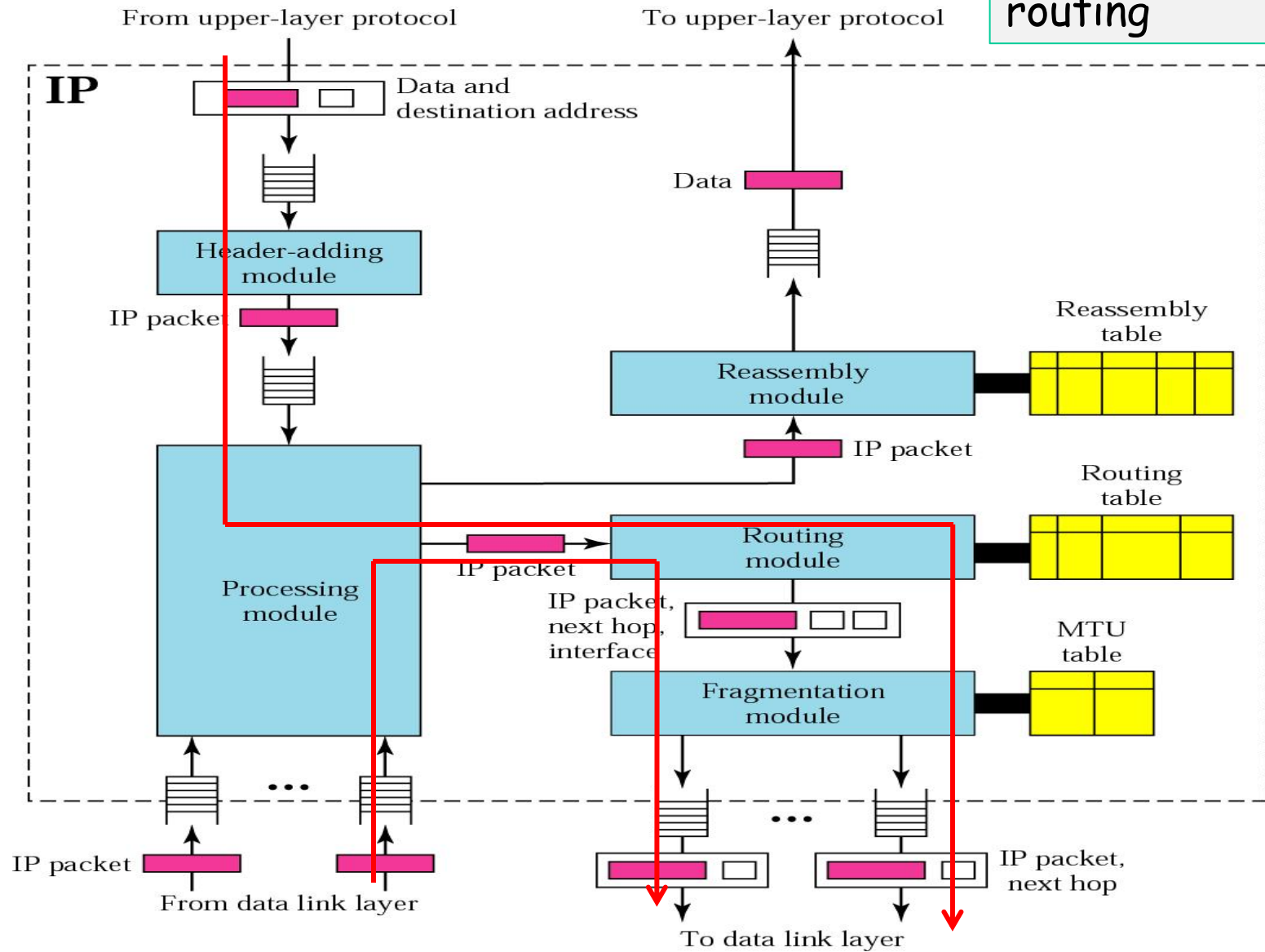
- ◆ Mask the destination address and comparing it to the table entry looking for a match
- ◆ It is possible that multiple entries (with different netmask lengths) match, in which case the longest mask is used
 - If there is a match for a /20 mask and a /24 mask, the /24 entry is used.

■ Implementation

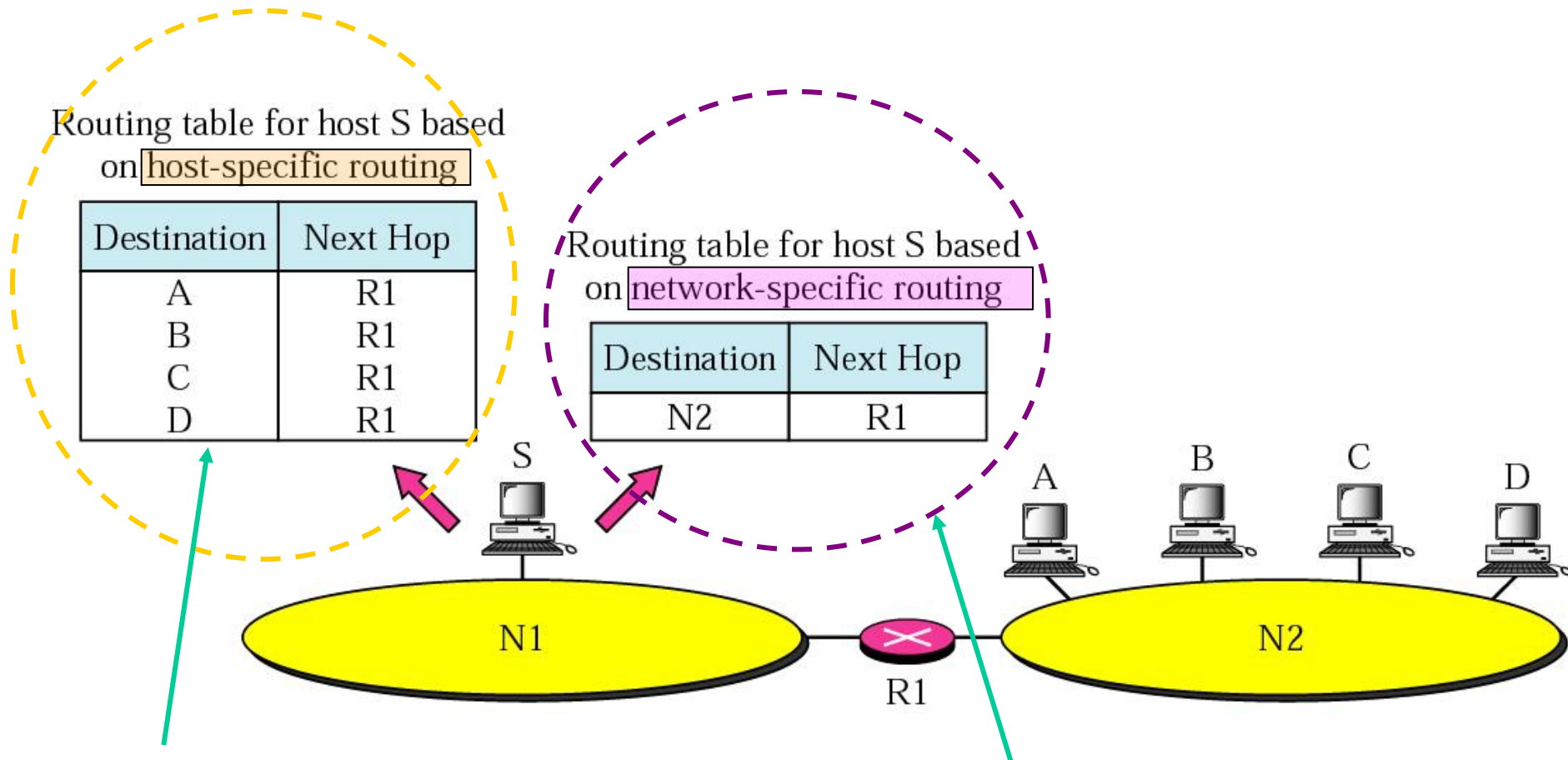
- ◆ Complex algorithms have been devised to speed up the address matching process
- ◆ commercial routers use custom VLSI (Very Large Scale Integration超大规模集成电路) chips with these algorithms embedded in hardware

IP components

Both **source** and **routers** consider routing



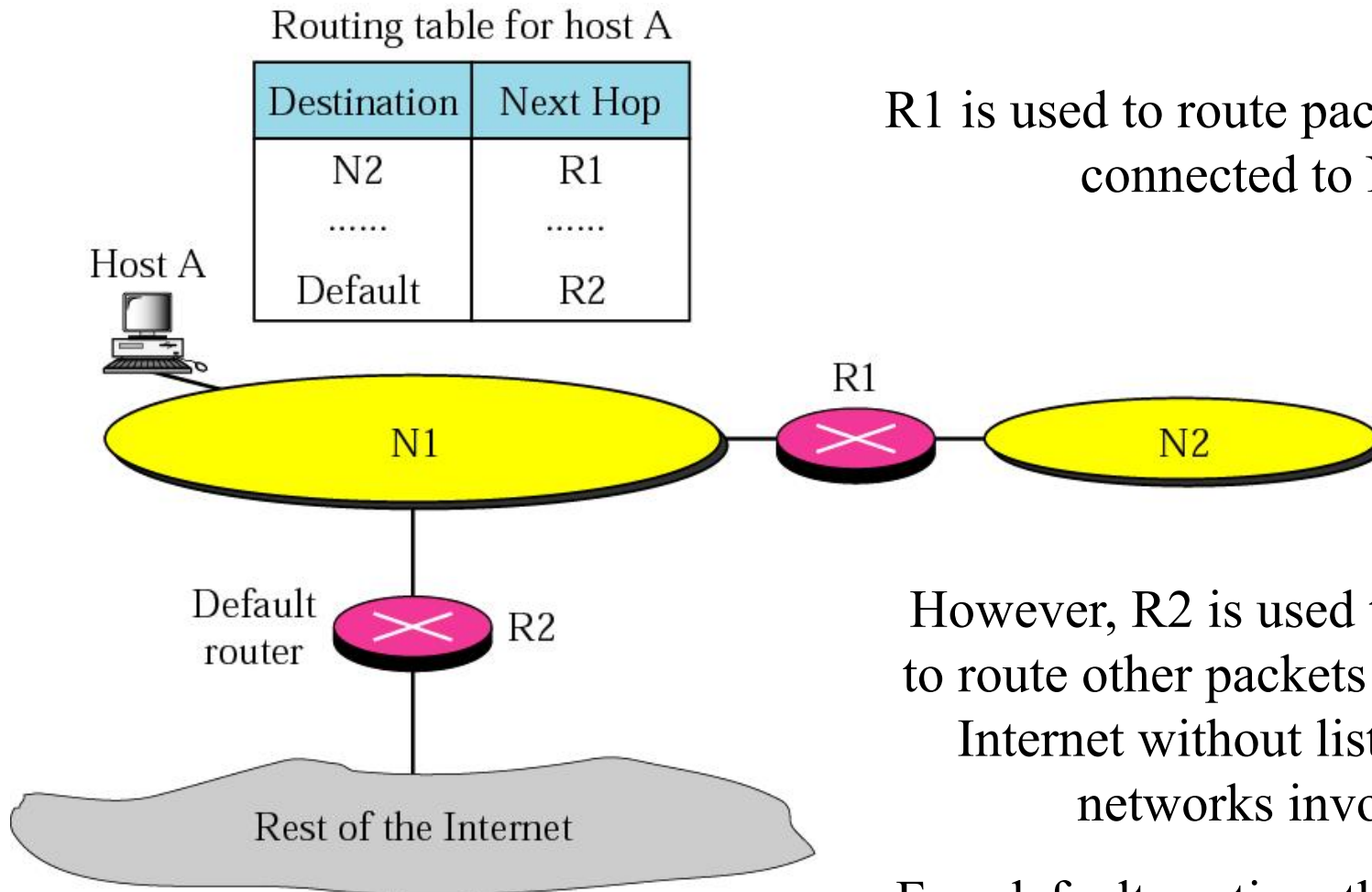
Network-specific & host-specific routing



The **destination host address** is given in the routing table; to have greater control over routing.

Only **one entry** is needed to defined the address of the **network** itself. All host connected to the same network as one single entity.

Default routing (默认/缺省路由)

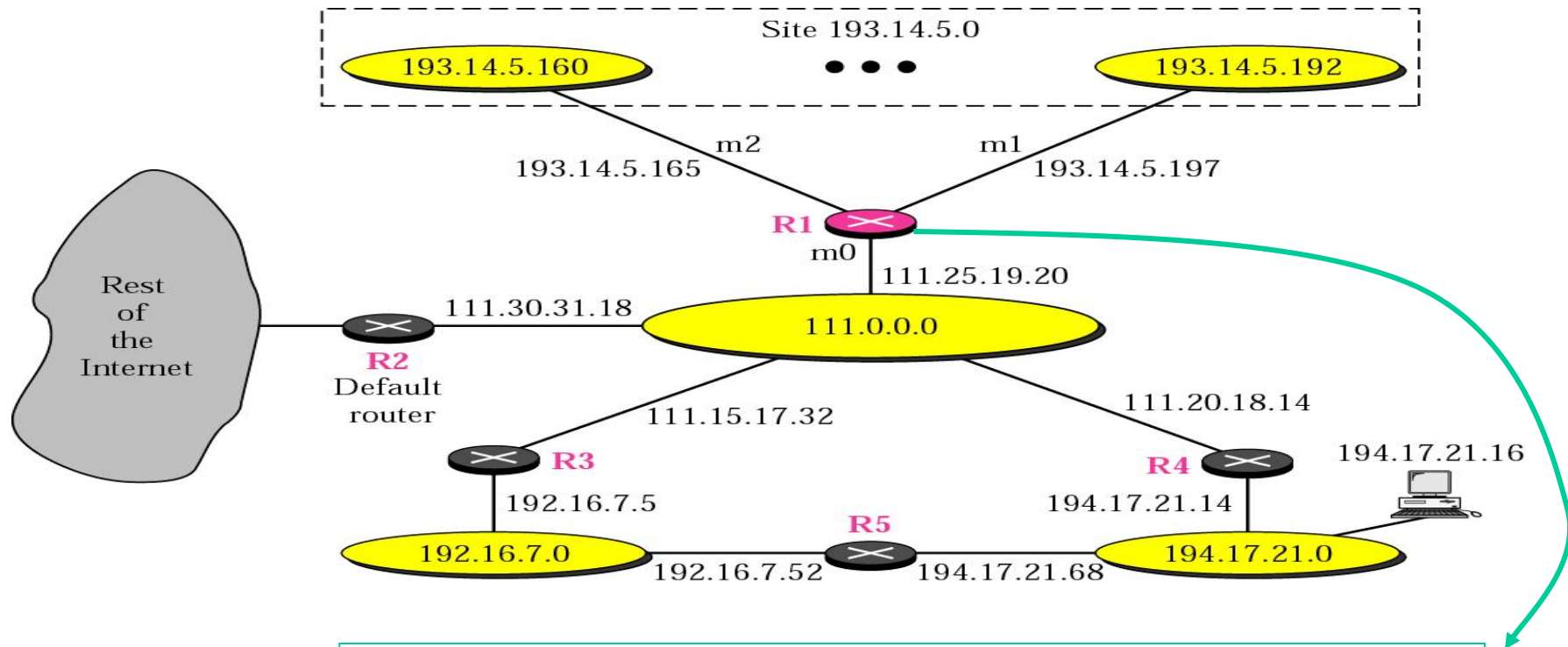


R1 is used to route packets to hosts connected to N2.

However, R2 is used to as **default** to route other packets to the rest of Internet without listing all the networks involved

For default routing, the destination network address is **0.0.0.0**

Configuration for routing example



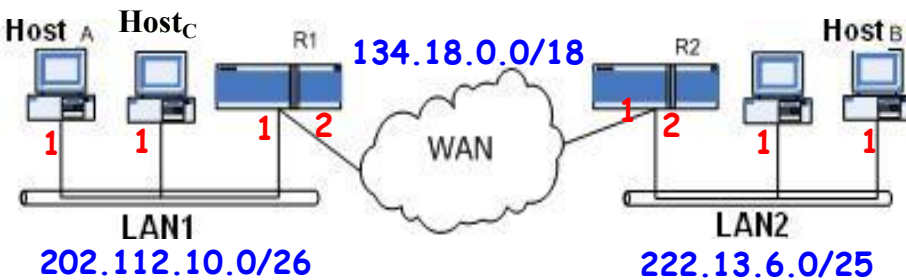
	<u>Mask</u>	<u>Dest</u>	<u>Next Hop</u>	<u>Int.</u>
Standard delivery	255.0.0.0	111.0.0.0	-	m0
	255.255.255.224	193.14.5.160	-	m2
	255.255.255.224	193.14.5.192	-	m1
Host-specific	255.255.255.255	194.17.21.16	111.20.18.14	m0
	255.255.255.0	192.16.7.0	111.15.17.32	m0
Network-specific	255.255.255.0	194.17.21.0	111.20.18.14	m0
	0.0.0.0	0.0.0.0	111.30.31.18	m0

How does Source make routing?

- Source host applies the **mask** to the destination address of the packet in order to find the corresponding **destination network address**.
 - ◆ If it is **local network**, send directly to **destination**;
 - ◆ Otherwise, send to the **local router**

Host A's routing table:

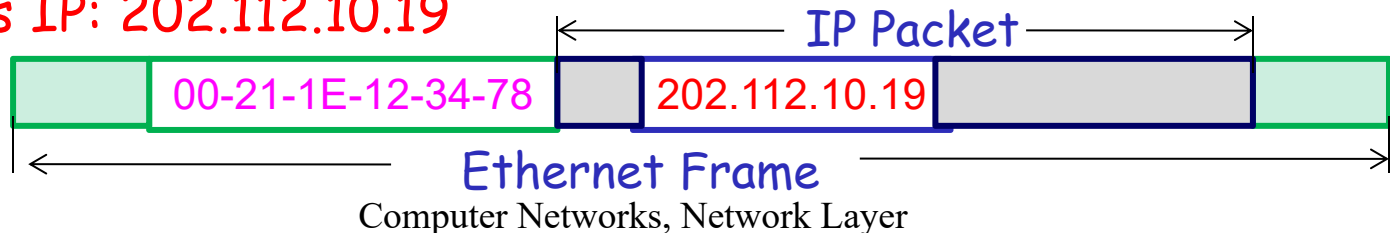
Des IP	mask	port	next hop
202.112.10.0	/26	1	--
0.0.0.0	/0	1	202.112.10.18



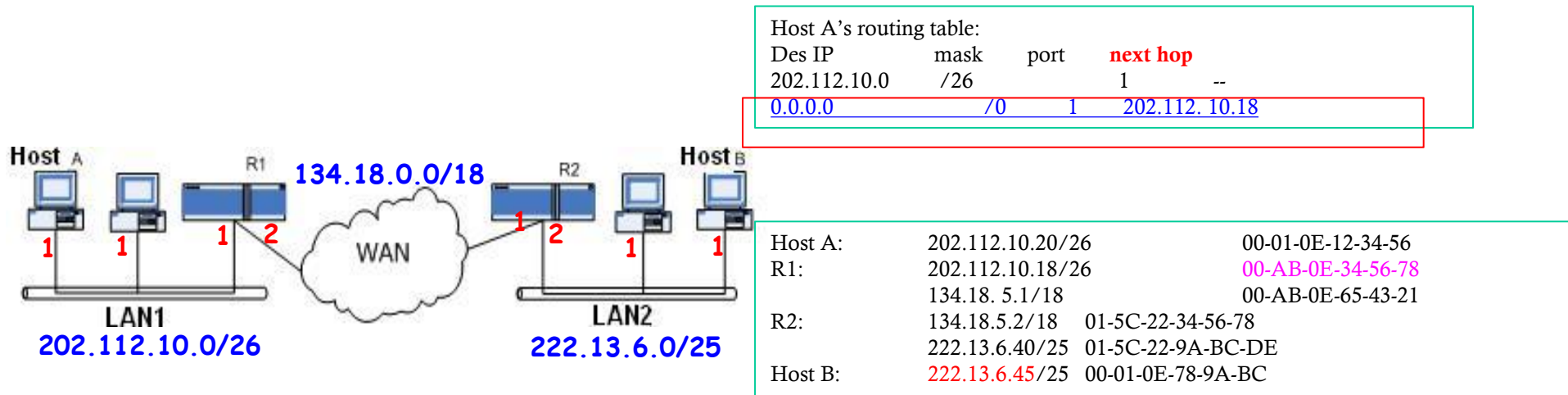
Host A:	202.112.10.20/26	00-01-0E-12-34-56
Host C:	202.112.10.19/26	00-21-1E-12-34-78
R1:	202.112.10.18/26	00-AB-0E-34-56-78
	134.18.5.1/18	00-AB-0E-65-43-21
R2:	134.18.5.2/18	01-5C-22-34-56-78
	222.13.6.40/25	01-5C-22-9A-BC-DE
Host B:	222.13.6.45/25	00-01-0E-78-9A-BC

Case 1: Des IP: 202.112.10.19

From A:

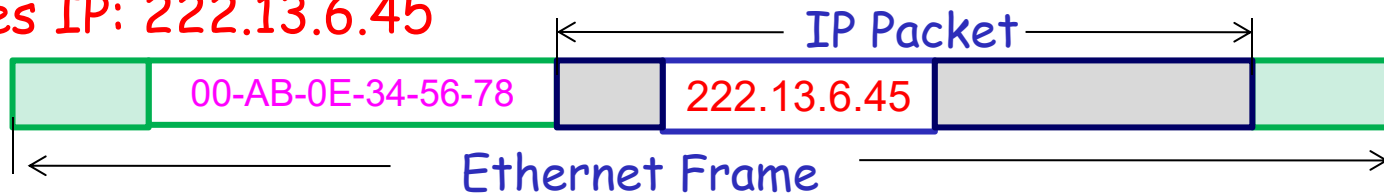


How does Source make routing?



Case 2: Des IP: 222.13.6.45

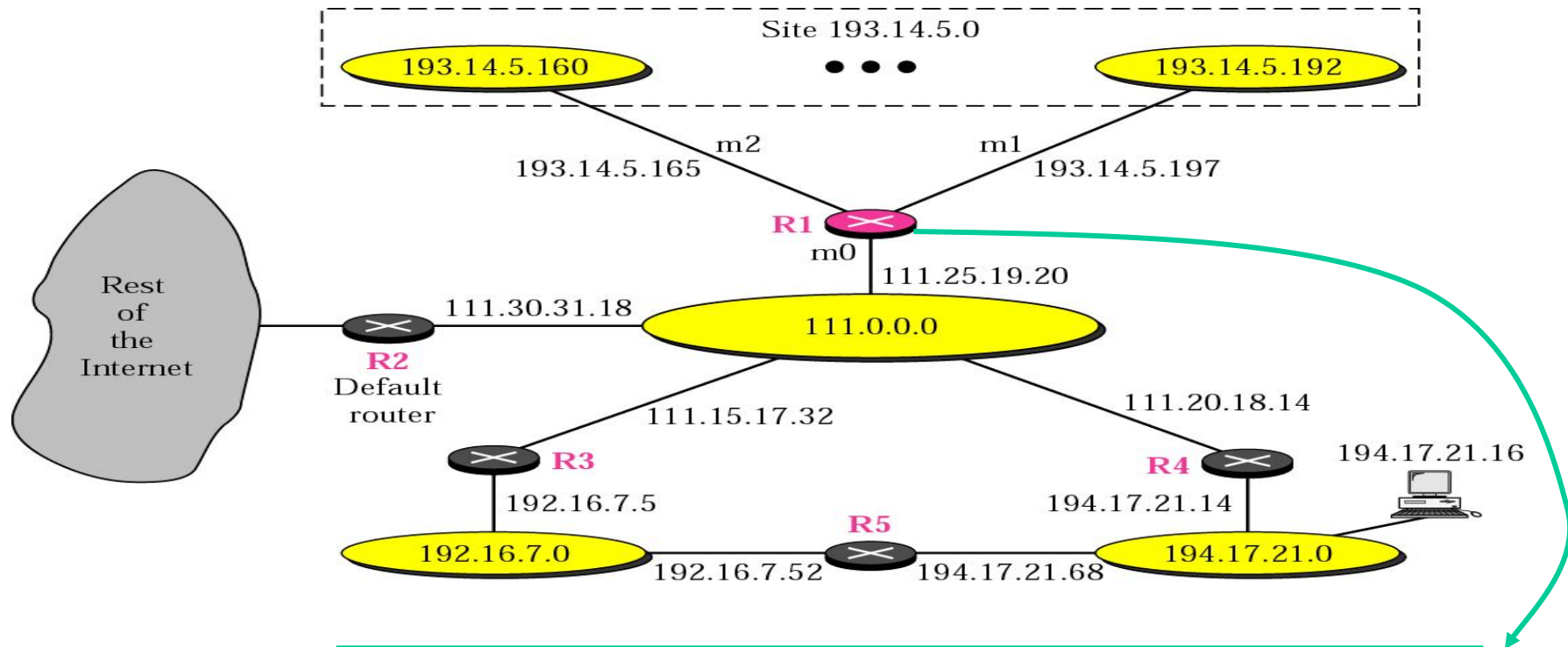
From A:



How do routers make routing?

- Generally, a routing table needs a minimum of 4 columns: **mask**, **destination network address**, **next hop address** and **outgoing interface**.
- When a packet arrives, the router applies the **mask** to the destination address of the packet in order to find the corresponding **destination network address**.
- If found **matching entry**(include default routing) in routing table, the packet is sent out from the corresponding **interface** in the table.
- If not found, the packet is discard and the router should send a **ICMP error report message** to the source host.

Configuration for routing example



	<u>Mask</u>	<u>Dest</u>	<u>Next Hop</u>	<u>Int.</u>
Standard delivery	255.0.0.0	111.0.0.0	-	m0
	255.255.255.224	193.14.5.160	-	m2
	255.255.255.224	193.14.5.192	-	m1
Host-specific	255.255.255.255	194.17.21.16	111.20.18.14	m0
	255.255.255.0	192.16.7.0	111.15.17.32	m0
Network-specific	255.255.255.0	194.17.21.0	111.20.18.14	m0
	0.0.0.0	0.0.0.0	111.30.31.18	m0

Example 1

Router R1 receives 500 packets for destination **192.16.7.14**; the algorithm applies the masks row by row to the destination address until a match is found:

Rule of thumb: Apply the individual mask (from Routing table) to the received destination address (row-by-row) and see if its matches any of the DEST address stated in its routing table. If match is found, then stop

Solution

	<u>Mask</u>	<u>Dest</u>	<u>Next Hop</u>	<u>Int.</u>
Standard delivery	255.0.0.0	111.0.0.0	-	m0
	255.255.255.224	193.14.5.160	-	m2
	255.255.255.224	193.14.5.192	-	m1
Host-specific	255.255.255.255	194.17.21.16	111.20.18.14	m0
Network-specific	255.255.255.0	192.16.7.0	111.15.17.32	m0
	255.255.255.0	194.17.21.0	111.20.18.14	m0
	0.0.0.0	0.0.0.0	111.30.31.18	m0

Example 1

Router R1 receives 500 packets for destination 192.16.7.14; the algorithm applies the masks row by row to the destination address until a match is found:

Rule of thumb: Apply the individual mask (from Routing table) to the received destination address (row-by-row) and see if its matches any of the DEST address stated in its routing table. If match is found, then stop

Solution

Direct delivery

192.16.7.14	&	255.0.0.0	→	192.0.0.0	no match	to 111.0.0.0
192.16.7.14	&	255.255.255.224	→	192.16.7.0	no match	to 193.14.5.160
192.16.7.14	&	255.255.255.224	→	192.16.7.	no match	to 193.14.5.192

Host-specific

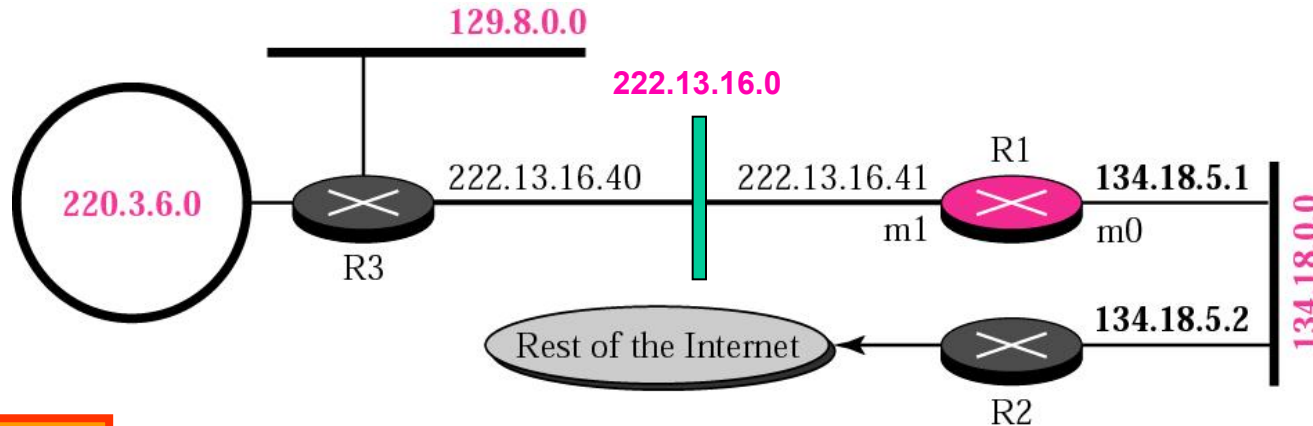
192.16.7.14	&	255.255.255.255	→	192.16.7.14	no match	to 194.17.21.16
-------------	---	-----------------	---	-------------	----------	-----------------

Network-specific

192.16.7.14	&	255.255.255.0	→	192.16.7.0	match	to 192.16.7.0
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Example 2

Make the routing table for router R1 in figure below



Solution

Mask	Destination	Next Hop	Int.
255.255.0.0	134.18.0.0	--	m0
255.255.255.0	222.13.16.0	--	m1
255.255.0.0	129.8.0.0	222.13.16.40	m1
255.255.255.0	220.3.6.0	222.13.16.40	m1
0.0.0.0	0.0.0.0	134.18.5.2	m0

Routing Tables in IP with CIDR

- Longest-Prefix Matching(最长前缀匹配): Longest mask is used for multiple matches
- For each entry in the routing table:
 - ◆ MaskedAddress := EntryMask (bitAND) IPDatagramDestinationAddress;
 - ◆ if (MaskedAddress == EntryDestination), **Mark** the entry

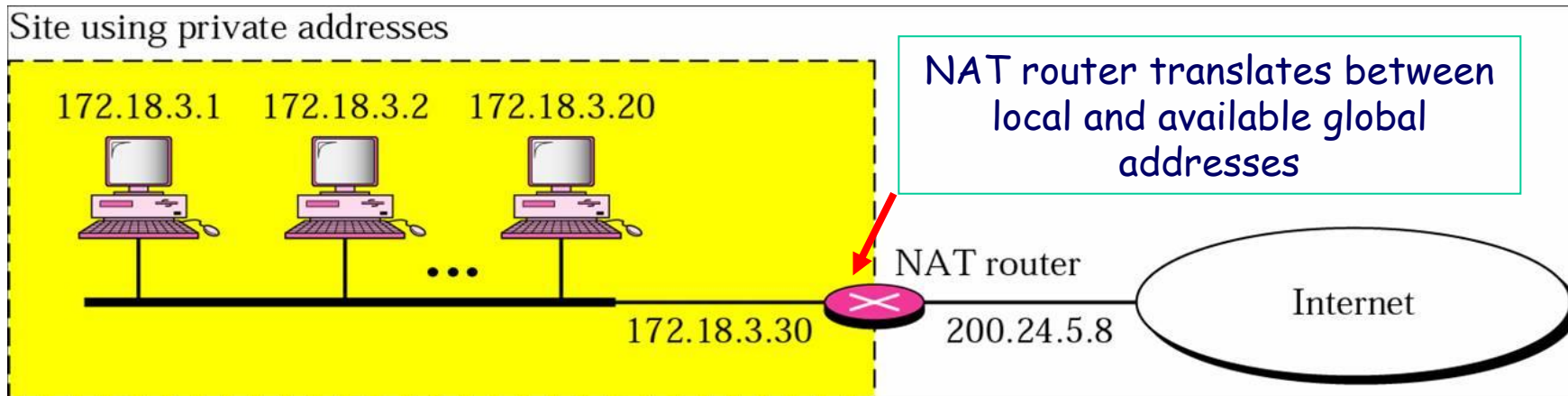
Mask	Destination	Next Hop
/12	128.96.0.0	145.12.56.29
/17	128.125.0.0	153.202.12.128
<u>/12</u>	<u>128.112.0.0</u>	<u>153.202.14.1</u>
/26	128.105.14.64	153.2.45.101
/32	128.105.14.66	153.2.45.101

- ◆ e.g. Destination: 128.125(0111 1101).1.3, what is next hop?
128.112(0111 0000).0.0

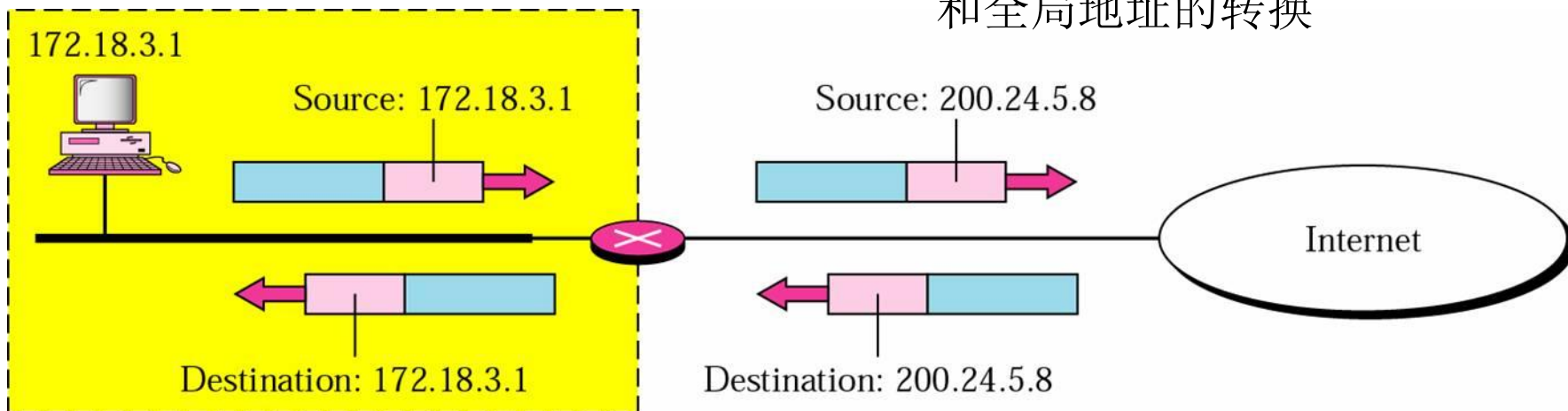
(9)NAT: Network Address Translation

- Problem: IP addresses are scarce
- Solution: using reserved Private IP addresses(私有/专用地址)
 - ◆ 10.0.0.0 - 10.255.255.255/8
 - ◆ 172.16.0.0 - 172.31.255.255/12 (16: 0001 0000)
 - ◆ 192.168.0.0 - 192.168.255.255/16
- NAT router converts internal IP source address to another global IP address
- NAT router is often combined with a firewall in a single device
- Source TCP/UDP port is also converted by NAT box

NAT Sample



NAT路由器负责私有地址
和全局地址的转换



[Forouzan]

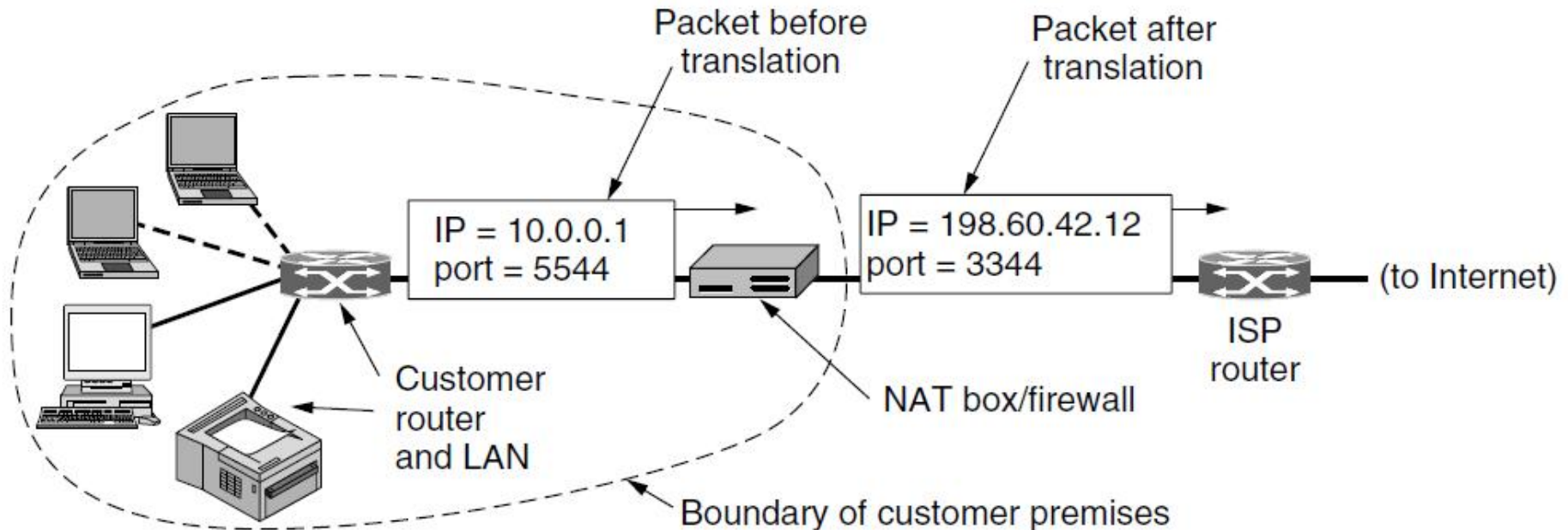
NAT (Network Address Translation)

■ Outgoing packet

- ◆ 10.x.y.z source address and TCP Source port are replaced
- ◆ Both the IP and TCP checksums are recomputed

■ Incoming packet

- ◆ Destination port => internal IP address and original TCP Source port, Recompute IP and TCP checksums
- ◆ Deliver the packet using the 10.x.y.z address



What if 2 local hosts talking to same destination?

- Using port number: Layering discipline is violated

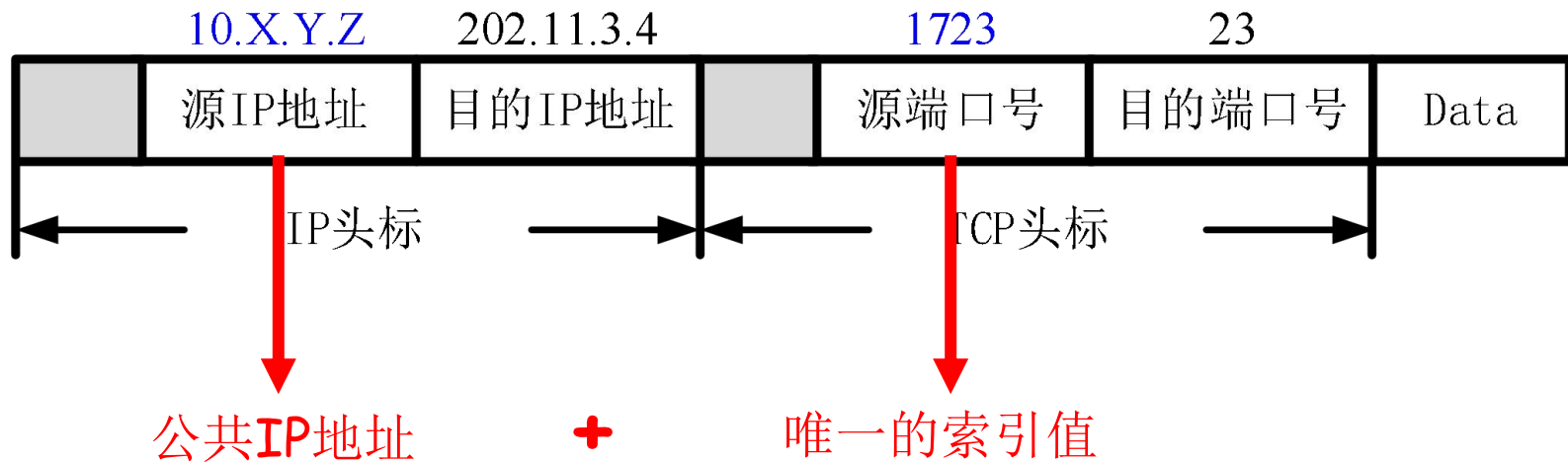
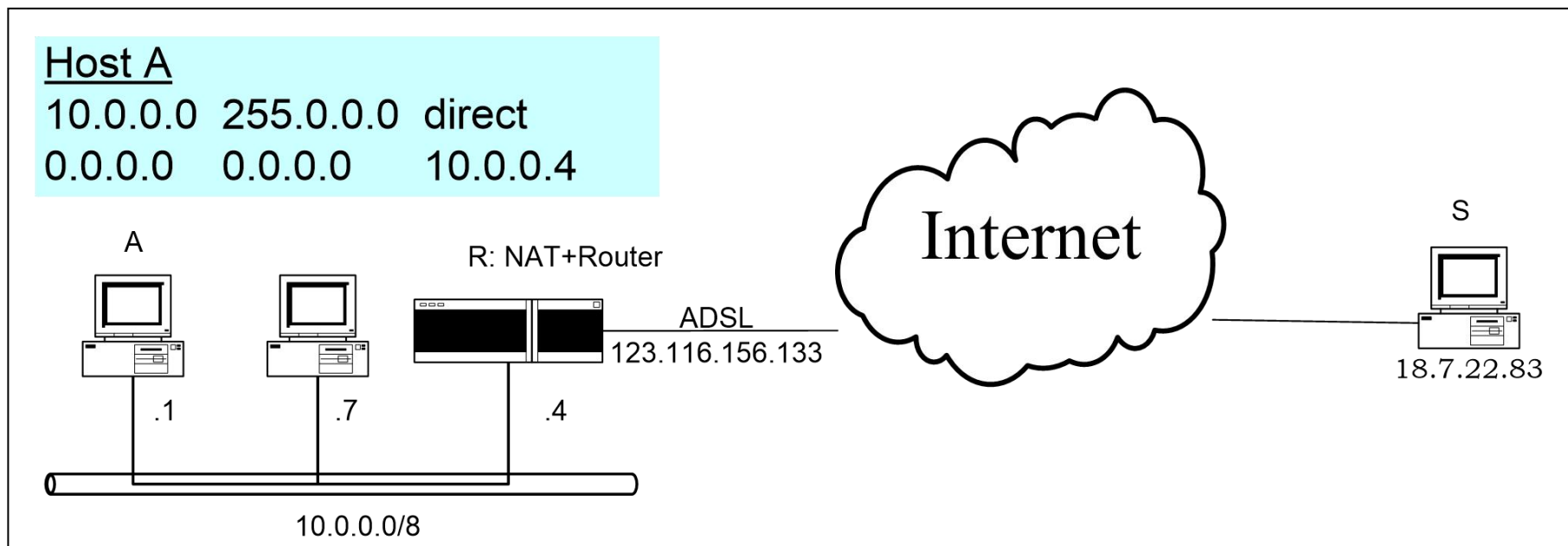


Table in NAT router

Peer	Local: IP+port		Global IP	Global port
202.11.3.4:23	10.1.1.2	1723	198.60.42.12	1723
202.11.3.4:23	10.1.1.3	1723	198.60.42.12	1924
202.11.3.4:23	10.1.1.4	1723	198.60.42.12	1492

入境IP: 198.60.42.12:1492 -> 10.1.1.4:1723

举例：动态NAT表项



■ A: 发送TCP报 10.0.0.1:3723->18.7.22.83:80

■ R: 路由器动态创建NAT表项:

(Local)10.0.0.1:3723—(Global)123.116.156.133:33120—(Peer)18.7.22.83:80

改TCP报为123.116.156.133:33120->18.7.22.83:80

■ S: 服务器S回数据18.7.22.83:80->123.116.156.133:33120

■ R: 查NAT表: 改TCP报为为18.7.22.83:80->10.0.0.1:3723

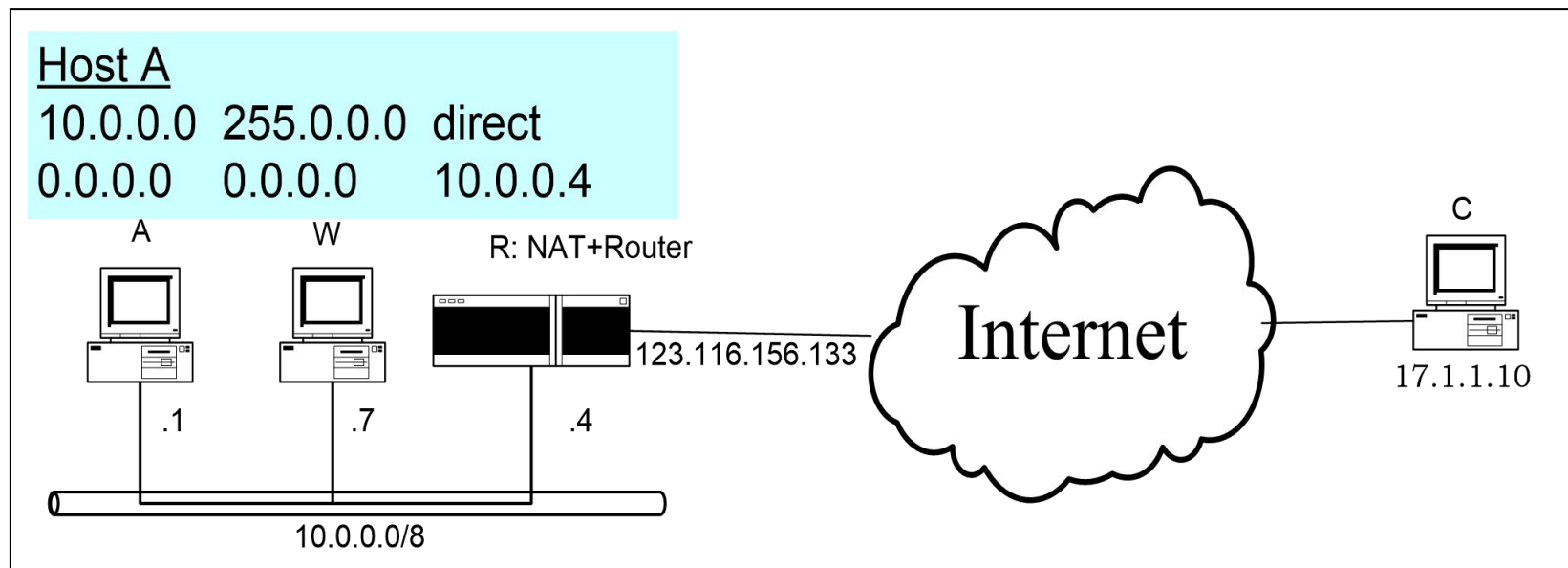
■ A: 收到数据包18.7.22.83:80->10.0.0.1:3723

(连接过程中使用该NAT表项, 当连接关闭时或长久未用的NAT表项, 删除)

OUT: (SRC)10.0.0.1:3723—(DST)18.7.22.83:80=>(SRC)123.116.156.133:33120

IN: (SRC)18.7.22.83:80—(DST)123.116.156.133:33120=>(DST)10.0.0.1:3723

举例：静态NAT表项



- R: 设10.0.0.7为Web服务器: **建立静态NAT表**
(Local)10.0.0.7:8080—(Global)123.116.156.133:80 —(Peer)*:*
- C: 因特网上的客户端C发送TCP数据包:
17.1.1.10:3598->123.116.156.133:80
- R: 查 NAT表, 改为17.1.1.10:3598->10.0.0.7:8080
- W: Web服务器回数据10.0.0.7:8080->17.1.1.10:3598
- R: 查 NAT表, 改为123.116.156.133:80->17.1.1.10:3598
- C: 收到返回TCP数据报123.116.156.133:80->17.1.1.10:3598

关于**IP**地址我们要掌握什么？

- “网络地址/前缀”的含义
- 已知网络需要的地址数（主机数），能够分配**IP**地址空间
- 把多个网络聚合成一个网络
- 子网掩码的作用：计算出网络地址
- 已知子网掩码(“/前缀”), 可以知道网络中有多少个**IP**地址，多少个可用地址
- 已知一个数据报的目的**IP**地址和子网掩码，可以获得目的网络地址，并根据该地址查路由表确定如何转发该**IP**包
- 注意**最长前缀匹配法**

The network layer in the Internet

- 5.6.1 IPv4 Packet format
- 5.6.2 IP Addressing
- 5.6.3 IPv6
- 5.6.4 ARP, DHCP and ICMP
- 5.6.6&5.6.7 Routing Protocols: OSPF and BGP

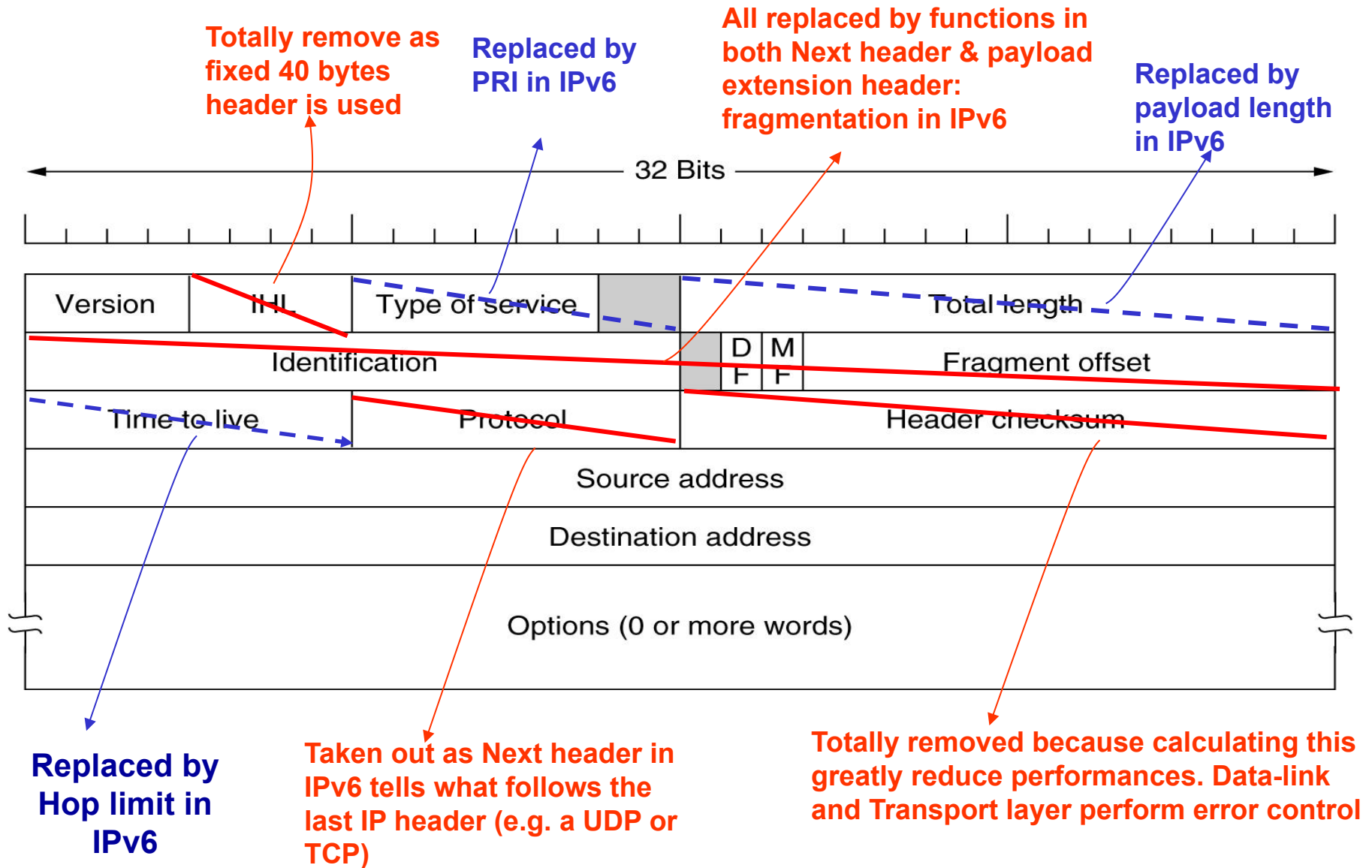
IPv6: Motivation

- IPv4 is becoming insufficient for the fast growing Internet and the needs:
 - ◆ The use of addressing space is becoming insufficient for conventional netid & hostid two-level addressing structure in Classful addressing
 - ◆ Real-time application i.e. video transmission requires minimum delay strategies and reservation of resource is not provided
 - ◆ Encryption and authentication of data and security mechanism is not provided
 - ◆ Limitation of flexibility and mobility supported

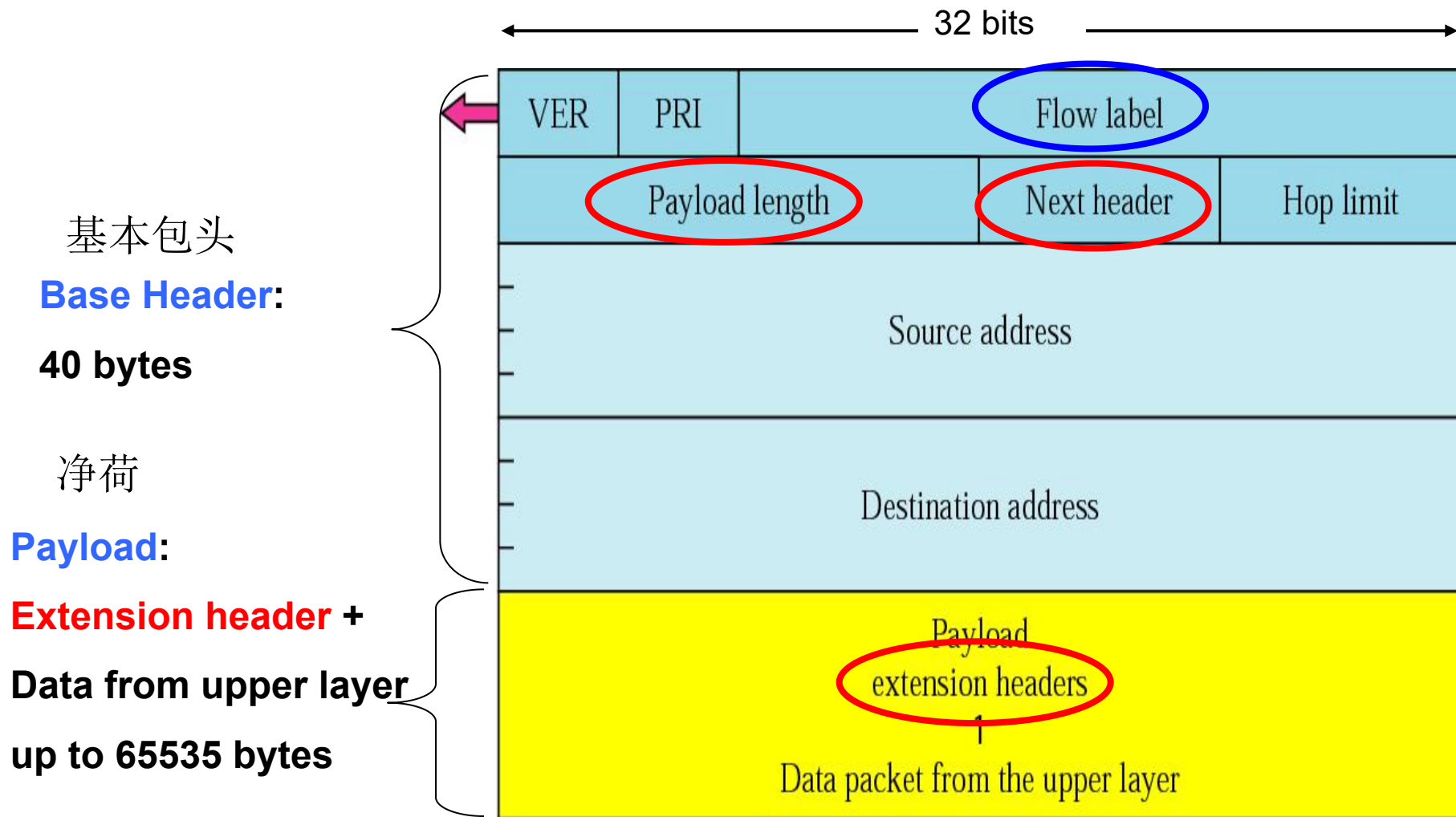
IPv6: Major Goals

- Address space
- Reduce the size of the routing tables
- Simplify the protocol, to allow routers to process packets faster
- Security
- Pay more attention to type of service, particularly for real-time data
- Multicasting
- Make it possible for a host to roam without changing its address
- Allow the protocol to evolve in the future
- Permit the old and new protocols to coexist for years

IPv6: Revised IPv4 Header



IPv6 Datagram



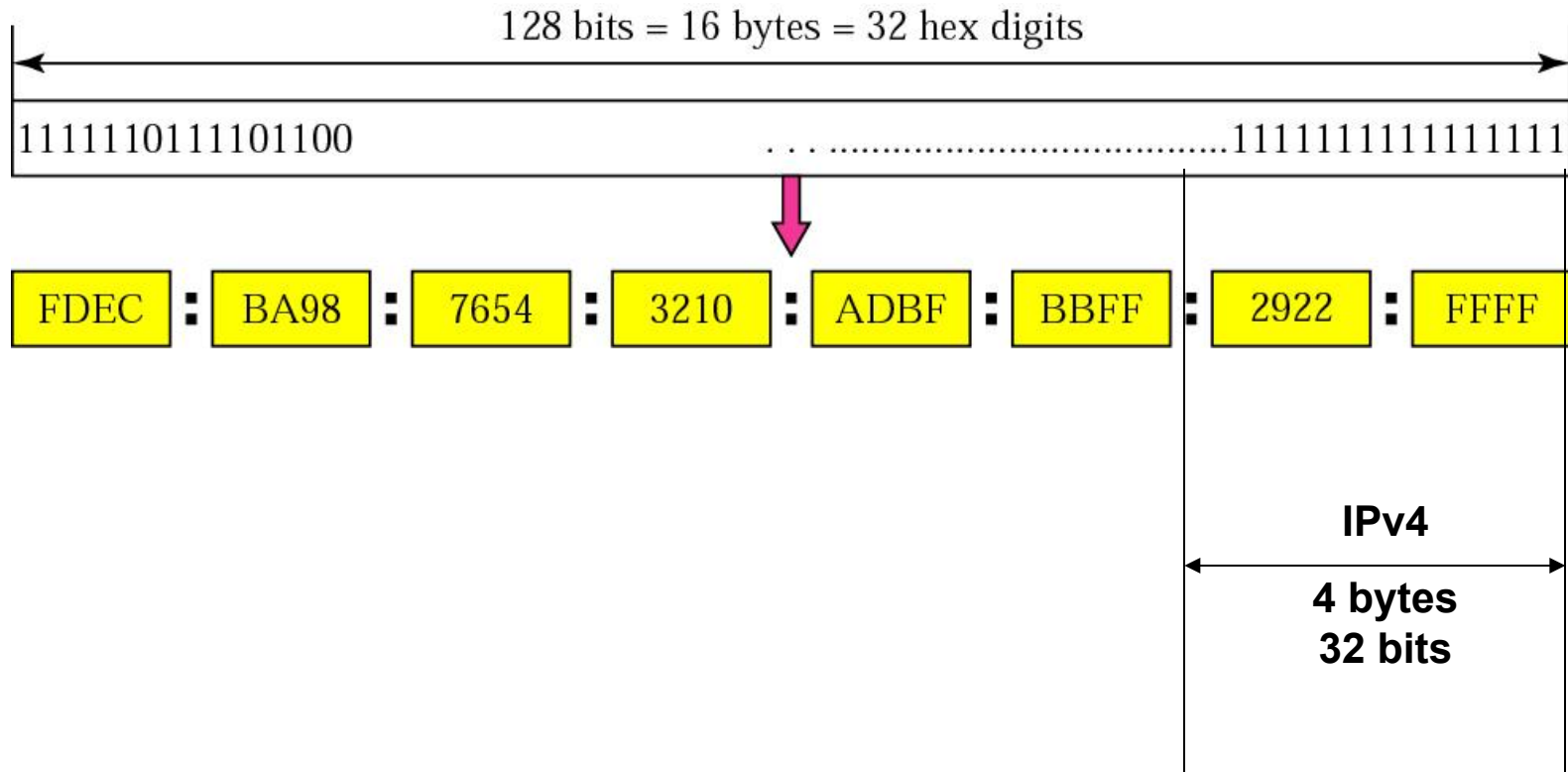
IPv6 Header: Fields

- **VER:** IP version; 4 for IPv4 and 6 for IPv6
- **PRI:** priority of packets; 0-7: for packets that can slow down in event of congestion; 8-15 for real-time traffic with constant sending rate - Audio & Video.
- **Flow label:** pseudo-connection between source and destination. for reserved bandwidth and stringent delay requirement. (Still experimental).
- **Payload length:** tells how many payload bytes follow the 40-bytes header
- **Next header:** the subsequent headers that follow the base header; either **optional extension headers** used by IP or **header** for an upper-layer protocol, e.g. UDP & TCP.
- **Hop limit:** serves the same purpose as the TTL in IPv4.
- **Source and Destination address field:** both 128 bits
- **Extension Headers:** 6 extension headers which is optional in IPv4.

IPv6: Improvement

- **Longer addresses** : 128 bits long > 32 bits currently in IPv4. That means a huge increase of addressing space.
- **Simplification in Header**: contains only 8 vital fields < 13 in IPv4. (speeds up routing process). – 5 fields becomes options.
- **New options**: to support additional functionalities
- **Security**: Encryption and authentication are key features of new IP
- **More Services**: to handle special/real-time multimedia traffic.

IPv6 Address(冒分十六进制)



[Forouzan]

Abbreviated Address

Unabbreviated

FDEC : BA98 : 0074 : 3210 : 000F : BBFF : 0000 : FFFF



FDEC : BA98 : 74 : 3210 : F : BBFF : 0 : FFFF

Abbreviated

Abbreviated

FDEC : 0 : 0 : 0 : 0 : BBFF : 0 : FFFF



FDEC :: BBFF : 0 : FFFF

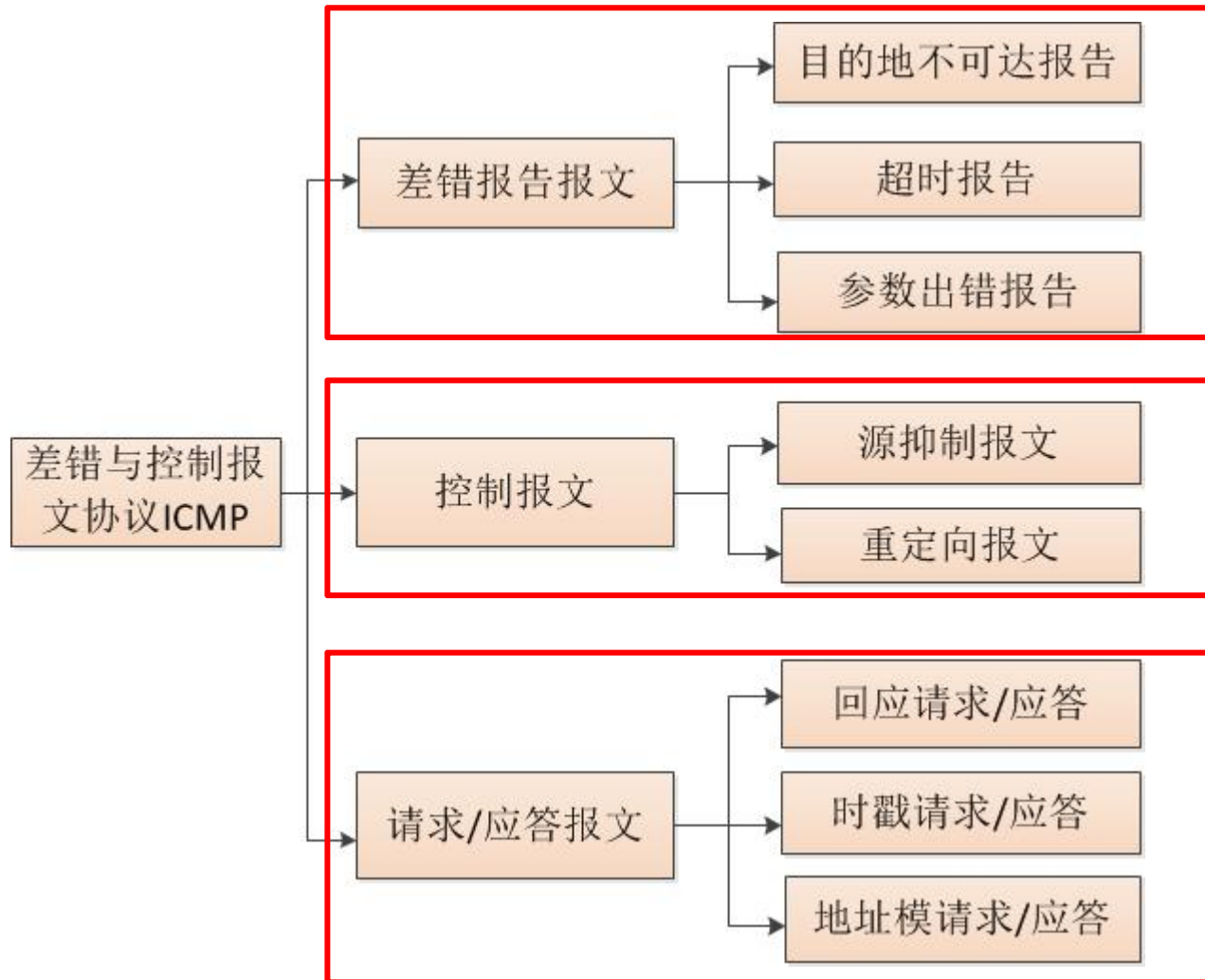
More Abbreviated

The network layer in the Internet

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- 5.6.3 IPv6
- 5.6.4 ARP, DHCP and ICMP
- 5.6.6&5.6.7 Routing Protocols: OSPF and BGP

(1) ICMP(因特网控制报文协议)Messages

- An ICMP message is encapsulated in an IP packet.
- The principal ICMP message types



ICMP(因特网控制报文协议)Messages

■ Destination Unreachable

- ◆ a router cannot locate the destination
- ◆ a packet with the **DF** bit cannot be delivered because a 'small-packet' network stands in the way.
- ◆ **path MTU discovery(PMTUD)**

■ Time Exceeded

- ◆ a packet is dropped because its counter has reached zero
- ◆ **Traceroute**

■ Parameter Problem

- ◆ A **illegal value** has been detected **in header field**.
- ◆ Indicate a bug in the sending host's IP software or possibly in the software of a router transited.

ICMP(因特网控制报文协议)Messages

■ Source Quench(源抑制)

- ◆ Congestion control
- ◆ When a host received this message, it was expected to slow down

■ Redirect

- ◆ a router notices that a packet seems to be routed incorrectly.
- ◆ It is used by the router to tell the sending host to **update to a better route.**

■ ECHO and ECHO Reply

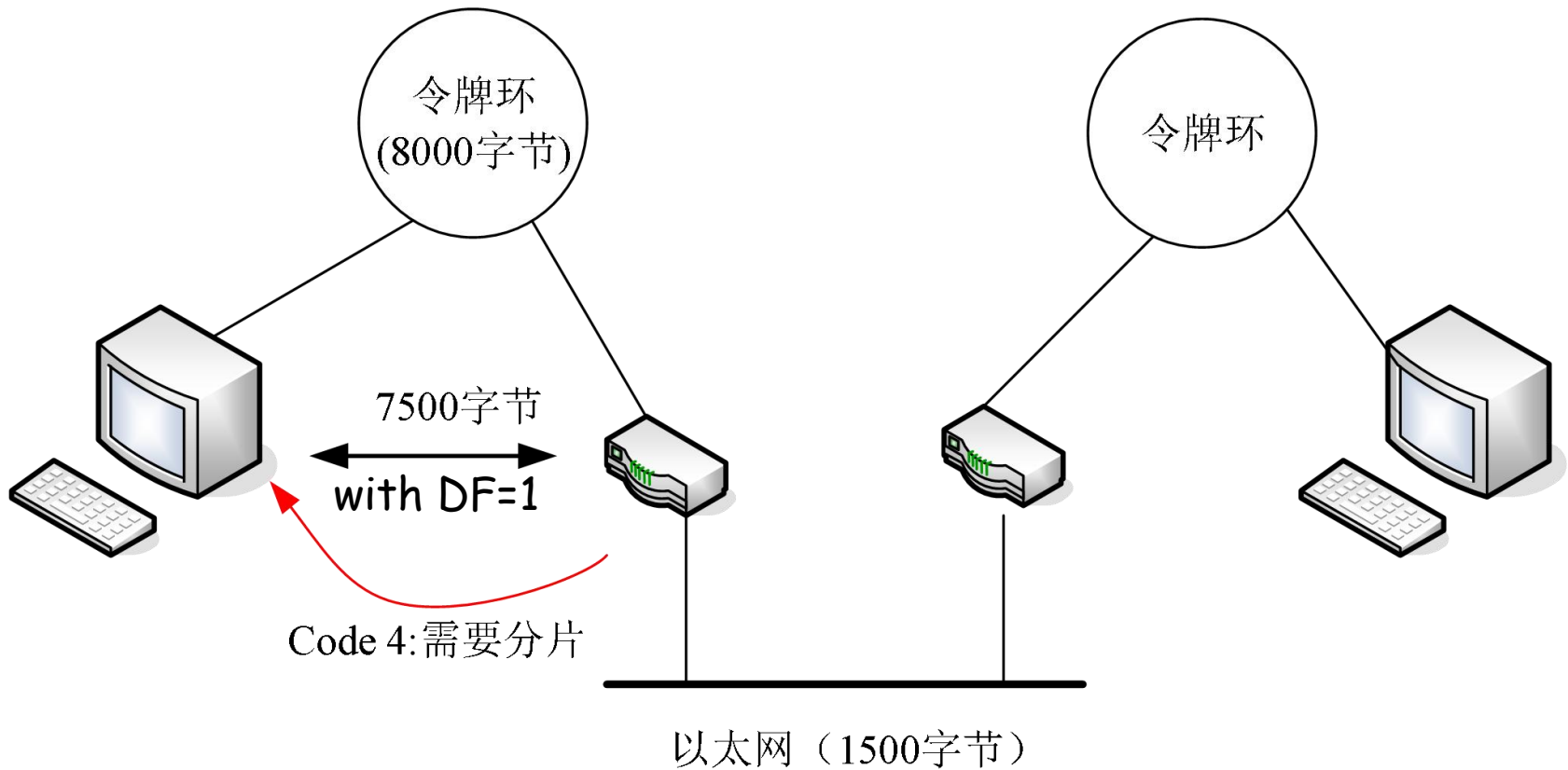
- ◆ Sent by hosts to see if a given destination **is reachable and currently alive.**
- ◆ Ping

■ Time Stamp Request and Reply

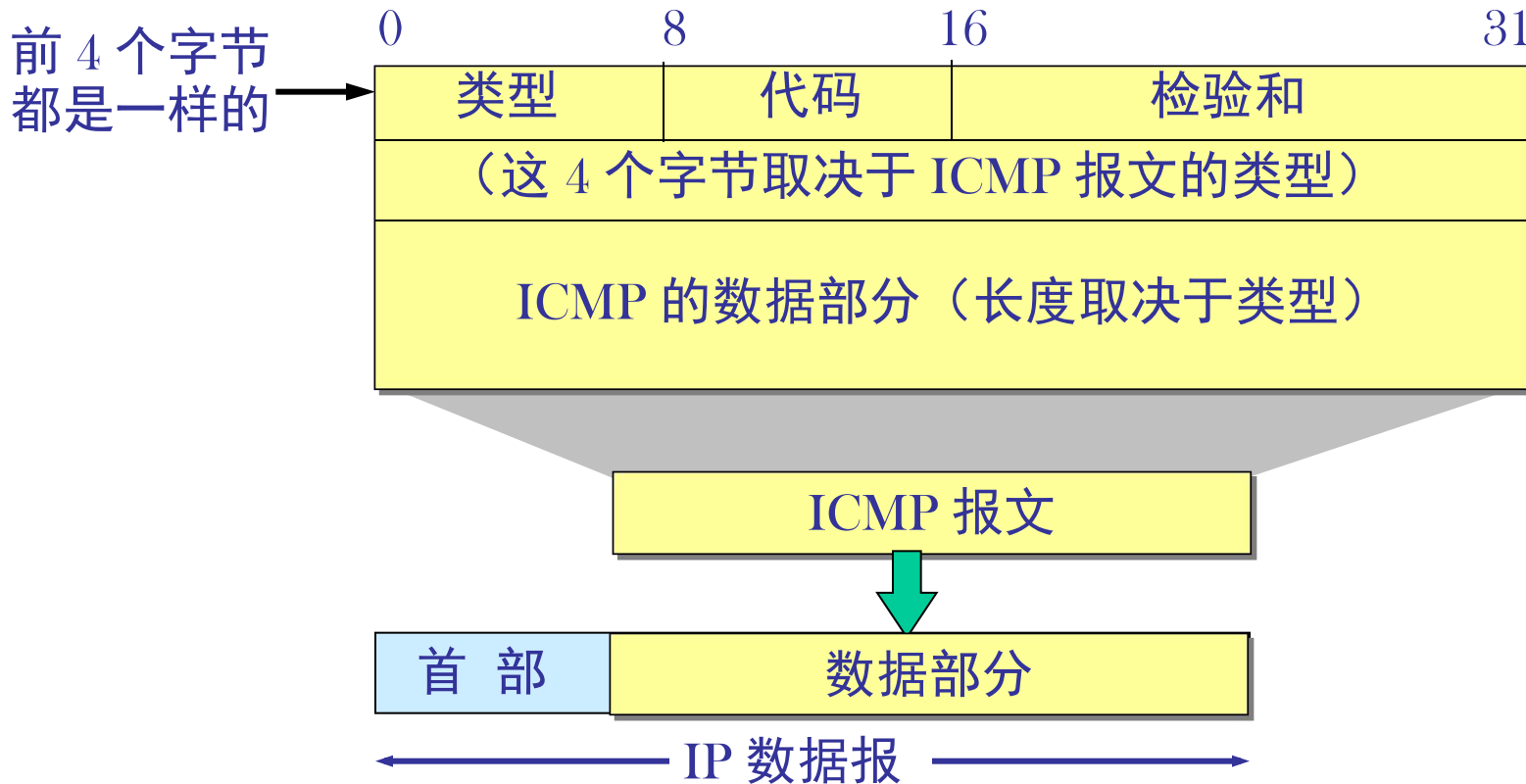
■ Router advertisement/solicitation

- ◆ Are used by hosts to find nearby router.

ICMP Example: Destination Unreachable



ICMP报文的格式

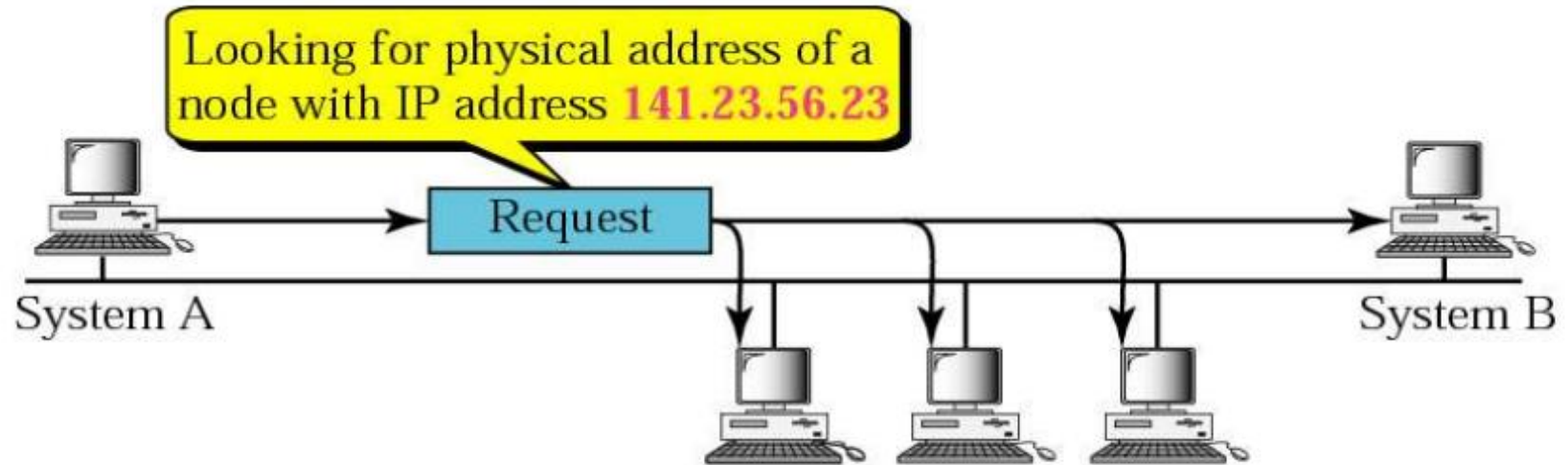


- ◆ ICMP 报文的前4个字节是统一的格式，共有三个字段：即类型、代码和检验和。接着的4个字节的内容与ICMP的类型有关
- ◆ ICMP报文分三类，即ICMP差错报告报文、控制报文和查询报文

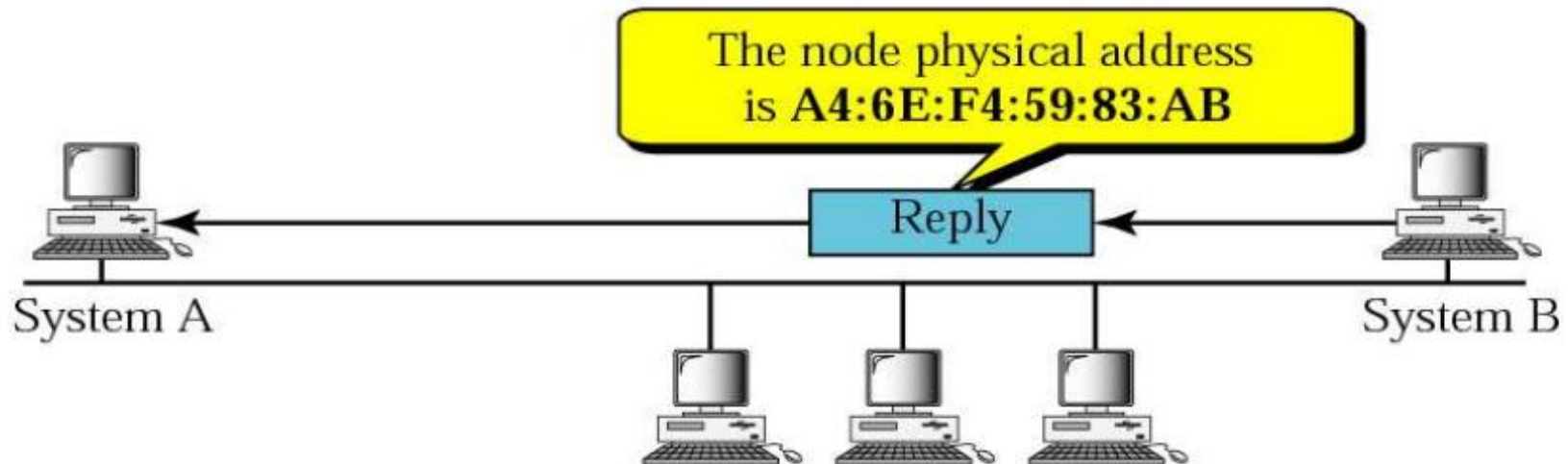
(2) ARP: Address Resolution Protocol (地址解析协议)

- How do IP addresses get mapped onto data link layer addresses, such as Ethernet?
 - ◆ One solution is to have a **configuration file** somewhere in the system that maps IP addresses onto Ethernet addresses
 - ◆ A better solution is to output a **broadcast** packet onto the Ethernet asking: Who owns IP address 192.31.65.5?

(2) ARP Operation

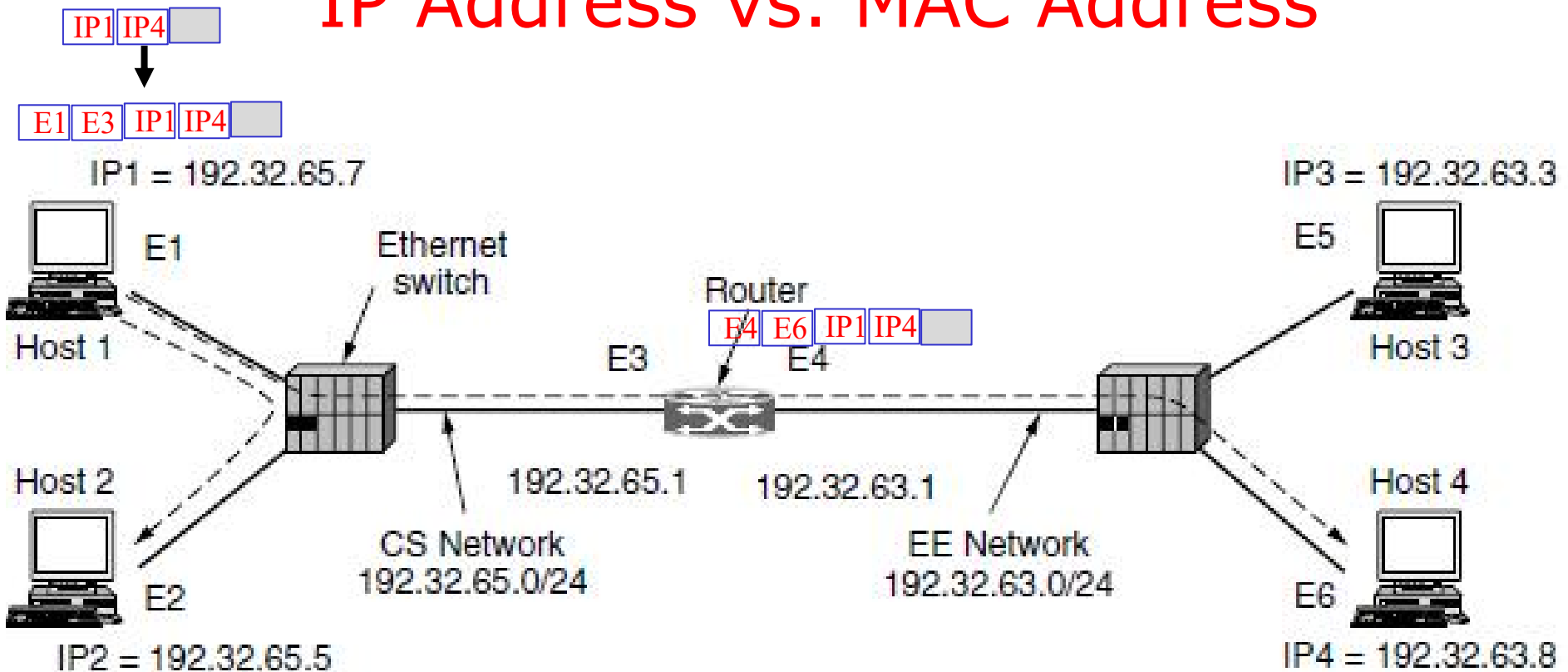


a. ARP request is broadcast



b. ARP reply is unicast

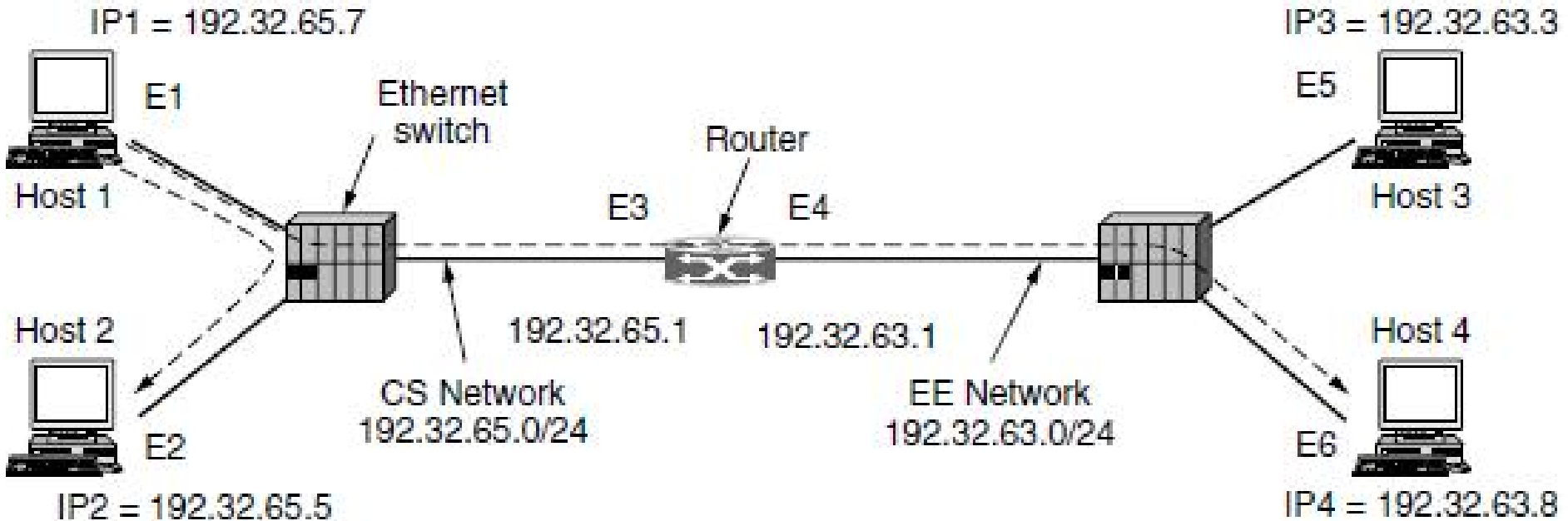
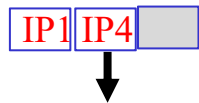
IP Address vs. MAC Address



Frame	Source IP	Source Eth.	Destination IP	Destination Eth.
Host 1 to 2, on CS net	IP1	E1	IP2	E2
Host 1 to 4, on CS net	IP1	E1	IP4	E3
Host 1 to 4, on EE net	IP1	E4	IP4	E6

IP地址全程有效，MAC只在一个子网内有效

IP Address vs. MAC Address



Frame	Source IP	Source Eth.	Destination IP	Destination Eth.
Host 1 to 2, on CS net	IP1	E1	IP2	E2
Host 1 to 4, on CS net	IP1	E1	IP4	E3
Host 1 to 4, on EE net	IP1	E4	IP4	E6



IP地址全程有效，**MAC**只在一个子网内有效

(3) RARP(逆向ARP)

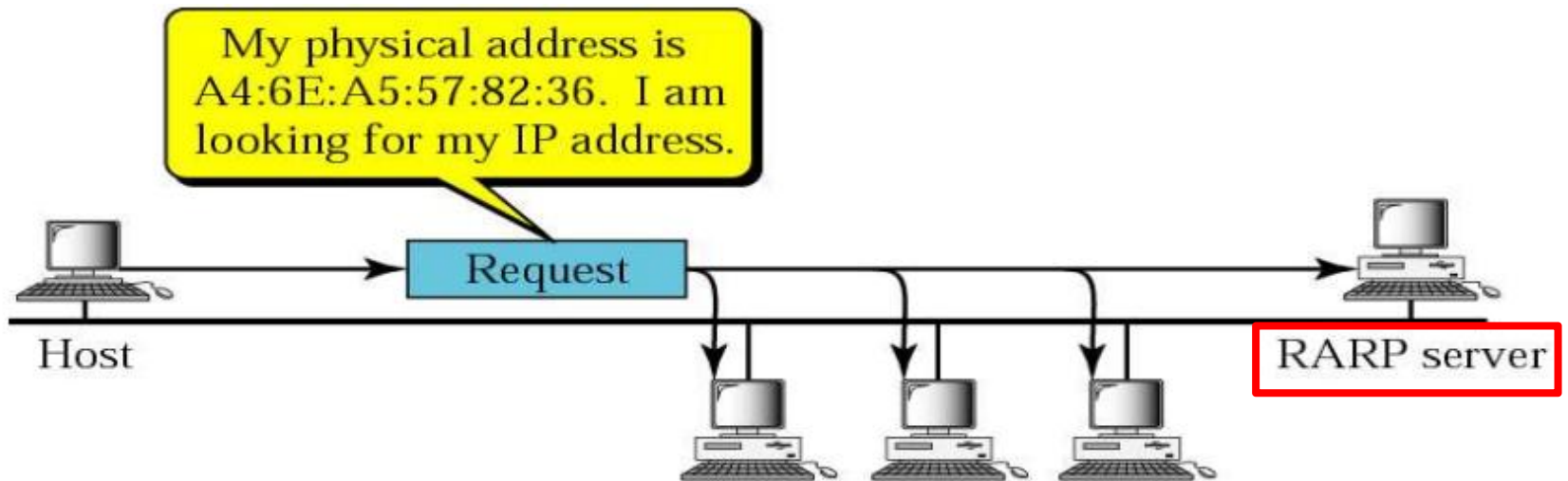
■ Requirement

- ◆ Given an Ethernet address, what is the corresponding IP address?
- ◆ Example: when a diskless workstation is booted
- ◆ Network layer protocol

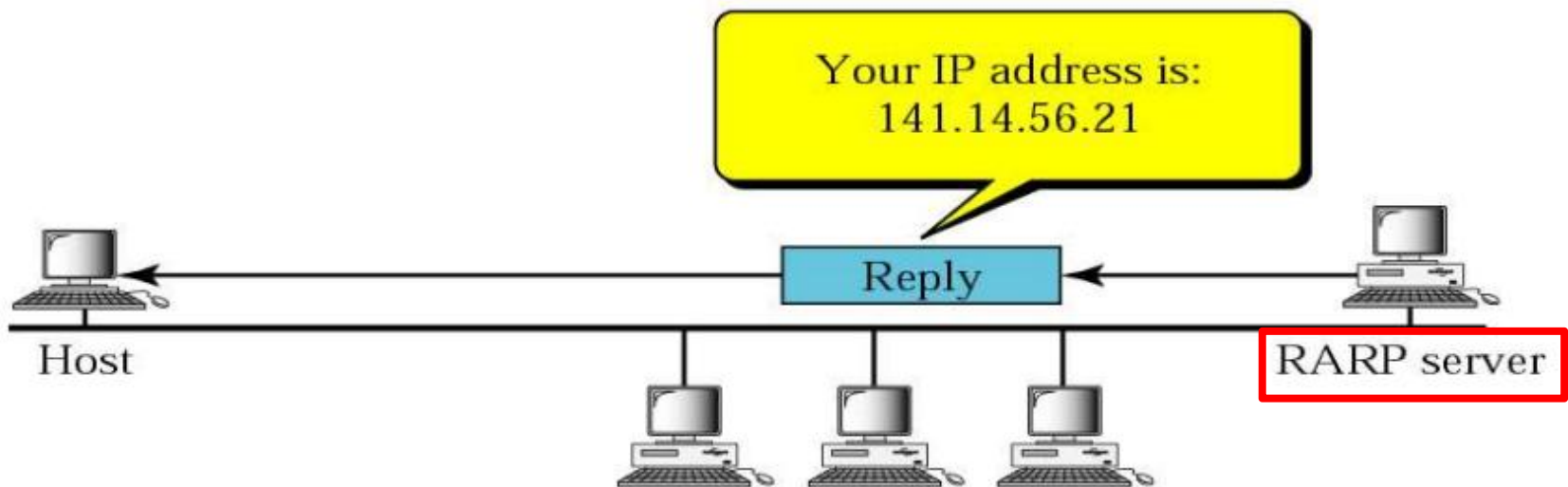
■ RARP (Reverse Address Resolution Protocol)

- ◆ Allows a workstation to broadcast on Ethernet:
My Ethernet address is 14-04-05-18-01-25. Does anyone out there know my IP address?

RARP Operation



a. RARP request is broadcast



b. RARP reply is unicast

(4) DHCP(动态主机配置协议)

■ Dynamic Host Configuration Protocol

- ◆ Provides **automatic configuration** of remote hosts including
 - IP address, Router's address, subnet mask
 - DNS server's address, file server...
- ◆ An application layer protocol, following **client-server paradigm**
- ◆ **DHCP packets**: Discover/Offer, Request/Ack

■ An extension to the BOOTP protocol

- ◆ It has largely replaced RARP and BOOTP

■ Leasing & renewal

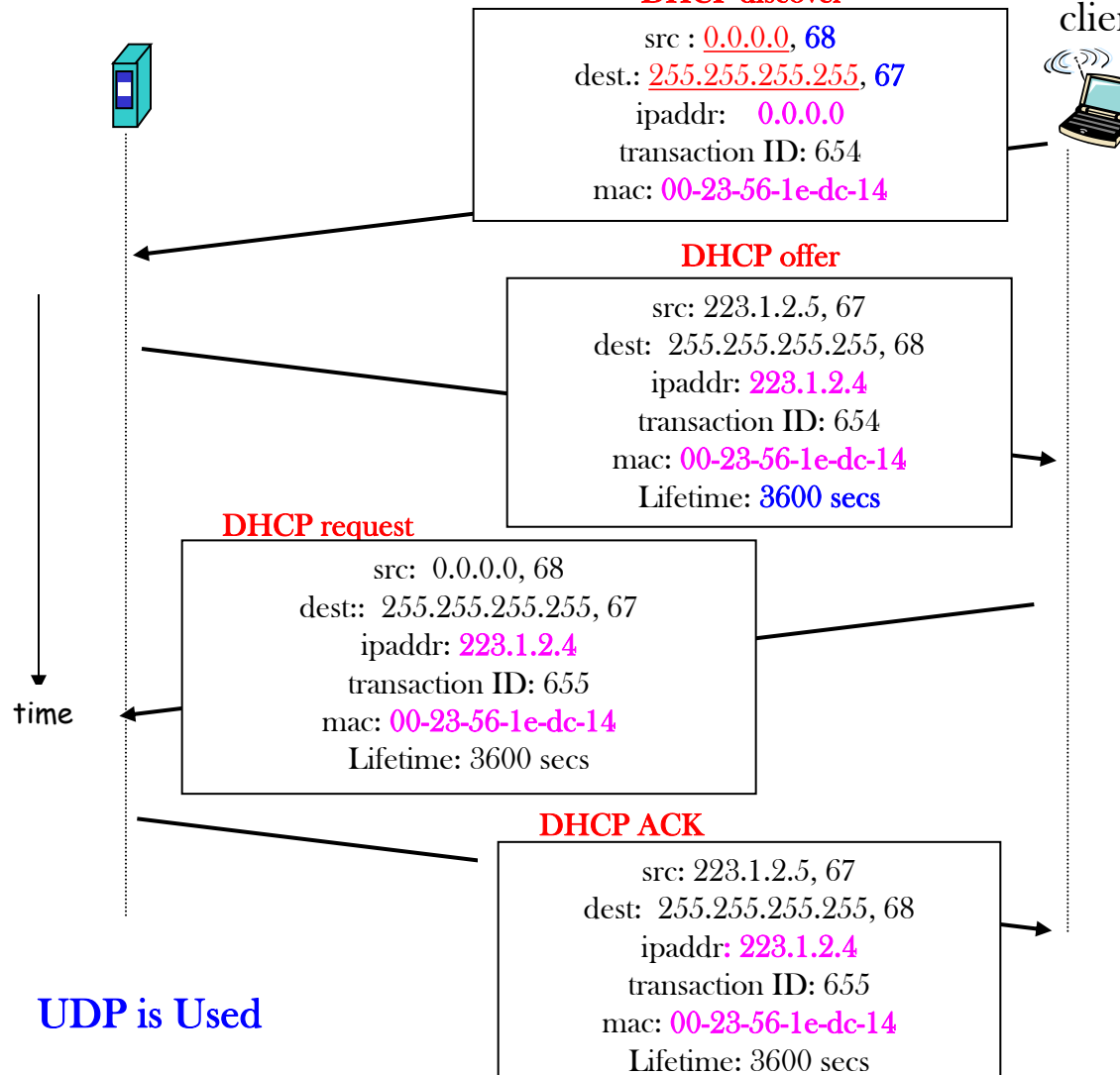
■ What's the difference of ARP and DHCP?

DHCP Client-Server Scenario

DHCP server: 223.1.2.5

Arriving
client

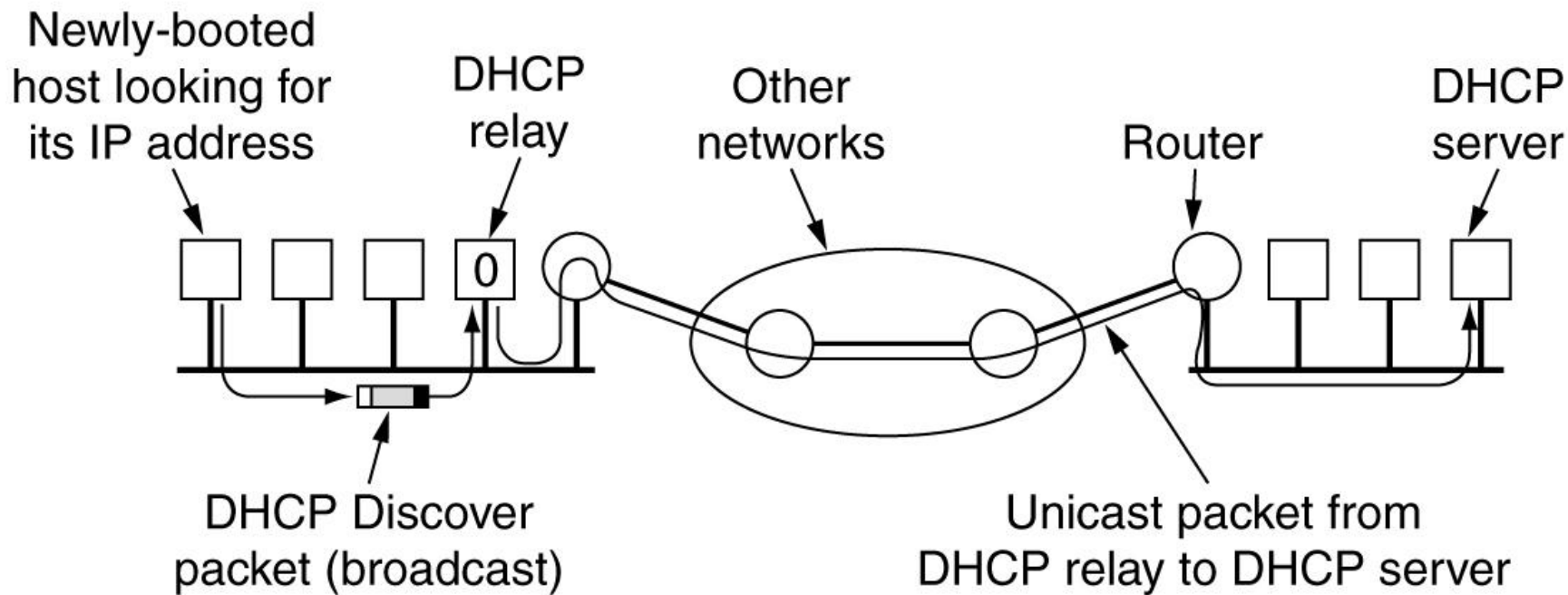
mac: 00-23-56-1e-dc-14



UDP is Used

DHCP Relay Agent(DHCP中继代理)

- When there is no DHCP server in the LAN, DHCP relay agent relays DHCP messages between DHCP clients and DHCP server



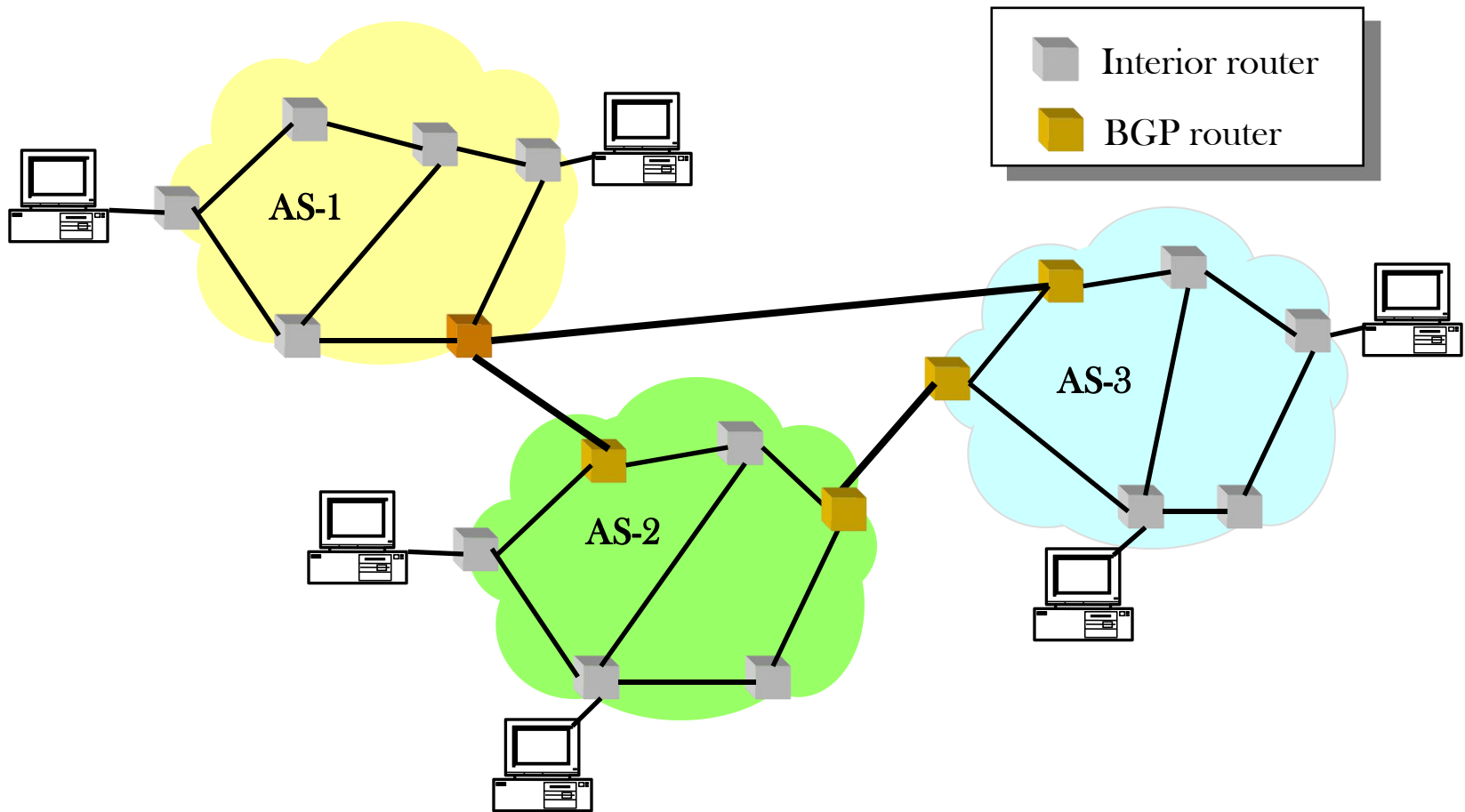
The network layer in the Internet

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- 5.6.6&5.6.7 Routing Protocols: OSPF and BGP

(1) Routing Architecture in the Internet

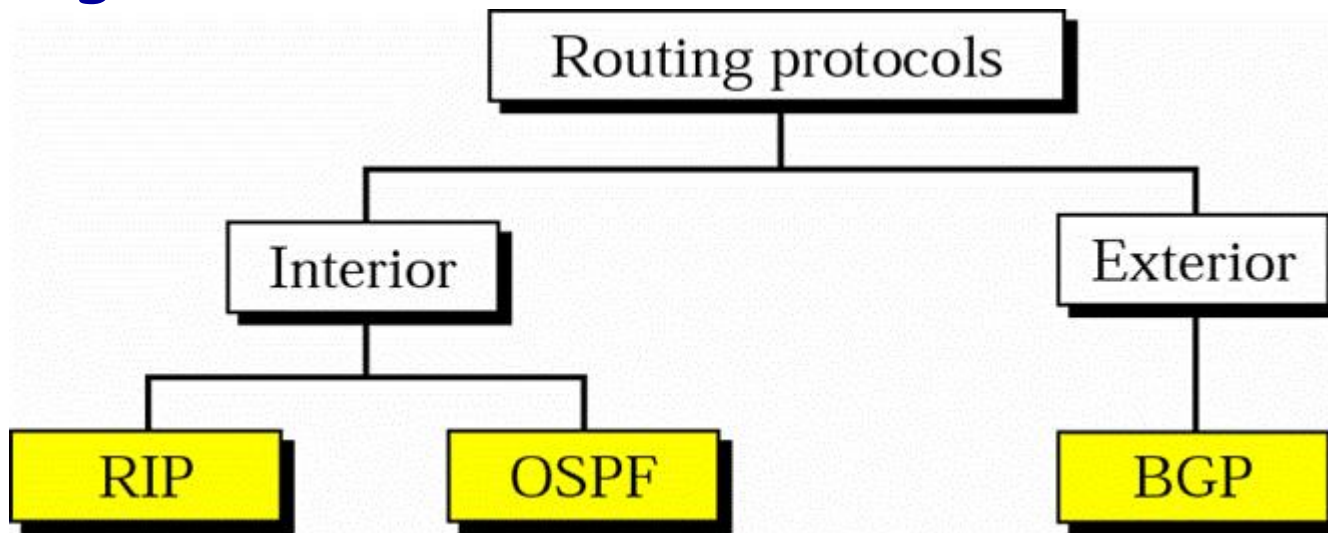
- Internet is divided in *Autonomous Systems (AS, 自治系统)* that are independently administered by individual organizations.
- Each administrative authority use its own routing protocol within the AS, i.e. *Interior Routing Protocol (Interior Gateway Protocol, IGP)*
- One routing protocol is used to exchange routing information among AS, i.e. *Exterior Routing Protocol (Exterior Gateway Protocol, EGP)*

Example



(2) Popular Routing Protocols in Internet

- **RIP**: Distance Vector Routing
- **OSPF, IS-IS**: Link State Routing
- **BGP**: Revised Distance Vector, Path Vector Routing



[Forouzan]

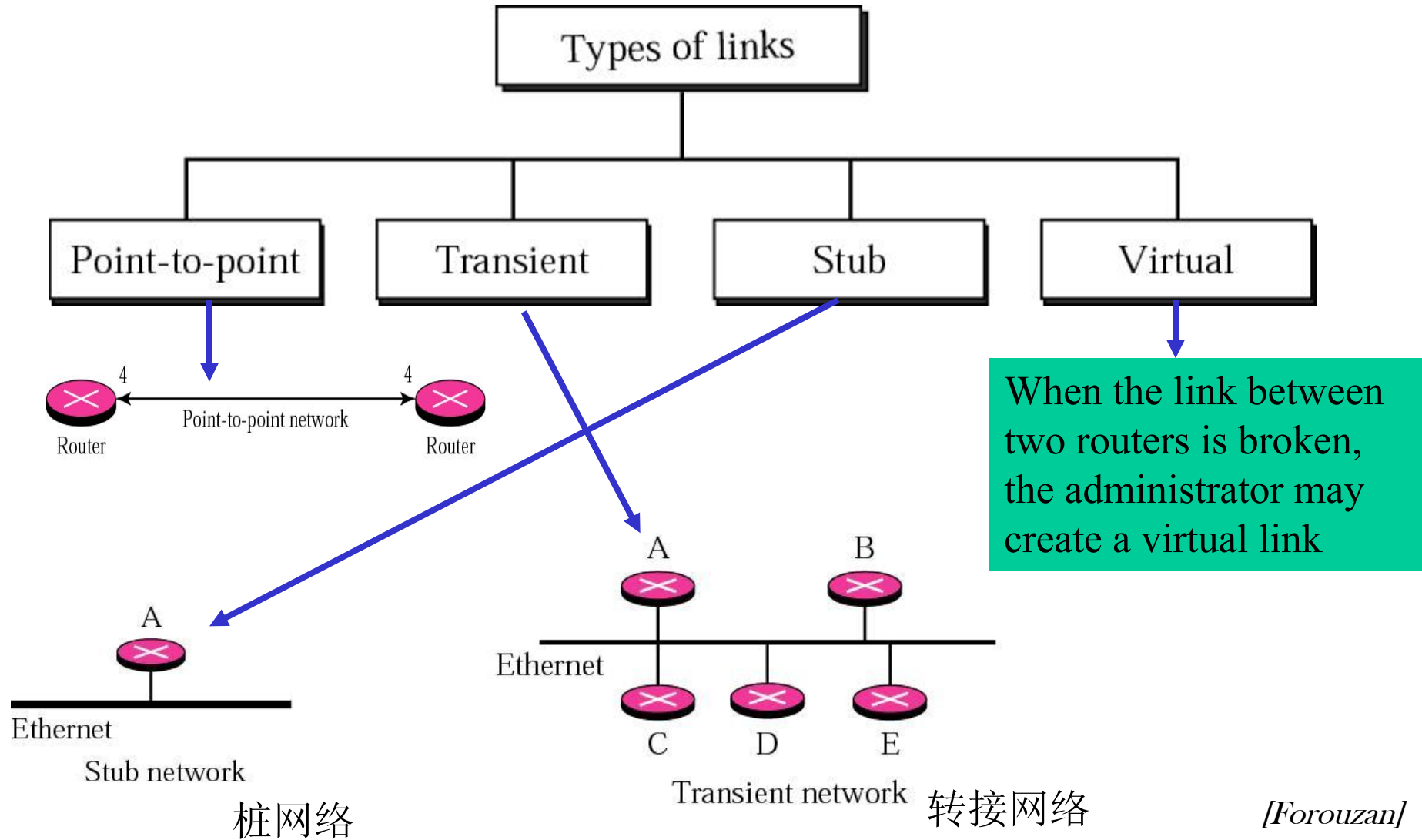
(3) RIP: Features

- Routing Information Protocol (路由信息协议)
- Direct implementation of Distance Vector routing
- Path costs are based on number of hops
 - ◆ RIP treats all network equals; the cost of passing thru a network is the same: one hop count per network
- Number of hops is limited to 16, lack of Scalability
- RIP messages are transmitted in UDP packets

(4) OSPF: Features

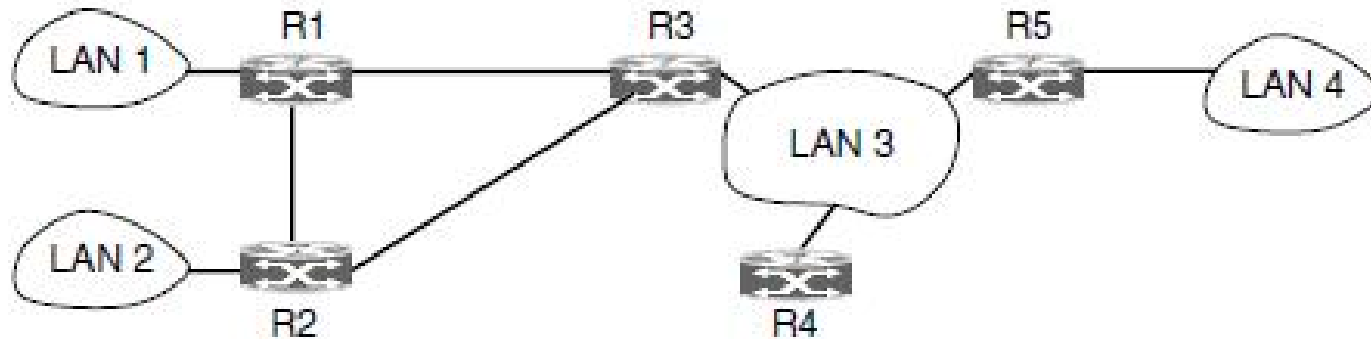
- Open Shortest Path First (开放最短路径优先)
- Based on Link State Routing
- OSPF allows the administrator to (only) assign a cost for passing through a network based on the type of service required.
- Each router should have the exact topology of the internet at every moment.
- OSPF standard supports novel concepts such as type of service routing, load balancing and authentication.
- OSPF messages are transported directly in IP packets

OSPF: Types of Links

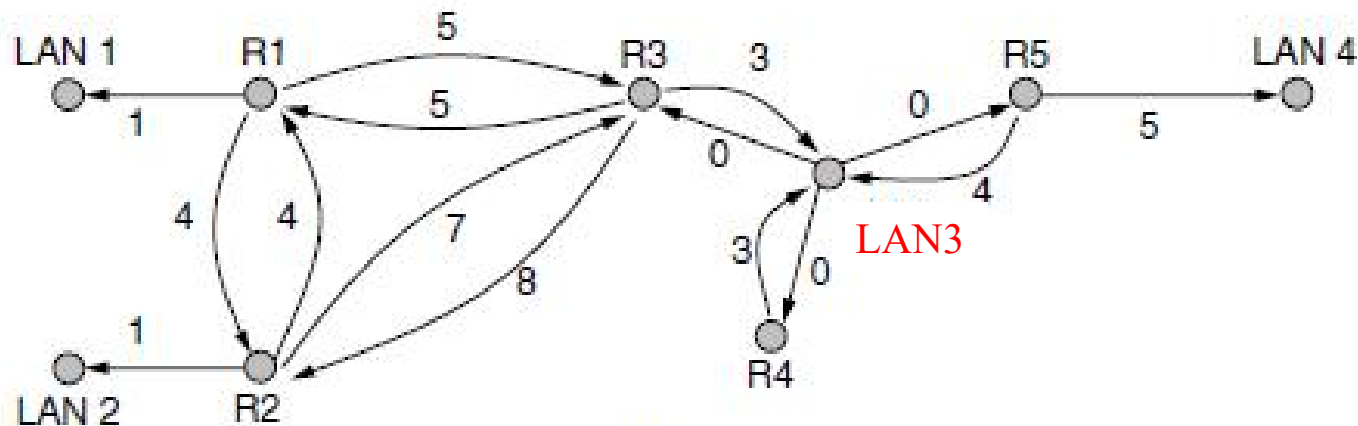


[Forouzan]

OSPF: Graph Representation

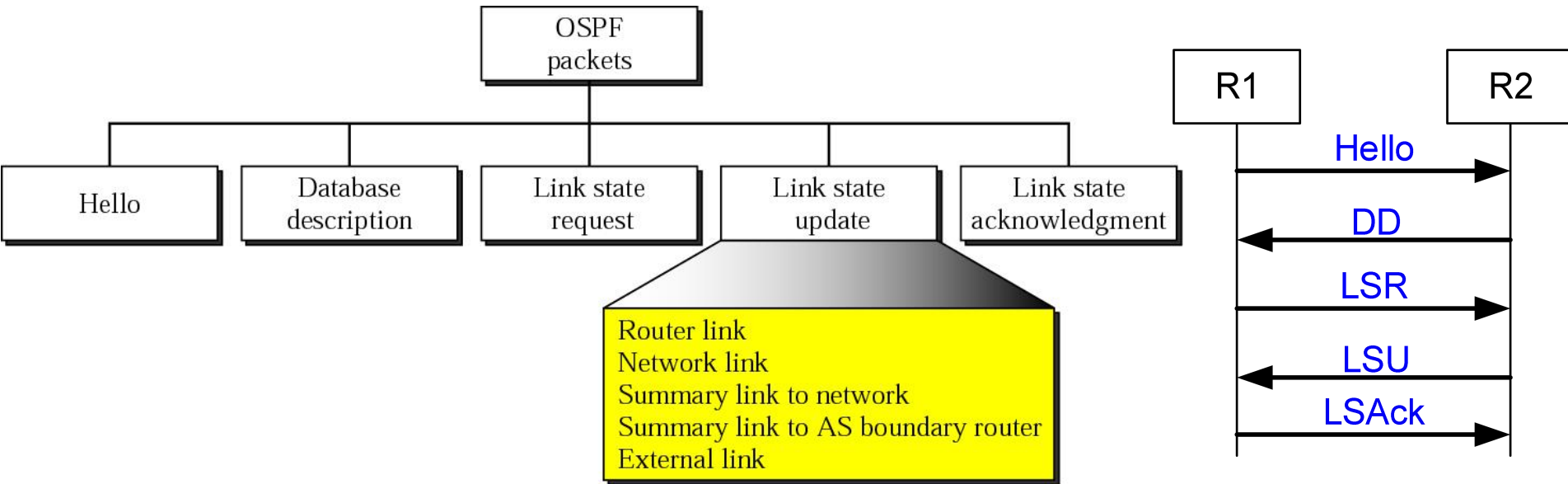


(a)



(b)

OSPF Packets



Message type	Description
Hello	Used to discover who the neighbors are
Link state update	Provides the sender's costs to its neighbors
Link state ack	Acknowledges link state update
Database description	Announces which updates the sender has
Link state request	Requests information from the partner

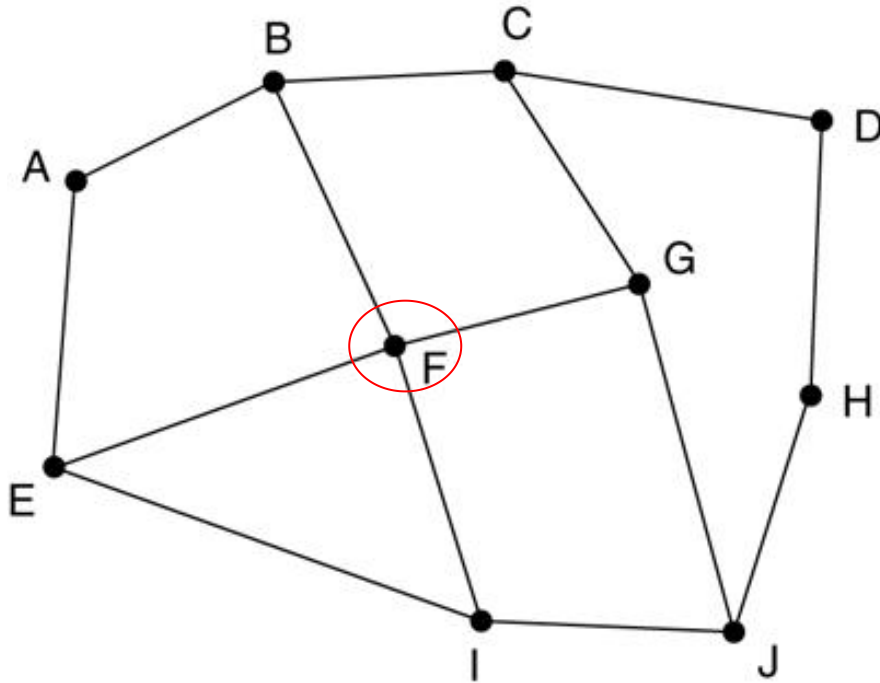
RIP vs. OSPF

Features	RIPv1	RIPv2	OSPF
RFC	RFC1058	RFC1723	RFC2178
Scheme	Distance-Vector		Link-state
Range of Metrics	16		65,535
Update Frequency	30s		Changes or 30mins
Dead interval	180s		Variable setting
Authentication	No	Yes	Yes
Convergence	Slow	slow	Fast
Support CIDR	No	Yes	Yes
Type of Service	No	No	Yes
Multiple paths	No	No	Yes
Ease of Use	Yes		Complex Setup

(5) BGP

- Border Gateway Protocol (边界网关协议)
- Path Vector Routing 路径矢量选路
- Distance vector algorithm with extra information
 - ◆ For each route, store the complete path (ASs)
 - ◆ No extra computation, just extra storage
- Advantages
 - ◆ Enables router to perform policy routing
 - E.g. avoid path to avoid transiting particular AS
 - E.g. link speed, capacity, tendency to become congested, and overall quality of operation, security
 - E.g. minimizing number of transit ASs
 - ◆ Easily avoid loops
- BGP using TCP to carry its messages, reliable delivery!

BGP Example



(a)

Information F receives
from its neighbors about D

From B: "I use BCD"
From G: "I use GCD"
From I: "I use IFGCD"
From E: "I use EFGCD"

(b)

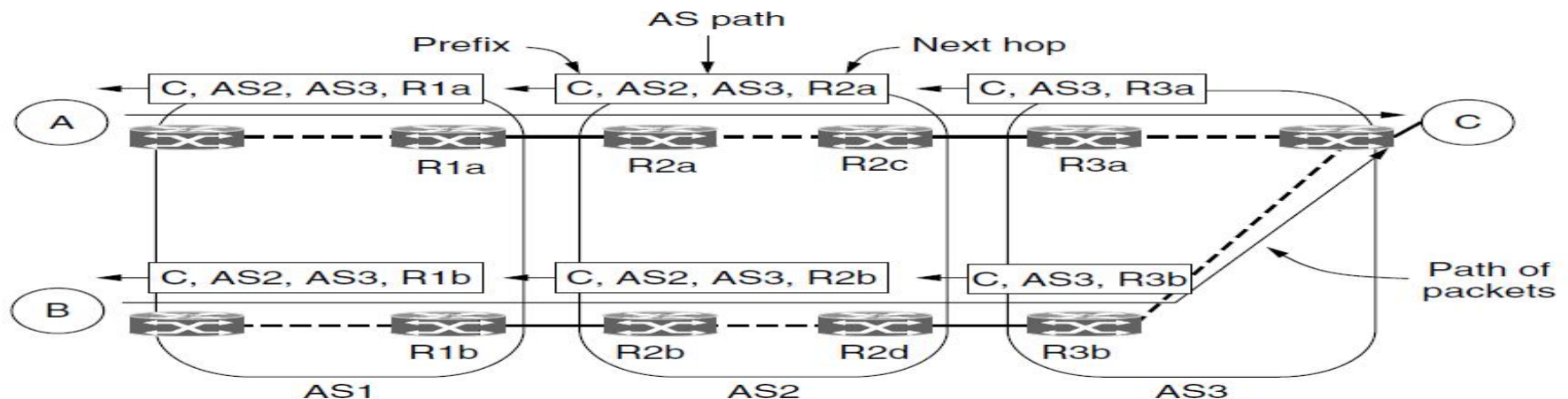
BGP: Routing Table Example

Network	Next Router	Path
N01	R01	AS14, AS23, AS67
N02	R05	AS22, AS67, AS05, AS89
N03	R06	AS67, AS89, AS09, AS34
N04	R12	AS62, AS02, AS09

[Forouzan]

How BGP Routers Choose Route

- Detect and break routing loop
- Routes via peered networks are chosen in preference to routes via transit providers
- Shorter **AS paths** are better
- Prefer the route that has the lowest cost within AS
 - ◆ known as early exit or hot-potato routing
 - ◆ It has the curious side effect of tending to make routes asymmetric



Key Points(1)

- Introduction to Network Layer
 - ◆ Position and functions of network layer
 - ◆ Virtual Circuit Switching vs. Datagram Routing
- Routing Algorithms
 - ◆ Static routing vs. Dynamic routing
 - ◆ Sink tree as benchmark
 - ◆ Flooding
 - ◆ Dijkstra algorithm
 - ◆ Distance Vector Routing (DVR): features, how to calculate routing table, drawbacks
 - ◆ Link State Routing (LSR): features, LSP vs. topology
 - ◆ Hierarchical Routing: idea

Key points(2)

■ Congestion Control:

- ◆ Virtual circuit network: Admission control
- ◆ Datagram network: find, notification and decrease sending rate
 - Choke packets, ECN, RED

■ QoS Issues: Token Bucket, Packet Scheduling

■ IP Addressing

- ◆ Classful address: class A, B and C
- ◆ Hierarchical address: Network-id + host-id
- ◆ Network address used in routing
- ◆ Subnetting
- ◆ Supernetting
- ◆ CIDR: slash notation
- ◆ How to calculate subnet mask

Key Points(3)

- IP Packet format
 - ◆ Function of each field
 - ◆ Checksum
 - ◆ Fragmentation
- Routing Protocols: RIP, OSPF and BGP
 - ◆ Features
- Other relative protocols
 - ◆ IPv6: features, addressing, header conversion
 - ◆ ICMP: error report and test
 - ◆ ARP: IP address → MAC address
 - ◆ DHCP: allocate IP address dynamically
 - ◆ IGMP: managing group members
 - ◆ Mobile Routing: idea

Terms

缩写	英文全称	中文
	adaptive algorithms	自适应算法
	addressing	寻址
	address resolution	地址解析
ARP	Address Resolution Protocol	地址解析协议
AS	Autonomous Systems	自治系统
	Admission control	接纳控制
BGP	Border Gateway Protocol	边界网关协议
	backward learning	逆向学习
	broadcast storm	广播风暴
	bridge	网桥
	broadcasting	广播
	checksum	校验和/检查和
	classful addressing	分类地址
	classless addressing	无类别地址
CIDR	Classless InterDomain Routing	无类别域间选路

缩写	英文全称	中文
	Choke Packet	抑制分组
	circuit switching	电路交换
	count-to-infinity	无穷计数
	congestion control	拥塞控制
	connectionless service	无连接服务
	connection-oriented service	面向连接的服务
	cut-through switching	直通交换
	datagram	数据报
	datagram network	数据报网络
DHCP	Dynamic Host Configuration Protocol	动态主机配置协议
	default mask	默认掩码
DVR	Distance Vector Routing	距离矢量选路
ECN	Explicit Congestion Notification	显式拥塞通知

缩写	英文全称	中文
	Ethernet	以太网
	Fair Queue	公平队列
	FIFO Queuing	先进先出队列
	Fixed Routing	固定查表选路
	flooding	洪泛
	flow control	流量控制
	forwarding	转发
	fragmentation	分段/分片
	fragmenting	分段/分片
	gateway	网关
	Hop-by-Hop	逐跳
	Hierarchical Routing	分级选路
ICMP	Internet Control Message Protocol	因特网控制报文协议
IGMP	Internet Group Management Protocol	因特网组管理协议

缩写	英文全称	中文
	implicit notification	隐式通知
	incoming interface	输入接口
	infinite loop	无限循环
	internet	互联网
	Internet	因特网
	internetworking	互通
IP	Internet Protocol	网际协议
	jitter	时延抖动
LAN	Local Area Network	局域网
	Leaky Buckets	漏桶
LSP	Link State Packet	链路状态包
LSR	Link State Routing	连路状态选路
	Longest-Prefix Matching	最长前缀匹配
MTU	Maximum Transmission Unit	最大传输单元
	multicasting	多播/组播

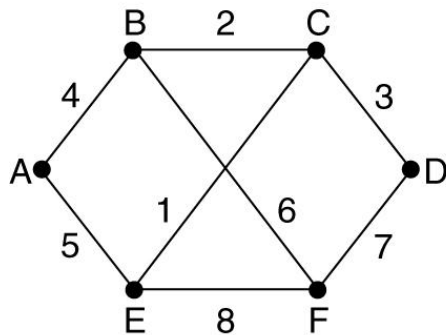
缩写	英文全称	中文
	nonadaptive algorithms	非自适应算法
	network address	网络地址
NAT	Network Address Translation	网络地址翻译
	next hop	下一跳
	outgoing interface	输出接口
OSPF	Open Shortest Path First	开放的最短路径优先
	packet	分组/包
	packetizing	打包
	Packet Scheduling	分组调度
	packet switching	分组交换
	prefix	前缀
	Priority Queuing	优先级队列
PVR	Path Vector Routing	路径矢量选路
QoS	Quality of Service	服务质量
	re-assembly	重装

缩写	英文全称	中文
RARP	Reverse Address Resolution Protocol	逆向地址解析协议
RED	Random Early Detection	随机早期检测
	Random Routing	随机选路
	Reverse Path Forwarding	逆向路径转发
	round robin	循环
	round-trip time	环回时间
	Route Aggregation	路由聚合/会聚
RIP	Routing Information Protocol	选路信息协议
	router	路由器
	routing	选路
	routing algorithm	选路算法
	routing table	路由表
	Resource Reservation	资源预留
	Sink Tree	宿树/汇集树
	spanning tree	生成树)
	static routing	静态选路

缩写	英文全称	中文
	store-and-forward	存储转发
	subnet	子网
	subnet mask	子网掩码
	subnetting	子网划分
	supernetting	超网划分
	supernetwork	超网
	switch	交换机
	Token Buckets	令牌桶
	Traffic Shaping	业务量(流量)整形
TTL	Time-to-Live	生命期
	Tunneling	隧道
VC	Virtual Circuit	虚电路
VCI	Virtual Circuit Identifier	虚电路标识符
	virtual circuit network	虚电路网络
WAN	Wide Area Network	广域网
WFQ	Weighted Fair Queuing	公平加权队列

Self-Study

LSP: Example



(a)

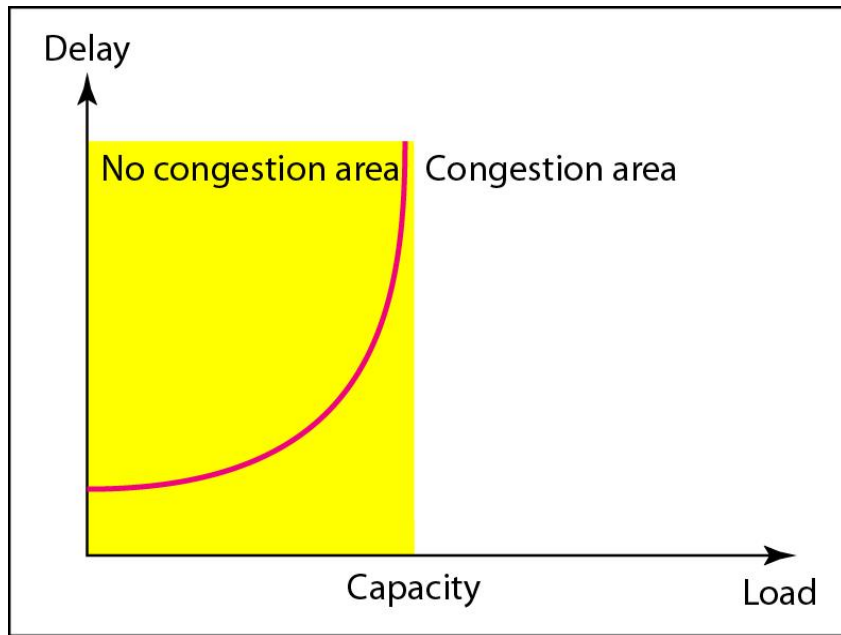
Link		State		Packets	
A		B		C	
Seq.		Seq.		Seq.	
Age		Age		Age	
B	4	A	4	B	2
E	5	C	2	D	3
		F	6	E	1

(b)

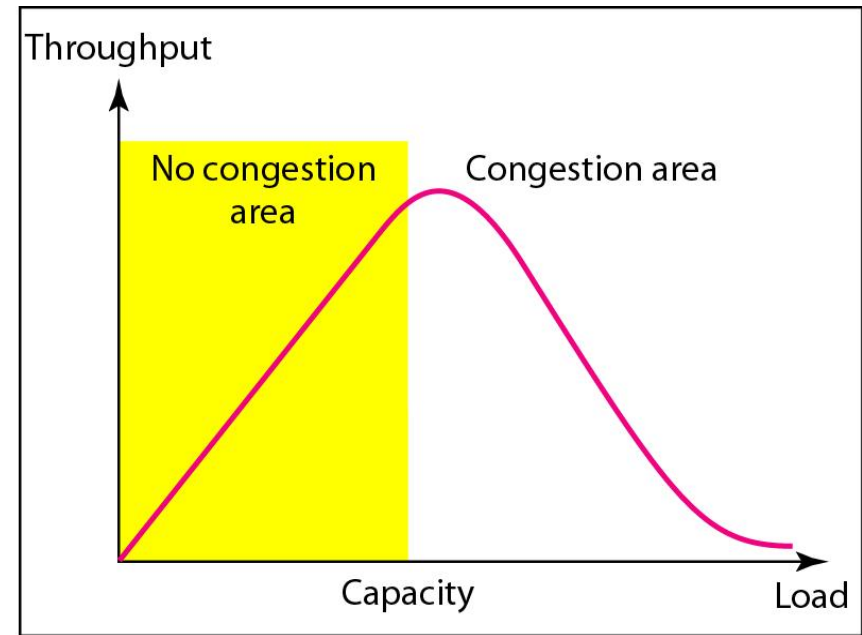
Source	Seq.	Age	Send flags			ACK flags			Data
			A	C	F	A	C	F	
A	21	60	0	1	1	1	0	0	
F	21	60	1	1	0	0	0	1	
E	21	59	0	1	0	1	0	1	
C	20	60	1	0	1	0	1	0	
D	21	59	1	0	0	0	1	1	

Record of LSPs received by B

Network performance: delay(时延) and throughput(吞吐量)



a. Delay as a function of load



b. Throughput as a function of load

■ Congestion Control involves two factors that measure the performance of a network:

- ◆ **Delay**: queuing due to processing delay and propagation delay
- ◆ **Throughput**: number of packets passing through in a unit of time

5.3.1 Network provisioning (网络提供)

- 规划、调整网络，以适应业务量(负载)
- To build a network that is well matched to the traffic that it carries
- Resources are added dynamically when there is serious congestion
- Links and routers that are regularly heavily utilized are upgraded at the earliest opportunity
- 成本高，时间长(约几个月的时间)

5.3.2 Traffic-aware Routing

■ The normal routing schemes

- ◆ Used **fixed link weights**, adapted to changes in topology, but **not** to changes in load

■ Traffic-aware Routing

- ◆ Set **the link weight** to be a function of the **link bandwidth** and **propagation delay** plus the **measured load** or **average queuing delay**
- ◆ **Danger**: **routing tables may oscillate wildly**, leading to erratic routing and many potential problems
- ◆ If load is ignored and only bandwidth and propagation delay are considered, this problem does not occur
- ◆ In **the Internet routing** protocols do **not** generally adjust their routes **depending on the load**

5.3.3 Admission control(接纳/准入控制)

- Widely used in virtual circuit networks
- Idea: do not set up a new virtual circuit unless the network can carry the added traffic without becoming congested. 没有足够资源，就不允许建立VC
- 关键问题：如何描述虚电路的流量特性
- Armed with traffic descriptions(by leaky bucket 漏桶 or token bucket 令牌桶), the network can decide whether to admit the new virtual circuit.
- Combined with traffic-aware routing: considering routes around traffic hotspots as part of the setup procedure

Techniques to improve QoS

- 5.4.2 Traffic shaping
- 5.4.3 Packet Scheduling
- *5.4.4 Admission control(自阉)*
- 5.4.5 Integrated Services(不讲)
- 5.4.6 Differentiated Services(不讲)

Resource Reservation(资源预留) & Admission Control(接纳控制)

■ Resource Reservation

- ◆ Bandwidth.
- ◆ Buffer space.
- ◆ CPU cycles.

Flow: 同一对源/目的主机之间的多个数据包

■ Flow negotiation

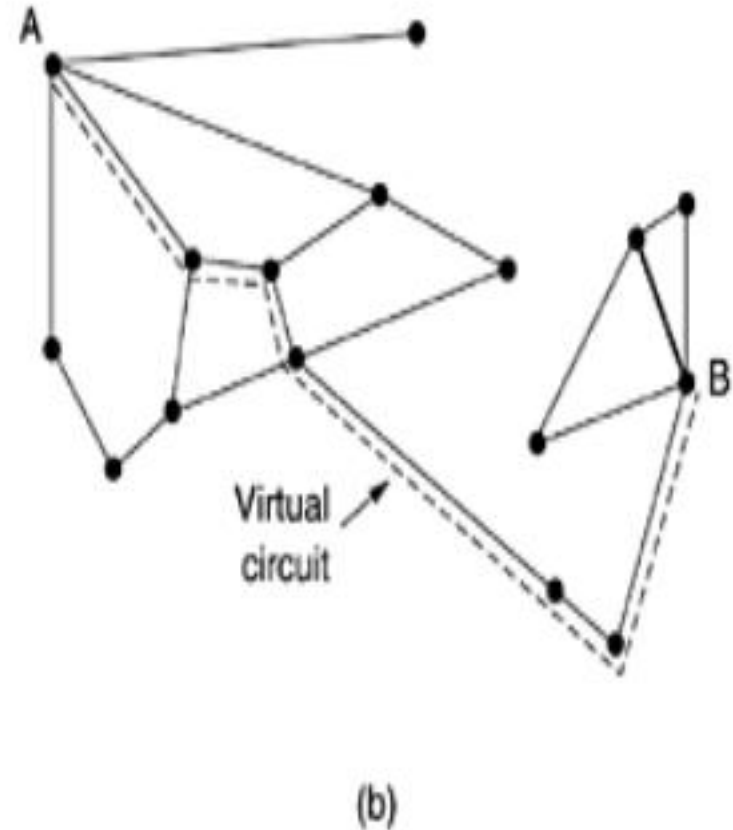
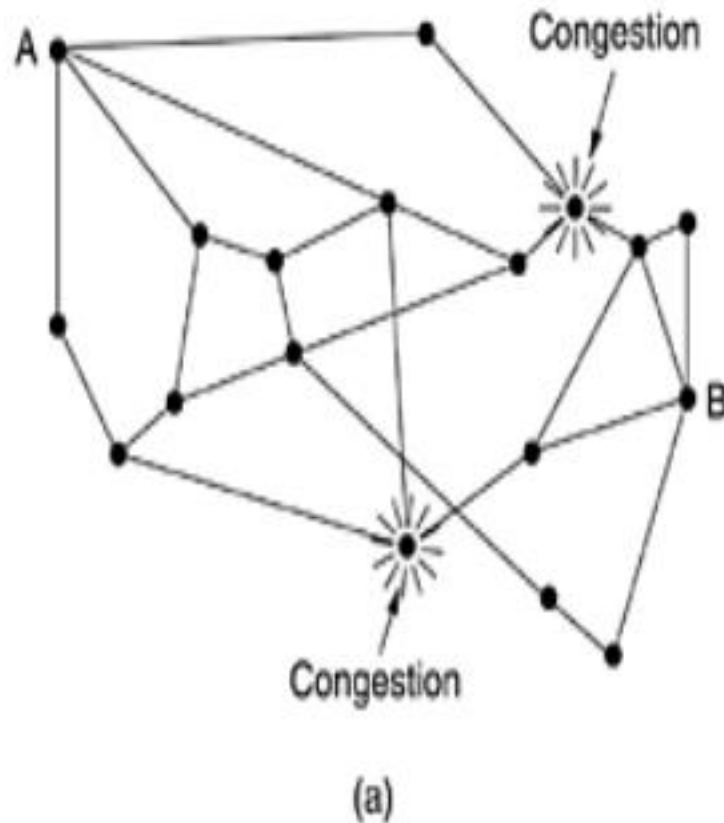
- ◆ Sender, receiver, and all routers along the path are involved in the flow negotiation

■ Flow specification (流描述)

- ◆ 描述一个流对网络的要求
- ◆ 源主机产生
- ◆ 沿途路由器和目的主机可修改

Parameter	Unit
Token bucket rate	Bytes/sec
Token bucket size	Bytes
Peak data rate	Bytes/sec
Minimum packet size	Bytes
Maximum packet size	Bytes

Route new VC around Congestion Areas



The IPv6 Extension Headers

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

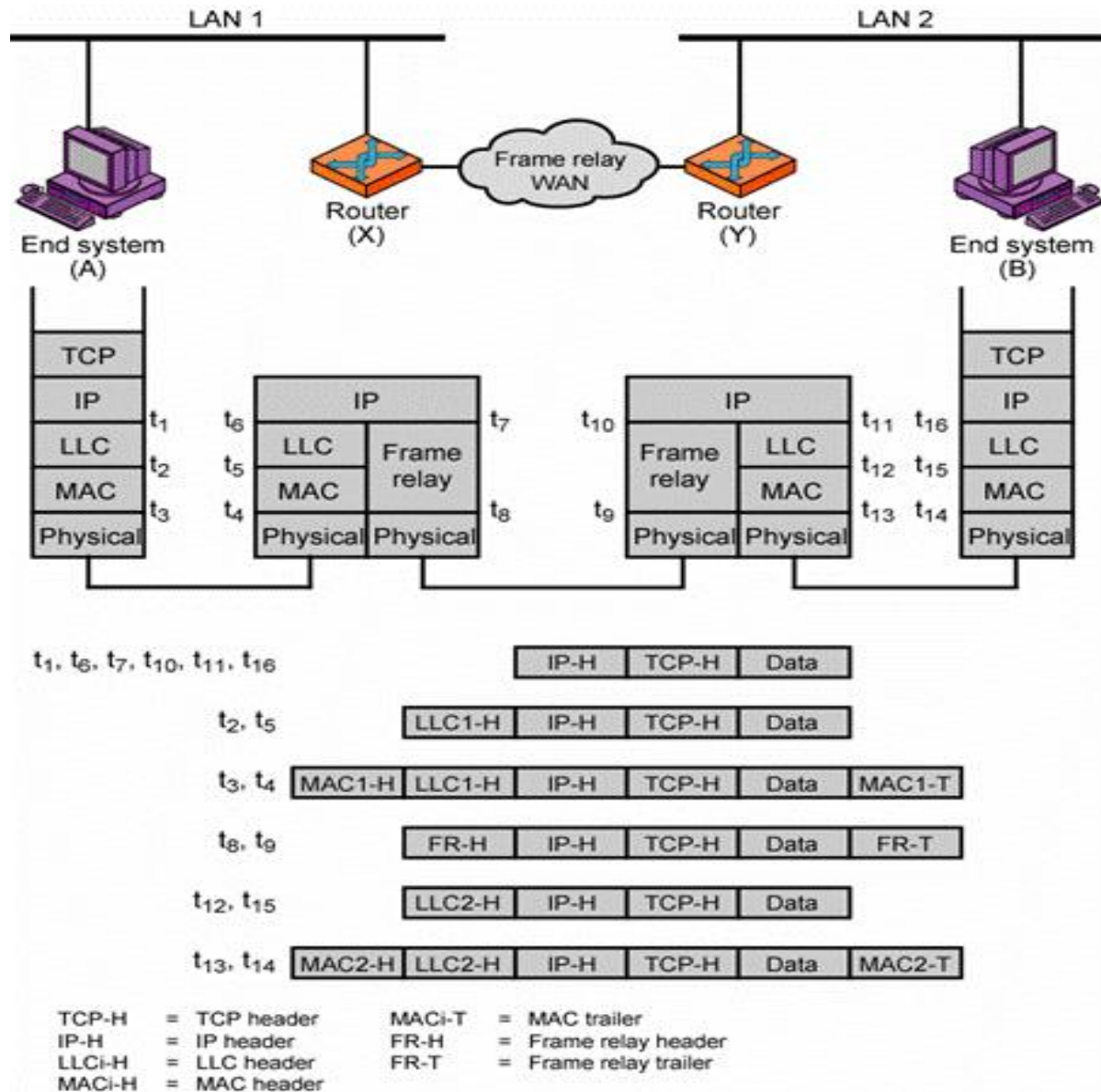
Design Issues of IP

- Design requirements: accommodate differences in constituting networks:
 - ◆ Different addressing schemes
 - ◆ Different maximum packet size
 - ◆ Different network access mechanism
 - ◆ Different timeouts
 - ◆ Different transmission modes (connection-oriented, connectionless)
 - ◆ Error control
 - ◆ Flow control

Different Network Access Mechanism

- Solved by abstracting networking functions in the Networking layer, and relying on Data Link Layer for networking access
- Routers handle the difference in network access mechanism

Operations of IP



[Stallings]

Different Timeout

- Problem: different networks use different timeout mechanisms.
- Solution: Introduce the concept of **Datagram Lifetime**
- Datagrams could loop indefinitely
 - ◆ Consumes resources
 - ◆ Transport protocol may need upper bound on datagram life
- Datagram marked with lifetime
 - ◆ **Time To Live (TTL, 生命期)** field in IP datagram
 - ◆ Once lifetime expires, datagram is discarded (not forwarded)
 - ◆ Hop count

Different Transmission Modes

- Connection-oriented and connectionless modes
- A **connectionless** architecture is adopted by IP

Error Control

- Minimal error control, error check only for the header
- Router should attempt to inform source if packet discarded (using **ICMP**, Internet Control Message Protocol, 因特网控制报文协议)
 - ◆ ICMP messages are encapsulated in IP
 - ◆ Error messages from a router or destination host
 - Destination host is unreachable,
 - TTL reached 0
 - Parameter error
 - ICMP redirect, which tells the source of a better way to a destination, ...
- No guaranteed delivery
- May inform higher layer protocol

Flow Control

- Allows routers and/or destination to limit rate of incoming data
- Limited in connectionless systems
- Send flow control packets
 - ◆ Requesting reduced flow
- Some **ICMP** messages are used for flow control
 - ◆ Source quench messages(源抑制报文, choke packet)

BOOTP(启动协议)

■ BOOTP

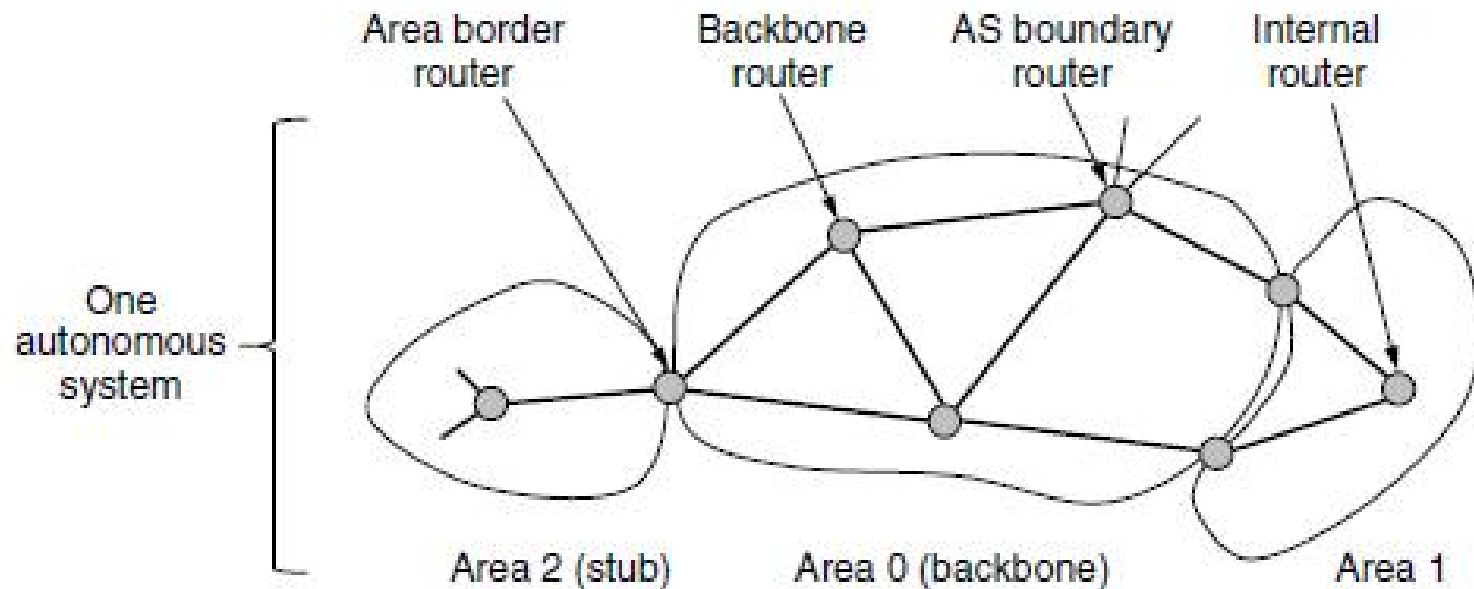
- ◆ Used to configure diskless workstations
- ◆ Unlike RARP, **BOOTP uses UDP messages**, which are forwarded over routers
- ◆ It also provides the **IP address of the file server**, the **IP address of the default router**, and the **subnet mask** to use

■ Problem

- ◆ It requires **manual** configuration of tables mapping IP address to Ethernet address

OSPF: Areas

- OSPF divides an AS(自治系统) into **areas(区域)**, a collection of network, routers all contained within an AS.
- Backbone Router, Boundary Router, Border Router and Internal Router



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